

Foundation Models and Reasoning LLMs for Scientific Multimodal Time Series: A Survey

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Abstract—Time series and spatio-temporal dynamics arising in natural systems—such as global weather, long-term climate, hydrological flood regimes, ocean circulation, earthquake wave propagation, satellite earth observations, and heliophysical radiation—differ fundamentally from generic tabular or urban traffic series. They are governed by continuous non-linear partial differential equations, multi-scale turbulent interactions, heterogeneous spatial geometries, and strict physical conservation laws. Recently, the emergence of pre-trained foundation models (PFMs) and scientific reasoning Large Language Models (LLMs) has instigated a paradigm shift across computational earth and physical sciences. This survey provides the first comprehensive, systematic review of foundation models and reasoning LLMs tailored for scientific multimodal time series. Guided by the PRISMA 2020 methodology, we review landmark architectures developed between 2021 and 2026 across major natural science domains. We introduce a systematic taxonomy structured along four orthogonal axes: (1) scientific domains and multimodal data representations, (2) foundation model architectural backbones (including 3D spherical transformers, multi-mesh graph neural networks, continuous neural operators, and generative diffusion ensembles), (3) levels of physical law integration (from purely data-driven to soft physical penalties and hybrid PDE solvers), and (4) functional roles of LLMs in scientific reasoning (as direct predictors, contextual enhancers, and autonomous tool-orchestrating agents). Furthermore, we curate standardized operational evaluation protocols (such as WeatherBench 2 and SciTS), contrast model scale against forecast horizon, and identify crucial open frontiers spanning universal cross-physics tokenization, physical conservation guarantees, and trustworthy autonomous scientific discovery.

Index Terms—Scientific Foundation Models, Multimodal Time Series, Spatio-Temporal Modeling, Large Language Models, Scientific Reasoning, Weather and Climate Forecasting, Physics-Informed Machine Learning.



1 Introduction

Temporal and spatio-temporal dynamics constitute the empirical foundation of the natural sciences. From the planetary atmospheric circulation driving global weather patterns to the sub-surface acoustic vibrations heralding seismic rupture, physical processes unfold across continuous temporal trajectories and multi-scale spatial coordinates. Over the past several decades, the modeling and prediction of these scientific phenomena have predominantly relied on Numerical Weather Prediction (NWP) and classical numerical differential equation solvers, such as the Integrated Forecasting System (IFS) maintained by the European Centre for Medium-Range Weather Forecasts (ECMWF) and finite-difference hydrodynamic models. While physically principled, numerical solvers incur colossal computational burdens, requiring massive supercomputing clusters to perform high-resolution discretization and step-by-step numerical integration.

In parallel, the broader machine learning community has witnessed a revolution driven by pre-trained foundation models (PFMs) and Large Language Models (LLMs). While foundational models initially transformed natural language processing and computer vision, their underlying principles—self-supervised pretraining on planetary-scale corpora, emergent scaling behaviors, and zero-shot downstream transferability—have rapidly migrated into temporal data mining. Earlier

general surveys, notably the seminal work by Jin et al. [1] and its position sequel [2], comprehensively charted large models for general time series analysis (LM4TS) and spatio-temporal data mining (LM4STD). However, as highlighted in Table 1, existing overviews focus heavily on urban computing, human mobility trajectories, traffic sensor networks, and generic 1D benchmark datasets (e.g., electricity, weather station benchmarks, exchange rates).

Natural physical systems diverge fundamentally from human-generated temporal sequences across several intrinsic dimensions:

- 1) **Governing Physical Laws and Conservation Constraints:** Physical time series are bounded by fundamental conservation laws of mass, energy, momentum, and moisture flux. Models that learn purely statistical correlations often generate unphysical states, such as negative precipitation or violation of energy conservation over extended rollouts.
- 2) **High-Dimensional Multimodal Representations:** Scientific investigations integrate heterogeneous observation modalities, including 3D atmospheric pressure tensors (e.g., ERA5 reanalysis with geopotential height, temperature, and wind fields), multi-spectral satellite imagery (e.g., Sentinel-2, Landsat), river gauge discharge records, and continuous seismic waveforms.
- 3) **Continuous Spatio-Temporal Geometries:** The Earth is a non-Euclidean spherical geometry with topological singularities at the poles, rendering standard 2D

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TABLE 1
Comparison of This Survey with Existing Literature and Related Temporal Surveys.

Survey / Study	Year	Scope Focus	Physical Grounding	Multimodal Modalities	LLM Reasoning	Core Domains Covered
Jin et al. [1]	2023	General Spatio-Temporal	Moderate	General Multimodal	Limited	Urban traffic, human trajectory, generic time series
Jin et al. [2]	2024	LLM for Time Series	Low	Text + 1D Series	Moderate (Position)	Benchmark time series forecasting, tokenization
Rasp et al. [3]	2024	Weather Benchmark	High (Meteorology)	Gridded NWP Reanalysis	None	Global numerical and ML weather forecasting
This Survey	2026	Scientific Multimodal	High (PDEs & Laws)	Gridded + Satellite + Waveforms + Text	Comprehensive (Agents & SciTS)	Weather/Climate, Earth Observation, Hydrology, Oceans, Seismology, Heliophysics

Euclidean convolutions and planar patch projections geometrically distorted.

- 4) The Need for Scientific Reasoning and Autonomous Discovery: Beyond numerical point prediction, domain scientists demand causal attribution, physical hypothesis testing, and interactive tool orchestration. This has catalyzed the emergence of scientific reasoning LLMs and autonomous scientific agents capable of operating over complex temporal data.

To bridge this literature gap, this survey delivers the first systematic, domain-science-centered investigation of foundation models and reasoning LLMs developed for scientific multimodal time series. We synthesize breakthroughs from 2021 through 2026 across atmospheric sciences, Earth observation, hydrology, oceanography, seismology, and heliophysics.

1.1 Contributions of This Survey

The primary contributions of this survey are summarized as follows:

- Domain-Science Centered Scoping: We delimit the survey strictly to natural physical systems, explicitly analyzing the coupling between continuous physics, governing equations, and deep representation architectures.
- Four-Dimensional Taxonomy Framework: We establish an orthogonal classification scheme covering (1) Scientific Domains and Modalities, (2) Foundation Model Architectures, (3) Physics Integration Levels, and (4) Functional Roles of Reasoning LLMs.
- Systematic Review Protocol: Adhering to PRISMA 2020 standards, every candidate study is screened against strict empirical criteria and validated via official academic API records.
- Synthesis of Evaluation Protocols and Frontiers: We analyze standardized scientific benchmarks (such as WeatherBench 2 and SciTS), map spatial resolution against maximum forecast lead time, and delineate critical open challenges in physical conservation, multi-physics tokenization, and autonomous scientific discovery.

2 Background and Preliminaries

2.1 Formal Data Modality Formulations

Scientific multimodal temporal datasets exhibit diverse mathematical structures depending on observation platforms, sensing geometries, and sampling frequencies. We formalize the four dominant scientific temporal modalities as follows:

2.1.1 In-Situ Station and Gauge Time Series

Discrete ground monitoring networks (e.g., weather stations, hydrological streamflow gauges, ocean buoys) record continuous physical variables at fixed geospatial coordinates (u_i, v_i) . An in-situ observation sequence is defined as:

$$\mathbf{X}^{(i)} = \{\mathbf{x}_t^{(i)} \in \mathbb{R}^{C_{\text{station}}}\}_{t=1}^T, \quad (1)$$

where T represents the temporal horizon and C_{station} denotes the recorded physical channels (e.g., precipitation, river discharge, 2-meter air temperature).

2.1.2 Gridded Global/Regional Reanalysis Tensors

Atmospheric and oceanic numerical simulations discretize the planetary sphere $\mathbb{S}^2$ into a regular latitude-longitude grid with H latitudinal points and W longitudinal points across L vertical vertical pressure levels. The state tensor at time step t is expressed as:

$$\mathcal{X}_t \in \mathbb{R}^{H \times W \times L \times C_{\text{grid}}}, \quad (2)$$

where C_{grid} incorporates upper-air variables (geopotential height z , temperature T , zonal and meridional wind components u, v , specific humidity q) alongside surface diagnostic fields (mean sea-level pressure msl , surface solar radiation).

2.1.3 Multi-Spectral Remote Sensing Cubes

Earth observation satellites (e.g., Sentinel-2, Landsat-8) capture surface reflectance over space, time, and electromagnetic spectral bands. A multi-temporal remote sensing observation cube is represented as:

$$\mathcal{S}_{1:T} \in \mathbb{R}^{T \times B \times H_s \times W_s}, \quad (3)$$

where B denotes the number of spectral bands (visible RGB, near-infrared, short-wave infrared, synthetic aperture radar polarization), and (H_s, W_s) denotes the spatial pixel footprint.

2.1.4 High-Frequency Continuous Waveforms

Geophysical and seismological networks deploy seismometers recording ground motion acceleration or velocity across three orthogonal physical axes (Vertical Z , North-South N , East-West E). A continuous seismic waveform segment is formalized as:

$$\mathbf{W} \in \mathbb{R}^{3 \times N_s}, \quad (4)$$

where $N_s = f_s \cdot \Delta\tau$ is the number of discrete digital samples collected at sampling frequency f_s (typically 50–100 Hz).

2.2 Core Problem Formulations

2.2.1 Deterministic Autoregressive Rollout Forecasting

$$\hat{\mathcal{X}}_{t+1:t+K} = \mathcal{M}_\theta(\mathcal{X}_{t-M+1:t}), \quad (5)$$

optimized via latitude-weighted multi-step rollout loss:

$$\mathcal{L}_{\text{det}} = \sum_{k=1}^K \lambda_k \|\hat{\mathcal{X}}_{t+k} - \mathcal{X}_{t+k}\|_{\mathbf{w}_{\text{lat}}}^2, \quad (6)$$

where $\|\cdot\|_{\mathbf{w}_{\text{lat}}}$ represents a latitude-weighted norm accounting for spherical grid cell area convergence toward the poles:

$$w_{\text{lat}}(i) = \frac{\cos(\phi_i)}{\frac{1}{H} \sum_{j=1}^H \cos(\phi_j)}. \quad (7)$$

2.2.2 Probabilistic Ensemble Generation

Because chaotic atmospheric and oceanic dynamics exhibit sensitivity to initial conditions (the butterfly effect), deterministic point forecasts blur at longer lead times. Probabilistic foundation models aim to sample from the conditional predictive distribution:

$$p_\theta(\mathcal{X}_{t+1:t+K} \mid \mathcal{X}_{t-M+1:t}), \quad (8)$$

minimized using Continuous Ranked Probability Score (CRPS) or diffusion score-matching loss objectives [4].

2.2.3 Scientific Question-Answering and Autonomous Reasoning

Let $\mathcal{Q}$ represent a natural language scientific inquiry, and $\mathcal{D}_{\text{time}}$ represent an associated temporal or spatio-temporal data sequence. A scientific reasoning LLM $\mathcal{F}_{\text{LLM}}$ generates explanatory reasoning chains and structured conclusions $\mathcal{Y}$:

$$\mathcal{Y} = \mathcal{F}_{\text{LLM}}(\mathcal{Q}, \mathcal{E}(\mathcal{D}_{\text{time}})), \quad (9)$$

where $\mathcal{E}$ is a modality-specific continuous tokenizer or latent temporal encoder [5].

3 Taxonomy and Overview

To systematically classify the rapidly expanding body of literature on scientific multimodal time series foundation models and reasoning LLMs, we construct a four-dimensional taxonomy framework, as illustrated in Fig. 1. Unlike general-purpose temporal surveys that divide models simply by task (e.g., forecasting vs. imputation), our categorization captures the unique physical, mathematical, and algorithmic requirements of scientific domain applications.

3.1 Dimension 1: Scientific Domains and Modalities

Scientific time series foundation models specialize across distinct physical environments:

- **Weather and Climate:** Characterized by 3D spherical atmospheric reanalysis fields (ERA5, MERRA-2, IFS HRES), satellite radiometer scans, boundary-layer turbulence profiles, and ground radar nowcasting sequences [6], [7], [8], [9], [10], [11], [12], [13].
- **Earth Observation and Remote Sensing:** Composed of irregular multi-spectral satellite revisit series (Sentinel-1 SAR, Sentinel-2 Optical, Landsat-8), high-resolution elevation maps, 3D spaceborne LiDAR waveforms, and

hyperspectral cubes [14], [15], [16], [17], [18], [19], [20], [21], [22].

- **Hydrology and Water Resources:** Integrating multi-basin river streamflow gauges, topological river drainage networks, and real-time urban hydrodynamic sensors for flood routing [23], [24], [25].
- **Oceanography and Marine Systems:** Involving global eddy-resolving reanalysis fields (e.g., GLORYS12V1), wide-swath interferometric radar altimetry (SWOT KaRIn), internal solitary wave signatures, sea surface height anomalies, temperature/salinity profiles, and acoustic sensor logs [26], [27], [28], [29], [30].
- **Geophysics and Seismology:** Continuous 3-component seismic waveforms, phase arrival triggers, fault slip observations, and GPS tectonic deformation velocities [31], [32], [33].
- **Space Weather and Heliophysics:** Continuous solar extreme ultraviolet (EUV) magnetograms, 3D coronal magnetohydrodynamics (MHD), and spherical heliospheric solar wind velocity streams [34], [35], [36].
- **Polar Cryosphere and Sea Ice Dynamics:** Involving satellite synthetic aperture radar (SAR) swaths, microwave radiometer brightness temperatures, ice-sheet altimetry, and continuous sea ice concentration fields governing planetary thermodynamic albedo [37], [38].
- **Geo-Energy and Subsurface Reservoirs:** Multiphase fluid flow in heterogeneous porous media, geological carbon sequestration (GCS) plume migration, geothermal reservoir temperature/pressure dynamics, and high-pressure equation-of-state (EOS) phase behavior [39], [40].
- **Cross-Domain Multimodal Foundations:** Universal time series representations bridging sensor networks, in-situ monitors, and multi-scale physical dynamics [41], [42].
- **Scientific Reasoning and Causal Discovery:** Natural language hypothesis generation, domain-guided causal graph estimation, teleconnection attribution, and multi-step symbolic-numerical reasoning [5], [43], [44], [45].

As visualized in the Domain $\times$ Modality Research Maturity Matrix (Fig. 2), while gridded weather reanalysis and multi-spectral satellite imagery have achieved high maturity, cross-modal integration (e.g., coupling continuous waveforms with satellite imagery, subsurface permeability with multi-phase plumes, spaceborne LiDAR with multi-spectral cubes, wide-swath radar altimetry with submesoscale ocean solvers, or ground gauges with scientific text logs) remains an active frontier.

3.2 Dimension 2: Foundation Model Architectures

We categorize foundation backbones by their spatial-temporal inductive biases:

- **3D Spatial and Spherical Transformers:** Employing 3D Earth-specific patch embeddings, hierarchical Swin-like windowed attention, 3D generative spectral pre-training, and vertical pressure-level encodings [6], [9], [46], [10], [21].
- **Multi-Mesh Graph Neural Networks:** Discretizing the spherical Earth via refined recursive icosahedral

Taxonomy Tree: Categorization of Scientific Multimodal Time Series Foundation Models



Fig. 1. Comprehensive taxonomy framework for scientific multimodal time series foundation models and reasoning LLMs, structured across four orthogonal dimensions: Scientific Domains & Modalities, Foundation Architectures, Physics Integration Levels, and Roles of Reasoning LLMs.

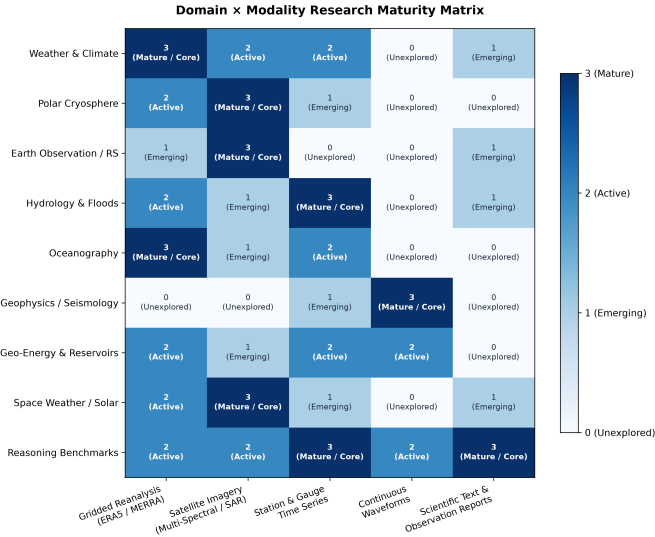


Fig. 2. Domain x Modality Research Maturity Matrix, highlighting established core areas and emerging multimodal frontiers across $N = 95$ verified studies.

meshes to ensure uniform spatial resolution without polar distortions [7].

- Continuous Neural Operators: Leveraging Fourier transforms and spherical harmonic convolutions (e.g.,

AFNO, SFNO) to learn discretization-invariant operators on Riemannian function spaces [8], [47], [36].

- Clifford Neural Operators and Geometric Algebras: Unifying scalar, vector, and rotational bivector fields via multivector representations and Clifford Fourier transforms for turbulent flows (CFNO) [12].
- Dual-Branch Global-Local Neural Operators: Decoupling macroscale topology from fine-scale reconnection dynamics via multi-resolution spectral branches for space weather magnetohydrodynamics (GL-FNO) [35].
- Deformation and Geometry-Adapted Neural Operators: Mapping complex non-Euclidean computational domains, irregular boundaries, and faulted subsurface formations into canonical coordinate spaces via learned coordinate diffeomorphisms (Geo-FNO) [48].
- Symplectic and Lie-Poisson Geometric Operators: Structuring operator layers according to canonical symplectic 2-forms or non-canonical Lie-Poisson brackets to guarantee exact preservation of Hamiltonian energy, Casimir invariants, and Kelvin's circulation theorem [49], [50].
- High-Order Tensor and Quantum-Classical Hybrid Operators: Factorizing infinite-dimensional spectral convolution weights via Tucker and Tensor-Train factorizations or embedding spectral transformation into parameterized quantum circuits (PQC) for scalable

PDE solving and wavefield tomography [51], [52], [53].

- Distribution-Free Conformal Uncertainty Wrappers: Constructing model-agnostic conformal prediction layers to provide statistically valid, finite-sample prediction intervals across spatial weather grids and forecasting lead times [54].
- Scale-Adaptive Multi-Modal Remote Sensing Architectures: Unifying multi-constellation SAR and optical imagery via cross-modal contrastive autoencoding with 2D-ALiBi spatial extrapolation and scale-conditioned joint embedding architectures (CROMA, AnySat) [55], [56].
- Scale-Translation Equivariant Networks for Ocean Waveforms: Injecting continuous scale-translation Lie symmetry to detect and invert oceanic internal solitary waves from spaceborne radar altimetry and SAR (STE-Net) [30].
- Asynchronous Coupled Earth System Emulators: Coupling multi-rate spherical neural operators for atmospheric, oceanic, and cryospheric dynamics across continuous interfacial flux boundary couplers (DLESyM, SamudrACE) [57], [58].
- Multimodal Masked Autoencoders and Continuous Hypernetworks: Extending masked autoencoding (MAE) across spatial patches, temporal revisit cycles, and dynamic wavelength hypernetworks [14], [15], [16], [17], [20], [42].
- Generative Diffusion and Lagrangian Assimilation Models: Utilizing score-based diffusion denoising to generate calibrated probabilistic ensembles and assimilate sparse, irregular Lagrangian drifter trajectories into dense Eulerian state estimates [4], [11], [59], [60].
- Simulation-Informed Generative Ensemble Models for Radar Topography: Combining wavelet-informed multiscale neural metrics with adversarial priors to denoise wide-swath SWOT radar interferometry (SIMP-GEN) [29].
- Unified Multi-Task Foundational Transformers: Applying prompt-conditioned masking and task tokens across heterogeneous time series topologies [41], [19].

3.3 Dimension 3: Physics Integration Levels

The physical fidelity of deep learning models ranges along four distinct levels:

- 1) Purely Data-Driven: Models optimize statistical reconstruction (e.g., mean squared error, MAE) over historical datasets without explicit differential constraints, relying solely on model capacity to reflect physical dynamics [6], [7], [19], [55], [56].
- 2) Soft Physics Loss Penalties: Physical conservation principles (e.g., kinetic energy conservation, mass balance, divergence-free constraints, vertical pressure weighting) are formulated as penalty terms within the training loss function [61], [8], [10], [11], [39], [57], [29].
- 3) Hard Architectural Conservation Constraints: Structural projections and geometric invariants (e.g., spherical harmonics basis expansion, divergence-free projection layers, canonical symplectic structure, Lie-Poisson Casimirs, multivector grade equivariance,

and exact terminal linear projection layers enforcing column mass, water, and enthalpy conservation to machine precision) guarantee that predictions strictly satisfy conservation laws by construction [47], [49], [50], [62], [63], [12], [30].

- 4) Hybrid PDE-Neural Solvers: Deep neural representations parameterize unresolved subgrid physical processes (convection, clouds, turbulence, wall-shear stresses, reservoir multiphase equilibria) coupled directly to differentiable numerical fluid dynamical cores or coupled atmosphere-ocean models [64], [65], [23], [40], [58], [13].

3.4 Dimension 4: Functional Roles of Reasoning LLMs

Large Language Models engage with scientific temporal data through four operational archetypes:

- Scientific Reasoner: Understanding physical time series tokens and generating natural language explanations, domain-expert conditional independence oracles for causal discovery, causal attribution, and hypothesis evaluation across scientific domains [5], [43], [66], [44], [45].
- Contextual Enhancer: Supplying auxiliary domain semantics, station metadata, and climate zone priors to guide numerical prediction backbones [67], [42].
- Autonomous Tool-Using Agent: Decomposing complex scientific inquiries, retrieving real-time sensor streams, executing scientific diagnostic libraries (e.g., CDO, xarray, SciPy), and orchestrating multi-agent emergency dispatch under extreme events [68], [28], [25].
- Interactive Natural Language Interface: Translating unstructured scientist inquiries into executable database queries and simulation configurations.

4 Core Foundation Model Architectures

4.1 3D Spatial and Spherical Transformers

Traditional 2D Vision Transformers process planar images by dividing them into uniform patches. However, planetary atmospheric states are inherently three-dimensional, structured across latitude, longitude, and geopotential pressure altitudes.

4.1.1 Pangu-Weather: 3D Earth-Specific Transformer

Bi et al. [6] introduced Pangu-Weather, the first deep learning weather forecasting system to demonstrably surpass top-tier operational NWP models (ECMWF IFS) in medium-range deterministic forecasting accuracy. Pangu-Weather formulates atmospheric prediction using a 3D Earth-Specific Transformer (3DEST) architecture. The state space is divided into 3D cuboid patches of size $2 \times 4 \times 4$ across vertical levels, latitude, and longitude. To account for inhomogeneous spatial dynamics (e.g., Coriolis force variations across latitudes and atmospheric lapse rates across altitudes), 3DEST replaces standard relative position biases with an Earth-Specific Positional Bias (ESPB) matrix indexed explicitly by pressure level and latitude:

$$\text{Attn}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d}} + B_{\text{Earth}}(l_i, l_j, \phi_i, \phi_j) \right) V. \quad (10)$$

To counteract rapid error accumulation during iterative autoregressive rollout, Pangu-Weather trains four distinct models with lead times of 1, 3, 6, and 24 hours. A greedy temporal aggregation algorithm chains these models to generate 7-day global forecasts in seconds.

4.1.2 FuXi: Cascade Forecasting System

Recognizing that predictive uncertainty compounds non-linearly with lead time, Chen et al. [46] developed FuXi, a cascade machine learning system consisting of three specialized Swin Transformer stages: FuXi-Short (0–42 hours), FuXi-Medium (42–120 hours), and FuXi-Long (120–360 hours). By dynamically switching between model weights optimized for different forecasting horizons, FuXi significantly curtails the blurring and spectral dissipation common in long-horizon neural forecasts.

4.1.3 Aurora: Scaling to Planetary Foundation Models

Bodnar et al. [9] demonstrated that scaling principles hold in Earth system modeling by introducing Aurora, a 1.3-billion parameter foundation model. Pretrained on over 1 petabyte of heterogeneous atmospheric data—including ERA5, ECMWF HRES, operational IFS analyses, and NOAA GFS—Aurora incorporates a 3D Perceiver architecture capable of accepting arbitrary combinations of vertical levels and surface variables at 0.1° operational resolution (10 km equivalent).

4.1.4 AIFS: ECMWF Operational GNN-Transformer Architecture

Lang et al. [69] developed the Artificial Intelligence Forecasting System (AIFS), ECMWF’s official operational data-driven model. AIFS marries the geometric flexibility of graph neural networks with the long-range receptive fields of sliding-window transformers. The architecture maps icosahedral multi-mesh node representations through a graph encoder, processes latent multi-scale interactions via a local sliding-window transformer that limits attention memory to $O(N \cdot W)$ tokens, and decodes back to the operational reduced Gaussian grid via a graph decoder. Pretrained on decades of ERA5 reanalysis and fine-tuned on ECMWF’s operational NWP analyses, AIFS runs in operational production at ECMWF four times daily alongside physics-based NWP systems.

4.2 Multi-Mesh Spherical Graph Neural Networks

4.2.1 GraphCast: Planetary Multi-Mesh Routing

Lam et al. [7] pioneered an alternative geometric approach by formulating global weather forecasting over a multi-resolution spherical graph. Standard equirectangular projections concentrate excessive grid cells near the poles while diluting them at the equator. GraphCast circumvents this singularity using an Encode-Process-Decode framework operating on a hierarchical icosahedral mesh:

- 1) **Grid-to-Mesh Encoder:** Maps 0.25° latitude-longitude grid variables onto the vertices of a 6-level refined icosahedral mesh M_6 (40,962 nodes, 122,880 spherical edges).
- 2) **Multi-Mesh Processor:** Employs 16 layers of homogeneous message passing across all hierarchical mesh levels (M_0 to M_6). Long-range edges in coarse mesh levels (M_0, M_1) facilitate rapid, global information

propagation across planetary teleconnection pathways in few message-passing hops.

- 3) **Mesh-to-Grid Decoder:** Projects refined mesh vertex features back to the 0.25° target grid to predict single-step 6-hour temporal increments $\Delta\mathcal{X}_t$.

GraphCast is trained with a 12-step autoregressive loss rollout, enabling robust, physically stable 10-day medium-range forecasts that outperform ECMWF HRES on over 90% of headline verification variables.

4.3 Continuous Neural Operators

4.3.1 FourCastNet: Adaptive Fourier Neural Operators

Pathak et al. [8] deployed Adaptive Fourier Neural Operators (AFNO) to solve planetary-scale atmospheric PDEs. Conventional spatial convolutions are intrinsically tied to fixed grid resolutions. In contrast, neural operators approximate non-linear mappings between infinite-dimensional function spaces. AFNO transforms tokenized feature representations into the frequency domain via 2D Fast Fourier Transforms (FFT), applies block-diagonal weight multiplications with soft-thresholding sparsity in the frequency domain, and returns to physical space via inverse FFT:

$$z_{\text{out}}(x) = \mathcal{F}^{-1}(\text{SoftThreshold}(\mathbf{W} \cdot \mathcal{F}(z_{\text{in}})(k))) + \mathbf{W}_s z_{\text{in}}(x). \quad (11)$$

This formulation provides computational complexity quasi-linear in spatial resolution while demonstrating remarkable zero-shot super-resolution and transferability.

4.3.2 SFNO: Spherical Fourier Neural Operators

While planar Fourier transforms accelerate spectral computation, standard 2D FFTs applied to equirectangular latitude-longitude coordinates suffer from severe metric distortions near the poles, where meridians converge. Bonev et al. [47] resolved this geometric bottleneck by formulating the Spherical Fourier Neural Operator (SFNO), which solves continuous PDEs directly on the Riemannian manifold of the unit sphere $\mathbb{S}^2 \subset \mathbb{R}^3$.

Let colatitude be $\theta \in [0, \pi]$ and longitude be $\lambda \in [0, 2\pi)$. Any square-integrable scalar atmospheric state $u \in \mathcal{L}^2(\mathbb{S}^2)$ admits an orthogonal expansion in terms of spherical harmonics $Y_\ell^m(\theta, \lambda)$:

$$u(\theta, \lambda) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} \hat{u}_\ell^m Y_\ell^m(\theta, \lambda), \quad (12)$$

where $\ell \in \mathbb{N}_0$ denotes the spherical degree, $m \in \{-\ell, \dots, \ell\}$ is the order, and the orthonormal basis functions are defined via associated Legendre polynomials P_ℓ^m :

$$Y_\ell^m(\theta, \lambda) = \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos \theta) e^{im\lambda}. \quad (13)$$

Continuous convolution on $\mathbb{S}^2$ with an invariant kernel κ is realized through spherical harmonic transforms $\mathcal{SHT}$ and inverse transforms $\mathcal{ISHT}$:

$$(\kappa \star_{\mathbb{S}^2} u)(\theta, \lambda) = \mathcal{ISHT}(\mathbf{W}_\ell \cdot \mathcal{SHT}(u)(\ell, m)), \quad (14)$$

where $\mathbf{W}_\ell \in \mathbb{C}^{d_{\text{out}} \times d_{\text{in}}}$ denotes degree-dependent weight tensors. By the convolution theorem on $\text{SO}(3)$, this spherical harmonic formulation provides exact rotational equivariance, eliminates high-latitude numerical artifacts, and ensures long-term spectral stability over multi-day atmospheric rollouts.

4.4 Multimodal Masked Autoencoders and Latent Alignments

In the remote sensing and Earth observation domains, observations arrive as asynchronous, multi-spectral satellite imagery cubes, irregular in-situ sensor readings, and gridded reanalysis tensors.

- SatMAE [14] introduced a temporal and multi-spectral Vision Transformer trained via masked autoencoding. By decoupling spatial, temporal, and spectral positional embeddings, SatMAE processes multi-temporal revisit sequences with high masking ratios (up to 75%), outperforming supervised baselines on land cover and crop yield estimation.
- Presto [15] developed a lightweight (1.2M parameters) transformer tailored for pixel-level time series. Presto natively ingests heterogeneous modalities—including Sentinel-1 SAR backscatter, Sentinel-2 multi-spectral bands, elevation, and ERA5 weather—yielding exceptional few-shot transfer performance.
- Galileo [16] expanded multi-sensor pretraining across spatial-temporal scales using dual global and local contrastive objectives, enabling unified geospatial segmentation and dynamic flood mapping.
- SkySense [70] established a billion-scale (1.9B parameter) multi-modal remote sensing foundation model. By formulating a factorized spatio-temporal cross-attention architecture, SkySense jointly pretrains on over 21.5 million multi-modal satellite temporal sequences (combining optical multi-spectral reflectance with synthetic aperture radar SAR backscatter). It establishes state-of-the-art generalist performance across 16 diverse Earth observation benchmarks, including fine-grained crop classification, building damage assessment, and temporal change detection.
- DOFA [20] introduced a Dynamic One-For-All foundation architecture inspired by neural plasticity. Recognizing that Earth observation platforms possess disparate band wavelengths and ground sampling distances (GSD), DOFA conditions dynamic weight hypernetworks directly on continuous physical wavelengths $\lambda \in [400 \text{ nm}, 14,000 \text{ nm}]$, unifying optical, SAR, and hyperspectral temporal modalities under a single pre-trained representation.
- GEO-Bench [71] established a standardized benchmark suite for Earth monitoring foundation models. Unifying multi-temporal Sentinel-1/2, SPOT, and m-CASIS data across regional land-use segmentation and multi-spectral classification, GEO-Bench provides rigorous zero-shot and fine-tuning evaluation protocols for multi-sensor satellite representations.
- EarthPT [19] developed a 700M-parameter autoregressive decoder foundation model specifically designed for Earth Observation time series. Treating multi-spectral reflectance measurements (400–2300 nm) across Copernicus Sentinel-2 pixel-year revisit sequences as dense temporal tokens, EarthPT predicts future surface reflectance evolutions with high physical fidelity.
- Prithvi [17] and its weather-climate counterpart Prithvi WxC [72] established open-source geospatial

foundation backbones jointly developed by NASA and IBM, scaling to 2.3 billion parameters.

4.4.1 Formal Alignment Paradigms across Heterogeneous Geometries

As illustrated in Fig. 3, cross-modal integration across disparate spatial geometries follows three dominant architectural paradigms:

- 1) Strategy A: Early Heterogeneous Patch Projection: Given heterogeneous modality inputs $\mathbf{X}^{(m)} \in \mathbb{R}^{T_m \times C_m \times H_m \times W_m}$ with disparate channel definitions and Ground Sampling Distances (GSD), modality-specific linear or convolutional patch embedding kernels $\mathbf{E}^{(m)}$ project local spatial-temporal patches into a shared hidden dimension D :

$$\mathbf{h}_i^{(m)} = \mathbf{E}^{(m)} \mathbf{x}_i^{(m)} + \mathbf{PE}(x_i, y_i, z_i, t_i), \quad (15)$$

where $\mathbf{PE}(\cdot)$ denotes a unified continuous 4D Fourier spatio-temporal positional encoding, ensuring coordinate harmonization across disparate satellite swaths and ground station coordinates (utilized in SatMAE [14] and Presto [15]).

- 2) Strategy B: Latent Cross-Attention Bottlenecks: When observation token counts $N_{\text{obs}} = \sum_m N_m$ fluctuate dramatically due to dynamic sensor availability or dense planetary grids, feeding all tokens directly into full self-attention scales quadratically as $O(N_{\text{obs}}^2)$. To circumvent this computational explosion, Perceiver-style architectures (Aurora [9], SkySense [70]) introduce a fixed bank of learnable latent query tokens $\mathbf{Z} \in \mathbb{R}^{M \times D}$ ($M \ll N_{\text{obs}}$) that attend to the irregular multi-sensor input tokens via cross-attention:

$$\mathbf{Z}_{\text{aligned}} = \text{Softmax} \left(\frac{\mathbf{Z} \mathbf{W}_Q (\mathbf{H}_{\text{obs}} \mathbf{W}_K)^T}{\sqrt{d}} \right) (\mathbf{H}_{\text{obs}} \mathbf{W}_V), \quad (16)$$

where $\mathbf{H}_{\text{obs}} = [\mathbf{H}^{(1)}; \dots; \mathbf{H}^{(K)}]$ concatenates all available sensor tokens. The spatio-temporal foundation backbone then processes only the constant-size latent state $\mathbf{Z}_{\text{aligned}}$ with linear complexity in input resolution, decoupling observation density from computation.

- 3) Strategy C: Dynamic Wavelength Hypernetworks: To support arbitrary spectral bands without modifying network weights, continuous wavelength hypernetworks (DOFA [20]) encode the physical central wavelength λ of each spectral channel via harmonic Fourier embeddings $\mathbf{e}_\lambda = [\sin(2^k \pi \lambda), \cos(2^k \pi \lambda)]_{k=0}^{K-1}$. A lightweight Multi-Layer Perceptron (MLP) hypernetwork $\mathcal{H}_\phi$ maps $\mathbf{e}_\lambda$ to generate the convolution or linear projection weights dynamically:

$$\mathbf{W}_{\text{patch}}(\lambda) = \mathcal{H}_\phi(\mathbf{e}_\lambda) \in \mathbb{R}^{D \times (P^2 \cdot C_{\text{in}})}. \quad (17)$$

This enables zero-shot sensor transfer across optical (e.g., Sentinel-2 10m/20m bands), thermal infrared (Landsat-8/TIRS), and radar backscatter (Sentinel-1 VV/VH) modalities within a unified model.

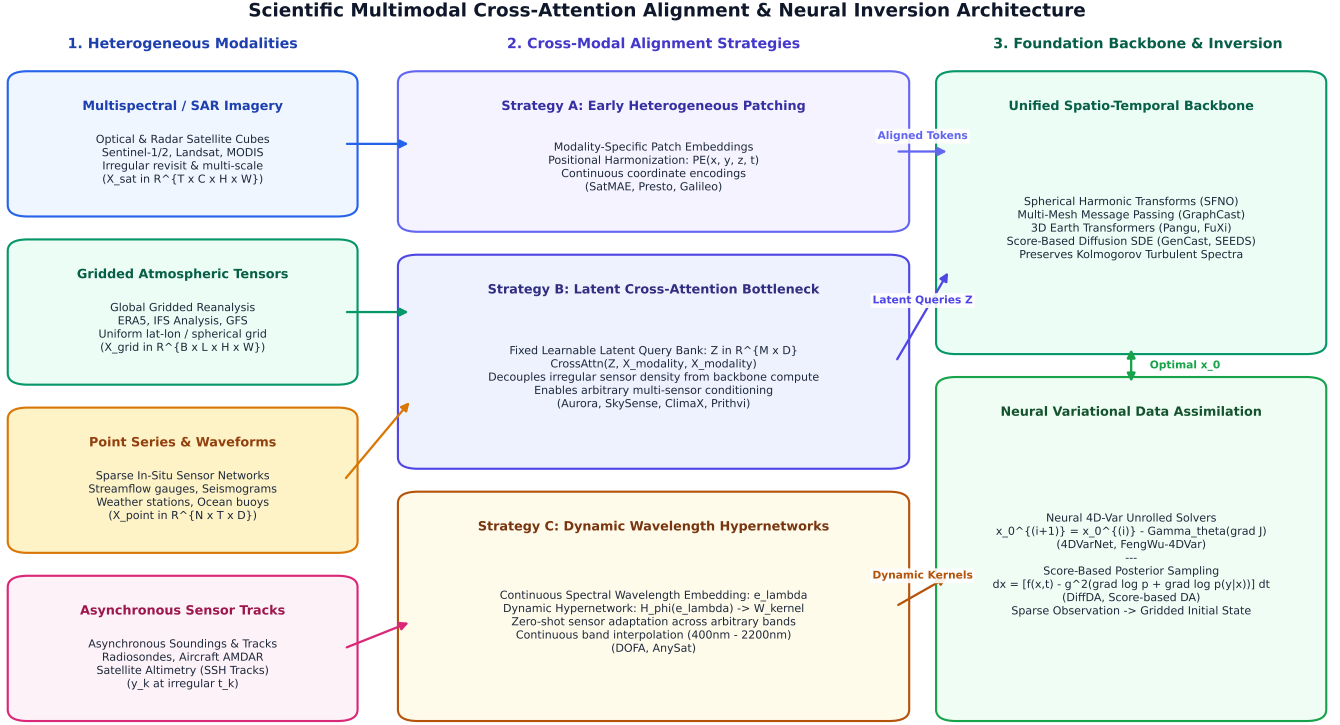


Fig. 3. Architectural taxonomy of scientific multimodal cross-attention alignment and neural variational data assimilation across heterogeneous spatial geometries. Left: Multi-source observational modalities spanning satellites, reanalyses, point sensor networks, and asynchronous tracks. Center: Three primary alignment paradigms (Early Heterogeneous Patching, Latent Cross-Attention Bottlenecks, and Dynamic Wavelength Hypernetworks). Right: Unified spatio-temporal foundation backbones and neural variational data assimilation solvers bridging sparse observation streams directly into gridded initial state representations.

4.5 Neural Data Assimilation and Sensor-to-Grid Variational Inversion

A fundamental gap separating operational numerical weather forecasting from purely data-driven foundation models is data assimilation (DA): the mathematical process of synthesizing sparse, asynchronous, and noisy physical observations into a physically consistent gridded initial atmospheric or oceanic state $\mathbf{x}_0$. While existing foundation models (e.g., Pangu-Weather, GraphCast) rely entirely on pre-computed gridded reanalyses (ERA5 or operational IFS analyses) produced by supercomputing NWP centers, emerging architectures introduce differentiable variational and score-based neural assimilation solvers to invert raw sensor streams directly into gridded model initial states.

4.5.1 Variational Formulation and Neural 4D-Var

In classical numerical weather prediction, four-dimensional variational data assimilation (4D-Var) optimizes an initial state $\mathbf{x}_0$ over an assimilation window $[t_0, t_K]$ to minimize a penalized least-squares cost function:

$$\mathcal{J}(\mathbf{x}_0) = \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \frac{1}{2} \sum_{k=0}^K \|\mathbf{y}_k - \mathcal{H}_k(\mathcal{M}_{0 \rightarrow k}(\mathbf{x}_0))\|_{\mathbf{R}_k^{-1}}^2, \quad (18)$$

where $\mathbf{x}_b$ is the background prior state, $\mathbf{B}$ is the background error covariance matrix, $\mathbf{y}_k$ denotes sparse observations at time t_k , $\mathbf{R}_k$ is the observation error covariance, $\mathcal{H}_k$ is the observation forward operator, and $\mathcal{M}_{0 \rightarrow k}$ is the non-linear

dynamical forward model. In operational NWP, minimizing Eq. (18) requires maintaining complex adjoint code $\mathcal{M}^*$ to compute gradients $\nabla_{\mathbf{x}_0} \mathcal{J}$, consuming up to 80% of operational supercomputing compute budgets.

To overcome this bottleneck, Fablet et al. [73] introduced 4DVarNet, reformulating variational data assimilation as an end-to-end learnable gradient-based solver. 4DVarNet replaces the analytical minimization loop with an unrolled recurrent neural architecture:

$$\mathbf{x}_0^{(i+1)} = \mathbf{x}_0^{(i)} - \Gamma_{\theta} \left(\nabla_{\mathbf{x}_0} \mathcal{J}(\mathbf{x}_0^{(i)}), \mathbf{h}^{(i)} \right), \quad (19)$$

where Γ_{θ} is a learned recurrent neural operator (e.g., ConvLSTM or U-Net solver), $\mathbf{h}^{(i)}$ represents internal solver memory states, and the gradient $\nabla_{\mathbf{x}_0} \mathcal{J}$ is computed via automated differentiation through the observation operator $\mathcal{H}_k$ and neural forward dynamics $\mathcal{M}_{\phi}$. Originally deployed for sea surface height (SSH) reconstruction from sparse satellite altimetry tracks in oceanography, 4DVarNet achieves orders-of-magnitude faster convergence than classical conjugate gradient descent while learning dynamic background error covariances directly from historical data.

Scaling neural data assimilation to global atmospheric forecasting, Xiao et al. [74] introduced FengWu-4DVar. FengWu-4DVar couples the FengWu multi-scale atmospheric transformer with an unrolled 4D variational optimization loop. By exploiting the end-to-end differentiability of the neural forecasting backbone, FengWu-4DVar directly backpropagates observation residuals from asynchronous in-situ

soundings and satellite radiances through 24-hour assimilation windows, generating initialized atmospheric states that match operational ECMWF IFS analysis quality while reducing assimilation wall-clock time from hours to seconds.

4.5.2 Score-Based and Diffusion Data Assimilation

Complementing deterministic variational solvers, generative diffusion models reframe data assimilation as conditional posterior sampling: drawing physical state trajectories $\mathbf{x}_{0:K} \sim p(\mathbf{x}_{0:K} | \mathbf{y}_{0:K})$.

Rozet and Louppe [75] formulated Score-based Data Assimilation (SDA), showing that continuous score-based stochastic differential equations can execute exact spatio-temporal smoothing and filtering over non-linear fluid dynamics without tangent-linear approximations. Given a trained unconditional foundation prior $s_\theta(\mathbf{x}_t, t) \approx \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$, SDA conditions the reverse-time generative SDE on sparse partial observations $\mathbf{y}$ via Tweedie’s guidance:

$$d\mathbf{x}_t = [\mathbf{f}(\mathbf{x}_t, t) - g(t)^2 (\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t))] dt + g(t) d\bar{\mathbf{w}}, \quad (20)$$

Under Gaussian observation noise $\mathbf{y} \sim \mathcal{N}(\mathbf{H}\mathbf{x}_0, \mathbf{R})$, the intractable conditional score $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t)$ is approximated analytically via the posterior expectation $\hat{\mathbf{x}}_0(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}}(\mathbf{x}_t - \sqrt{1 - \alpha_t}\epsilon_\theta(\mathbf{x}_t, t))$:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t) \approx \frac{1}{\sqrt{\alpha_t}} \mathbf{H}^T (\mathbf{H} \Sigma_t \mathbf{H}^T + \mathbf{R})^{-1} (\mathbf{y} - \mathbf{H} \hat{\mathbf{x}}_0(\mathbf{x}_t)). \quad (21)$$

Scaling diffusion assimilation to weather-scale operational state estimation, Huang et al. [76] introduced DiffDA. DiffDA trains a conditional denoising diffusion probabilistic model (DDPM) conditioned on sparse surface weather station observations and radiosonde soundings. Operating directly on physical observation coordinates, DiffDA samples ensemble analyses that accurately reproduce fine-scale frontal boundaries, atmospheric rivers, and convective storms, demonstrating that neural generative assimilation can bridge heterogeneous sensor observations directly into gridded foundation state spaces.

4.6 Probabilistic Diffusion and Extreme-Event Ensembles

A pervasive theoretical limitation of deterministic forecasting models trained under mean squared error (L_2) or mean absolute error (L_1) losses is their vulnerability to spectral blurring and variance dissipation at extended forecast horizons. Formally, given an observation history $\mathbf{y}_{1:t}$ and future atmospheric target $\mathbf{x}_{t+\tau}$, optimizing an L_2 regression objective $\min_\theta \mathbb{E}[\|\mathbf{x}_{t+\tau} - f_\theta(\mathbf{y}_{1:t})\|^2]$ provably converges to the conditional posterior mean:

$$f_\theta^*(\mathbf{y}_{1:t}) = \mathbb{E}[\mathbf{x}_{t+\tau} | \mathbf{y}_{1:t}] = \int \mathbf{x} p(\mathbf{x} | \mathbf{y}_{1:t}) d\mathbf{x}. \quad (22)$$

Because atmospheric flow exhibits non-linear chaotic dynamics with positive Lyapunov exponents, the true posterior probability density $p(\mathbf{x} | \mathbf{y}_{1:t})$ bifurcates into multi-modal trajectories at extended lead times ($\tau \geq 5$ days). Averaging over these distinct physical realizations washes out high-wavenumber turbulent eddies, severely dampens the kinetic energy power spectrum $E(k) \propto k^{-5/3}$ or k^{-3} , and underestimates extreme-event tail probabilities $P(\|\mathbf{x}\| > \tau_{\text{threshold}})$.

To resolve this limitation, generative diffusion models reframe weather forecasting as stochastic sampling from the true conditional distribution $p(\mathbf{x} | \mathbf{y})$.

4.6.1 Continuous Score-Based Diffusion Formulation

Let the atmospheric state trajectory follow a continuous forward stochastic differential equation (SDE) over diffusion time $s \in [0, 1]$:

$$d\mathbf{x}_s = \mathbf{f}(\mathbf{x}_s, s) ds + g(s) d\mathbf{w}, \quad (23)$$

where $\mathbf{w}$ is standard Brownian motion, $\mathbf{f}$ represents drift, and $g(s)$ governs noise dispersion. By Anderson’s reverse-time theorem, the generative reverse trajectory is governed by:

$$d\mathbf{x}_s = [\mathbf{f}(\mathbf{x}_s, s) - g(s)^2 \nabla_{\mathbf{x}_s} \log p_s(\mathbf{x}_s | \mathbf{y})] ds + g(s) d\bar{\mathbf{w}}, \quad (24)$$

where $d\bar{\mathbf{w}}$ denotes backward Brownian motion, and the score function $s_\theta(\mathbf{x}_s, s, \mathbf{y}) \approx \nabla_{\mathbf{x}_s} \log p_s(\mathbf{x}_s | \mathbf{y})$ is parameterized by a deep neural operator or graph network trained via denoising score matching.

4.6.2 GenCast, SEEDS, and FuXi-Extreme: Operational Diffusion Ensembling

Price et al. [4] operationalized this paradigm in GenCast, conditioning a score-matching network on spherical multi-mesh graphs using historical 6-hour atmospheric increments $(\mathbf{x}_{t-6h}, \mathbf{x}_t)$. By sampling 50 stochastic trajectories, GenCast accurately generates medium-range ensemble forecasts that maintain realistic energy spectra up to Day 15, outperforming ECMWF ENS in Continuous Ranked Probability Scores (CRPS) across over 80% of headline atmospheric variables and drastically improving tropical cyclone track and intensity predictions.

Concurrently, Li et al. [11] introduced SEEDS (Scalable Ensemble Envelope Diffusion Sampler), demonstrating that generative diffusion can emulate operational numerical weather forecast ensembles (such as NOAA GEFS) conditioned on only 1 or 2 deterministic forecast trajectories. SEEDS generates large ensembles (e.g., 50 members) at 1° operational resolution orders of magnitude faster than numerical ensembles, precisely reproducing the spread-skill relationship and capturing extreme heatwave and storm tail events without numerical energy collapse.

Extending diffusion specifically to hazard extremes, Zhong et al. [77] formulated FuXi-Extreme. While deterministic weather models drastically underestimate tail precipitation rates and gale-force wind gusts, FuXi-Extreme couples a cascade Swin Transformer with a conditional latent diffusion model. Trained on high-resolution ERA5 extreme events, it generates realistic sharp spatial gradients for extreme precipitation (> 50 mm/day) and 10-meter wind gusts, resolving the longstanding spectral smoothing dilemma for operational disaster early warnings.

4.7 Universal Tokenization and Cross-Domain Temporal Transferability

A major architectural obstacle in scientific foundation models is the profound heterogeneity of physical data representations: regular lat-lon reanalysis grids (ERA5), irregular unstructured meshes (finite-volume ocean models), sparse in-situ station networks (hydrology streamflow gauges), 1D continuous

high-frequency waveforms (seismology), and irregular multi-spectral satellite revisits (remote sensing). Recent research develops universal tokenization mechanisms to bridge this divide:

4.7.1 Unified Multi-Task Architectures: UniTS

Gao et al. [41] developed UniTS, a unified multi-task foundation model for time series. Unlike domain-specific architectures requiring separate parameterizations for each dataset, UniTS processes arbitrary numbers of variables and sequences by introducing:

- 1) Sequence-Variable Tokenization: Decouples temporal sequence patches and variable channels into modular tokens via lightweight linear projections.
- 2) Prompt-Conditioned Task Tokens: Ingests task specification tokens (e.g., forecasting horizon, classification labels, imputation mask) into self-attention layers, conditioning a single shared transformer backbone to execute forecasting, anomaly detection, imputation, and classification across 38 diverse datasets without architectural modification.

4.7.2 Language Model Tokenization for Time Series: Chronos

Ansari et al. [78] introduced Chronos, demonstrating that classical language model tokenization principles translate directly to continuous scientific and physical time series. Chronos discretizes real-valued time series through affine mean-variance normalization followed by uniform quantile binning into a fixed vocabulary of V discrete tokens (e.g., 4096 bins). Pretrained on diverse time-series corpora using standard cross-entropy auto-regression over T5 architectures, Chronos demonstrates that language model pretraining objectives yield remarkable zero-shot probabilistic forecasting capabilities across diverse physical and socio-economic systems.

4.7.3 Autoregressive Probabilistic Foundations: Lag-Llama

Rasul et al. [79] formulated Lag-Llama, a dedicated probabilistic foundation model utilizing a decoder-only LLaMA-style transformer architecture. Ingesting historical lagged temporal features and tokenized continuous values, Lag-Llama applies rotary position embeddings (RoPE) and outputs parametric Student- t distribution parameters (μ, σ, ν) at each step. This formulation natively captures fat-tailed physical distributions and extreme deviations across physical sensor streams.

4.7.4 Open Foundation Models: MOMENT

Goswami et al. [42] introduced MOMENT, an open-source family of foundation models trained on the Time-series Pile, a curated corpus of over 13 million diverse sequences spanning earth science sensors, industrial machinery, and physical processes. Using a patched T5-style masked autoencoder architecture, MOMENT establishes consistent zero-shot and few-shot transfer capabilities across diverse scientific temporal benchmarks.

4.7.5 Atmospheric Patch Randomization: Stormer

Nguyen et al. [10] introduced Stormer, demonstrating that a standard Vision Transformer backbone can match or exceed specialized weather forecasting architectures when coupled with two critical physical inductive biases:

- 1) Randomized Dynamics Objective: During pretraining, Stormer forecasts atmospheric states across randomized lead times $\Delta t \sim \mathcal{U}(6h, 120h)$, forcing the model to learn multi-scale continuous temporal dynamics rather than overfitting to fixed autoregressive step sizes.
- 2) Pressure-Weighted Vertical Loss: Weights atmospheric layer losses proportional to geopotential layer thickness and density, aligning optimization with physical mass distribution.

4.7.6 Standardized Waveform Tokenization: SeisBench

In geophysics, Woollam et al. [33] established SeisBench, a standardized framework and benchmarking suite unifying 3-component continuous seismic waveform datasets (ETHZ, GEOFON, INSTANCE, SCEDC, STEAD, comprising over 30 million waveforms). By standardizing sampling rates (100 Hz), coordinate rotations, and trace tokenization, SeisBench enables cross-regional foundation pretraining and transferability for phase arrival picking, earthquake detection, and fault rupture localization.

4.8 Physical Conservation Constraints and Differentiable Hybrid Solvers

A persistent vulnerability of purely data-driven black-box neural networks is their tendency to violate fundamental physical conservation principles over prolonged autoregressive rollouts, leading to unphysical mass dissipation, kinetic energy damping, or moisture divergence. Recent advances address this through distinct levels of physical integration:

- 1) Soft Physics Regularization: Incorporates auxiliary loss penalties that penalize physical inconsistencies during training:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{conservation}}, \quad (25)$$

where $\mathcal{L}_{\text{conservation}}$ penalizes deviations in global total dry mass, vertical geostrophic balance, or spectral kinetic energy distributions (employed in FourCastNet [8], FengWu [61], and Stormer [10]).

- 2) Hard Architectural Invariance: Enforces conservation laws and symmetry directly into neural layer transformations. For instance, SFNO [47] guarantees SO(3) rotational equivariance on the unit sphere $\mathbb{S}^2$, preventing polar singularities and high-latitude numerical distortions by construction.
- 3) Hybrid Multiscale Subgrid Parameterization: ClimSim: Yu et al. [65] established ClimSim, the largest multiscale dataset and benchmark for training deep learning subgrid physics emulators. In multiscale climate modeling (e.g., E3SM-MMF), macro-scale fluid dynamics are integrated by numerical PDE solvers on coarse grids, while complex micro-scale processes (moist convection, cloud formation, turbulence, and

radiative transfer) are parameterized by neural networks. ClimSim provides 5.7 billion high-resolution atmospheric column profiles, benchmarking physics-informed ML architectures on strict conservation of total column water mass and dry static energy.

- 4) Differentiable Hybrid Solvers: NeuralGCM: Kochkov et al. [64] coupled a fully differentiable atmospheric dynamical core with neural network parameterizations. The resolved large-scale hydrodynamics are integrated explicitly by an analytical spectral PDE solver that strictly enforces conservation of mass, momentum, and moisture:

$$\frac{\partial \mathbf{X}}{\partial t} = \mathcal{F}_{\text{dynamical}}(\mathbf{X}) + \mathcal{N}_{\theta}(\mathbf{X}), \quad (26)$$

where $\mathcal{F}_{\text{dynamical}}$ executes analytical fluid dynamics, while the neural network $\mathcal{N}_{\theta}$ parameterizes sub-grid physical processes. NeuralGCM demonstrates unprecedented multi-decadal stability without climate drift, matching the operational deterministic accuracy of ECMWF IFS HRES at 1–10 day weather horizons while preserving global physical conservation laws for decadal climate simulations.

- 5) Continuous Physics-Informed Neural ODE Dynamics: ClimODE: Verma et al. [80] pioneered ClimODE, fundamentally rethinking scientific time series modeling by replacing discrete multi-step mapping $\mathbf{x}_{t+\Delta t} = \mathcal{M}_{\theta}(\mathbf{x}_t)$ with continuous differential flow trajectories grounded in fluid mechanics. By explicitly embedding continuity and advective constraints into neural ordinary differential equations, ClimODE achieves global weather forecasting with only 0.3M parameters—over two orders of magnitude smaller than discrete vision transformers—while natively supporting arbitrary continuous-time evaluation intervals.

4.9 Continuous-Discrete Physics Invariants, Lie Symmetries, and Geometric Deep Learning

A central mathematical frontier in scientific foundation models is reconciling discrete computational architectures (e.g., attention over tokenized patches, autoregressive discrete roll-outs) with the continuous, symmetry-preserving nature of physical laws governing the Earth system.

4.9.1 Continuous Fluid Advection and Neural Differential Flows

Classical discrete autoregressive architectures forecast atmospheric states via discrete recurrence:

$$\mathbf{x}_{k+1} = \mathcal{M}_{\theta}(\mathbf{x}_k), \quad k \in \{0, 1, \dots, K\}, \quad (27)$$

where $\Delta t = t_{k+1} - t_k$ is strictly locked to a predefined sampling cadence (e.g., 6h in ERA5). This discrete formulation suffers from two fundamental limitations: (1) time-step rigidity, precluding direct assimilation or forecasting at arbitrary asynchronous timestamps, and (2) compounding discretization error, where unconstrained discrete updates accumulate unphysical drift.

In contrast, continuous-time formulation models the planetary atmosphere as a non-linear dynamical system evolving over the continuous manifold $\mathcal{M} = \mathbb{S}^2 \times \mathbb{R}^+$:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}_{\theta}(\mathbf{x}(t), t), \quad \mathbf{x}(t) = \mathbf{x}(t_0) + \int_{t_0}^t \mathbf{f}_{\theta}(\mathbf{x}(\tau), \tau) d\tau. \quad (28)$$

To enforce physical consistency, ClimODE [80] constrains the learned vector field $\mathbf{f}_{\theta}$ using the Navier–Stokes transport theorem. The continuous conservation of atmospheric mass density ρ is governed by the continuity PDE:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}_{\theta}) = 0, \quad (29)$$

where $\mathbf{v}_{\theta}(\mathbf{s}, t) \in T_{\mathbf{s}}\mathbb{S}^2$ is the neural velocity field over spherical coordinates $\mathbf{s} = (\lambda, \theta)$. Similarly, any scalar tracer field $\phi \in \{T, q, Z\}$ (temperature, specific humidity, geopotential) satisfies the continuous material advection-diffusion relation:

$$\frac{D\phi}{Dt} \equiv \frac{\partial \phi}{\partial t} + (\mathbf{v}_{\theta} \cdot \nabla)\phi = \mathcal{D}_{\theta}(\phi) + \mathcal{S}_{\theta}(\phi, \mathbf{s}, t), \quad (30)$$

where $\mathcal{D}_{\theta}$ denotes turbulent sub-grid diffusion and $\mathcal{S}_{\theta}$ encapsulates radiative source/sink fluxes. Integrating Eq. (28) with adaptive-step numerical ODE solvers (e.g., Dormand–Prince 5(4) or adaptive Runge–Kutta) guarantees that the flow remains bounded along physical trajectories, yielding superior long-term stability and eliminating the need for empirical post-hoc smoothing.

4.9.2 Lie Group Symmetries and Spherical Coordinate Invariance

The fundamental equations of geophysical fluid dynamics (primitive equations, shallow-water equations) are invariant under the continuous Lie group of spatial rotations $\text{SO}(3)$ and time translation $\mathbb{R}$. Let $g \in \text{SO}(3)$ represent an arbitrary rigid rotation of the planetary sphere $\mathbb{S}^2$. A scientific foundation operator $\mathcal{F}_{\theta} : \mathcal{X}(\mathbb{S}^2) \rightarrow \mathcal{X}(\mathbb{S}^2)$ is said to be $\text{SO}(3)$ -equivariant if and only if:

$$\mathcal{F}_{\theta}(\pi_g[\mathbf{X}]) = \pi_g[\mathcal{F}_{\theta}(\mathbf{X})], \quad \forall g \in \text{SO}(3), \quad (31)$$

where π_g denotes the group representation acting on the field state $\mathbf{X}$. Crucially, the transformation laws depend on the tensor rank of the physical variables:

- Scalar Tracers (Geopotential Φ , Surface Temperature T_{2m} , Pressure p): Act as rank-0 scalar invariants under coordinate rotation:

$$\pi_g[\Phi](\mathbf{s}) = \Phi(g^{-1}\mathbf{s}). \quad (32)$$

- Vector Velocity Fields (Horizontal Wind $\mathbf{u} = (u, v)^T$): Belong to the tangent bundle $T_{\mathbf{s}}\mathbb{S}^2$ and undergo an orthogonal frame transformation via the differential push-forward:

$$\pi_g[\mathbf{u}](\mathbf{s}) = \mathbf{J}_g(g^{-1}\mathbf{s}) \mathbf{u}(g^{-1}\mathbf{s}), \quad (33)$$

where $\mathbf{J}_g \in \text{SO}(2)$ denotes the local Jacobian transition matrix between tangent spaces.

Standard planar convolutional networks or naive Vision Transformers applied to equirectangular latitude-longitude projections violate Eq. (31) due to the non-uniform metric tensor:

$$g_{\mathbb{S}^2} = R^2 (d\theta^2 + \cos^2 \theta d\lambda^2), \quad (34)$$

which compresses grid cell longitudinal distances towards the poles ($\Delta x \rightarrow 0$ as $\theta \rightarrow \pm\pi/2$), causing severe polar singularities and high-frequency numerical artifacts. In contrast, geometric foundation models resolve this through two elegant paradigms:

- 1) Spherical Harmonic Harmonic Operators: SFNO [47] projects spatial states into continuous spherical harmonics $Y_\ell^m(\theta, \lambda)$, executing spectral convolutions in the frequency domain where $SO(3)$ acts block-diagonally via Wigner D -matrices:

$$\hat{\mathbf{X}}_\ell^m = \int_{\mathbb{S}^2} \mathbf{X}(\mathbf{s}) \overline{Y_\ell^m(\mathbf{s})} d\Omega(\mathbf{s}), \quad (35)$$

strictly preserving planetary rotational symmetry and eliminating polar distortion by construction.

- 2) Quasi-Uniform Spherical Meshes: GraphCast [7] and AIFS [69] map irregular spherical domains onto recursive icosahedral multi-mesh tessellations $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Message-passing edge attributes encode true Riemannian geodesic distances and relative orientation angles, achieving approximate isometry and continuous spatial invariance across the entire planetary sphere.

4.10 Extreme Value Theory (EVT) and Heavy-Tailed Uncertainty Quantification

A profound pathology in contemporary data-driven weather and climate foundation models is their systemic underforecasting of rare, high-impact perils (e.g., flash flood downpours, hurricane wind gusts, extreme heatwaves). Under standard training objectives, neural networks minimize L_2 error:

$$\hat{\mathbf{y}} = \arg \min_{\mathbf{f}} \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} [\|\mathbf{y} - \mathbf{f}(\mathbf{x})\|_2^2] = \mathbb{E}[\mathbf{y} \mid \mathbf{x}]. \quad (36)$$

Because the conditional expectation $\mathbb{E}[\mathbf{y} \mid \mathbf{x}]$ acts as a statistical mean-seeking operator, minimizing symmetric L_2 or L_1 loss over climatological distributions inevitably biases predictions toward the bulk mean. Consequently, when confronted with extreme values located in the heavy tails of the distribution, standard foundation backbones suffer from severe amplitude blunting, failing to capture 100-year return period perils.

4.10.1 Generalized Extreme Value (GEV) Formulations

In statistical climatology, the asymptotic distribution of block maxima (e.g., annual maximum daily precipitation or peak gust speed) $M_n = \max\{X_1, \dots, X_n\}$ is uniquely characterized by the Fisher–Tippett–Gnedenko theorem via the Generalized Extreme Value (GEV) distribution:

$$G(z; \mu, \sigma, \xi) = \exp \left\{ - \left[1 + \xi \left(\frac{z - \mu}{\sigma} \right) \right]_+^{-1/\xi} \right\}, \quad (37)$$

where $\mu \in \mathbb{R}$ is the location parameter, $\sigma > 0$ is the scale parameter, and $\xi \in \mathbb{R}$ denotes the shape parameter (tail index), with $[a]_+ \equiv \max(a, 0)$. The tail index ξ determines the qualitative asymptotic decay of natural extremes:

- Fréchet Heavy Tail ($\xi > 0$): Governs convective precipitation bursts, flash floods, and atmospheric rivers. The survival probability exhibits power-law decay $P(Z > z) \sim z^{-1/\xi}$, meaning extremes occur

orders of magnitude more frequently than predicted by light-tailed Gaussian models.

- Gumbel Exponential Tail ($\xi = 0$): Describes moderate wind speed distributions and synoptic temperature fluctuations with exponential decay $P(Z > z) \sim e^{-z/\sigma}$.
- Weibull Bounded Tail ($\xi < 0$): Characterizes bounded atmospheric variables (e.g., relative humidity $\leq 100\%$, upper limits on geostrophic wind shear).

4.10.2 ExtremeCast: Asymmetric Extreme Loss and Uncertainty Boosting

To break the mean-seeking regression trap without degrading bulk synoptic accuracy, Xu et al. [59] introduced ExtremeCast. ExtremeCast mathematically proves that symmetric loss functions inherently penalize rare overpredictions symmetrically with severe underpredictions, heavily suppressing tail predictions due to the extreme class imbalance ($P(Z > \tau) < 0.01$).

ExtremeCast formulates an asymmetric extreme loss function $\mathcal{L}_{\text{Ex}}$ that dynamically amplifies gradient penalties when true targets exceed high quantile thresholds τ :

$$\mathcal{L}_{\text{Ex}}(\mathbf{y}, \hat{\mathbf{y}}) = \mathcal{L}_{\text{base}}(\mathbf{y}, \hat{\mathbf{y}}) + \alpha \sum_{i=1}^N w(y_i) \cdot \max(0, y_i - \hat{y}_i)^\gamma, \quad (38)$$

where $\mathcal{L}_{\text{base}}$ preserves standard MSE/MAE optimization over bulk dynamics, $\alpha > 0$ balances tail prioritization, and $\gamma \geq 1$ enforces super-linear penalties against underforecasting. The weighting factor $w(y_i)$ is formulated using extreme value theory:

$$w(y_i) = \mathbb{I}(y_i \geq \tau) \cdot \left(\frac{y_i - \tau}{\sigma_\tau} \right)^\beta, \quad (39)$$

where τ is set to the 95th or 99th climatological percentile and σ_τ represents the tail scale. Furthermore, ExtremeCast incorporates ExBooster, a training-free uncertainty ensemble module that perturbs latent feature representations using calibrated extreme-value noise samples, generating diverse probabilistic realizations that significantly elevate the Critical Success Index (CSI) and Symmetric Extremal Dependence Index (SEDI) for category-5 typhoons, localized wind gusts, and severe heat extremes.

4.11 High-Order Tensor Decompositions and Quantum-Classical Hybrid Operators

While continuous neural operators (such as FNO and SFNO) offer discretization-invariant PDE solution capabilities, their practical deployment in 3D atmospheric columns, 3D subsurface seismic imaging, and mantle convection encounters severe memory saturation. A dense Fourier neural operator layer parameterizes spectral weights via a multidimensional complex tensor $\mathcal{W} \in \mathbb{C}^{d_{\text{in}} \times d_{\text{out}} \times K_1 \times \dots \times K_d}$. In high-resolution planetary volumes ($1024 \times 1024 \times 100$), storing and computing $\mathcal{W}$ requires tens of gigabytes of GPU VRAM per layer, creating a major hardware bottleneck.

4.11.1 Multi-Grid Tensorized Fourier Neural Operators (MG-TFNO)

To alleviate this memory complexity, Kossaifi et al. [51] introduced the Multi-Grid Tensorized Fourier Neural Operator (MG-TFNO). MG-TFNO parameterizes the spectral

convolution weights in a high-order latent subspace using tensor factorizations, specifically Tucker and Tensor-Train (Matrix Product State, MPS) decompositions:

$$\mathcal{W}(i_1, i_2, k_1, \dots, k_d) \approx \sum_{r_1=1}^{R_1} \cdots \sum_{r_{d+2}=1}^{R_{d+2}} \mathcal{G}(r_1, \dots, r_{d+2}) \prod_{m=1}^{d+2} \mathbf{U}^{(m)}(i_m, r_m), \quad (40)$$

where $\mathcal{G} \in \mathbb{C}^{R_1 \times \dots \times R_{d+2}}$ is a low-rank core tensor with ranks $R_m \ll \min(d_m, K_m)$, and $\mathbf{U}^{(m)}$ are orthogonal factor matrices. Coupled with multi-grid domain decomposition that parallelizes local and global frequency representations, MG-TFNO achieves over $150\times$ parameter reduction (compressing over 120M dense parameters to $< 0.8\text{M}$) while improving zero-shot super-resolution and out-of-distribution numerical stability on 2D/3D Navier–Stokes turbulence and Helmholtz acoustic wavefields.

4.11.2 WaveFNO: Continuous Seismic Wavefield Modeling and Full-Waveform Inversion

Applying continuous neural operators to solid-Earth geophysics, Yang et al. [52] pioneered WaveFNO for seismic wave propagation and full-waveform inversion (FWI). Seismic wave propagation in heterogeneous media is governed by the elastodynamic wave equation:

$$\rho(\mathbf{x}) \frac{\partial^2 \mathbf{u}(\mathbf{x}, t)}{\partial t^2} = \nabla \cdot \boldsymbol{\sigma}(\mathbf{x}, t) + \mathbf{f}(\mathbf{x}, t), \quad \boldsymbol{\sigma} = \mathbf{C}(\mathbf{x}) : \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T), \quad (41)$$

where $\mathbf{u}(\mathbf{x}, t)$ is the continuous displacement wavefield, ρ is density, $\boldsymbol{\sigma}$ is the stress tensor, $\mathbf{C}$ is the elastic stiffness tensor, and $\mathbf{f}$ represents tectonic or exploration source terms. WaveFNO directly learns the non-linear solution operator $\mathcal{G}_\theta : c(\mathbf{x}) \mapsto \mathbf{u}(\mathbf{x}, t)$ mapping subsurface velocity models $c(\mathbf{x})$ to continuous wavefield trajectories. Because the learned operator $\mathcal{G}_\theta$ is continuously differentiable with respect to its inputs, seismic full-waveform inversion is executed via reverse-mode automatic differentiation:

$$c^* = \arg \min_c \frac{1}{2} \sum_{r=1}^R \int_0^T \|\mathbf{u}_{\text{obs}}^{(r)}(t) - \mathcal{G}_\theta(c)(\mathbf{x}_r, t)\|_2^2 dt + \lambda \mathcal{R}(c), \quad (42)$$

computing exact subsurface acoustic velocity updates $\nabla_c \mathcal{J}$ in seconds on a single GPU. This avoids the hundreds of computationally prohibitive numerical finite-difference wavefield solves required by classical adjoint-state FWI workflows while avoiding cycle-skipping traps.

4.11.3 Partitioned Hybrid Quantum Fourier Neural Operators (PHQFNO)

At the outer algorithmic frontier, Marcandelli et al. [53] introduced the Partitioned Hybrid Quantum Fourier Neural Operator (PHQFNO), establishing the first continuous neural operator architecture combining classical tensor decomposition with Parameterized Quantum Circuits (PQC). PHQFNO partitions the spectral transformation across classical tensor operators and variational quantum circuits executed on Noisy Intermediate-Scale Quantum (NISQ) devices. Input spectral modes are embedded into multi-qubit quantum states $|\psi(k)\rangle$ using quantum amplitude encoding, followed by parameterized unitary evolution:

$$U(\boldsymbol{\theta}) = \prod_{l=1}^L \exp(-i\theta_l H_l) W_{\text{entangle}}, \quad (43)$$

where $H_l \in \{X, Y, Z\}^{\otimes n}$ represent multi-qubit Pauli observables and W_{entangle} is a non-local entangling layer. PHQFNO provides an architectural roadmap for continuous scientific machine learning, demonstrating that quantum entanglement can bypass classical tensor contraction bottlenecks when solving high-dimensional multi-physics differential equations.

4.12 Symplectic Manifolds and Lie-Poisson Geometric Neural Operators

Standard autoregressive forecasting models (whether based on Transformers, GNNs, or unconstrained Fourier neural operators) suffer from progressive secular drift: as rollout horizons extend past hundreds of autoregressive cycles, numerical dissipation inevitably damps physical energy, or unconstrained error accumulation triggers catastrophic numerical blowup. In conservative geophysical fluids, orbital astrodynamics, and wave mechanics, this failure stems from the omission of underlying symplectic geometry.

4.12.1 Infinite-Dimensional Hamiltonian Mechanics and Symplectic Preservation

Conservative continuum mechanics can be formulated as infinite-dimensional Hamiltonian partial differential equations on a phase space $\mathcal{M}$ endowed with a canonical symplectic two-form ω :

$$\frac{\partial \mathbf{u}}{\partial t} = \mathcal{J} \frac{\delta \mathcal{H}}{\delta \mathbf{u}}, \quad (44)$$

where $\mathbf{u} = (\mathbf{q}, \mathbf{p})^T$ denotes canonical generalized coordinates and momenta, $\mathcal{H}[\mathbf{u}] = \int_\Omega \mathcal{E}(\mathbf{q}, \mathbf{p}, \nabla \mathbf{q}, \nabla \mathbf{p}) dx$ represents the total energy Hamiltonian functional, and $\mathcal{J}$ is the canonical Poisson operator:

$$\mathcal{J} = \begin{bmatrix} 0 & \mathbf{I} \\ -\mathbf{I} & 0 \end{bmatrix}, \quad \mathcal{J}^* = -\mathcal{J}. \quad (45)$$

By Liouville’s theorem, the continuous flow map $\Phi_t : \mathbf{u}(0) \mapsto \mathbf{u}(t)$ preserves the differential two-form $\omega = \int_\Omega \delta \mathbf{q} \wedge \delta \mathbf{p} dx$:

$$\Phi_t^* \omega = \omega \iff (D\Phi_t)^T \mathcal{J}^{-1} (D\Phi_t) = \mathcal{J}^{-1}, \quad (46)$$

which strictly enforces phase-space volume conservation. Makara et al. [49] established Symplectic Neural Operators (SNO), proving that standard Fourier neural operators violate Eq. (46), causing monotonic artificial dissipation in wave and fluid systems. SNO enforces canonical symplectic preservation by structuring the operator kernel as the composition of generating functionals:

$$\mathbf{p}^{n+1} = \mathbf{p}^n - \Delta t \frac{\delta \mathcal{H}_\theta}{\delta \mathbf{q}^{n+1}}, \quad \mathbf{q}^{n+1} = \mathbf{q}^n + \Delta t \frac{\delta \mathcal{H}_\theta}{\delta \mathbf{p}^n}, \quad (47)$$

where $\mathcal{H}_\theta$ is parameterized by a non-local neural operator. By construction, the resulting discrete update operator exactly conserves a shadow Hamiltonian $\tilde{\mathcal{H}} = \mathcal{H} + \mathcal{O}(\Delta t^2)$, eliminating long-term amplitude decay across $> 10^4$ simulation time steps.

4.12.2 Lie-Poisson Symmetries, Casimir Invariants, and Kelvin’s Circulation

In rotating planetary atmospheres and oceanic gyres, fluid equations are non-canonical Hamiltonian systems defined on

the dual of a Lie algebra $\mathfrak{g}^*$ (e.g., the algebra of divergence-free vector fields on $\mathbb{S}^2$). The dynamics of relative vorticity ω are governed by the Lie-Poisson equation:

$$\frac{\partial \omega}{\partial t} = \{\omega, \mathcal{H}\} = -\text{ad}_{\frac{\delta \mathcal{H}}{\delta \omega}}^* \omega = -\mathbf{u} \cdot \nabla \omega, \quad (48)$$

where ad^* is the coadjoint operator. A defining mathematical feature of Lie-Poisson systems is the presence of Casimir invariants $\mathcal{C}[\omega]$, functionals that Poisson-commute with every arbitrary Hamiltonian $\{\mathcal{C}, \mathcal{F}\} = 0$ for all $\mathcal{F}$. In 2D and quasi-geostrophic shallow-water atmospheres, every smooth function of vorticity produces a conserved Casimir functional:

$$\mathcal{C}_\Phi[\omega] = \int_\Omega \Phi(\omega) dx = \text{const}, \quad (49)$$

including total circulation ($\Phi(\omega) = \omega$) and potential enstrophy ($\Phi(\omega) = \omega^2$). Furthermore, by Kelvin's circulation theorem, the circulation around any fluid loop $\Gamma(t)$ advected by the velocity field is invariant:

$$\frac{d}{dt} \oint_{\Gamma(t)} \mathbf{u} \cdot d\mathbf{x} = 0. \quad (50)$$

Standard data-driven models inevitably destroy Casimir conservation, causing artificial enstrophy cascades and non-physical vortex distortion. Eldred et al. [50] developed Lie-Poisson Neural Networks (LPNets), enforcing geometric coadjoint orbit preservation:

$$\frac{d\omega}{dt} = \mathbb{B}(\omega) \nabla_\omega \mathcal{H}_\theta(\omega), \quad (51)$$

where $\mathbb{B}(\omega)$ is an algebraically constructed skew-symmetric Lie-Poisson matrix operator satisfying the Jacobi identity by design. LPNets strictly preserve all Casimir invariants and circulation invariants independently of the neural network weights θ , maintaining stable planetary jet streams, Rossby waves, and atmospheric blocking patterns over multi-month rollouts without empirical damping.

4.12.3 Learned Coordinate Diffeomorphisms for Complex Domains (Geo-FNO)

To apply neural operators across non-Euclidean computational domains, irregular boundaries, and faulted geological formations, Li et al. [48] proposed Geo-FNO. Classical FNO assumes uniform Cartesian grids in Euclidean space $\mathbb{R}^d$, failing on irregular physical geometries. Geo-FNO learns a bijective coordinate diffeomorphism:

$$\phi : \Omega \rightarrow [0, 1]^d, \quad \boldsymbol{\xi} = \phi(\mathbf{x}), \quad (52)$$

which deforms the irregular physical domain Ω into a canonical periodic unit cube $[0, 1]^d$. Spectral convolution is executed in the latent canonical coordinate space $\boldsymbol{\xi}$, while physical differential operators are transformed via the metric pullback tensor $\mathbf{g}(\mathbf{x}) = (D\phi)^T(D\phi)$. Geo-FNO enables continuous operator learning on arbitrary 3D geological reservoirs, faulted subsurfaces, and aerodynamic boundary layers without requiring conformal remeshing.

4.13 Generative Lagrangian Continuous Data Assimilation for Sparse Observables

While operational weather models rely on gridded satellite radiances and reanalysis fields, oceanic and environmental monitoring is overwhelmingly constrained by the extreme spatial sparsity and irregular sampling of autonomous in-situ platforms (e.g., Argo profiling floats, drifting surface buoys, underwater gliders). These platforms record physical states along Lagrangian trajectories:

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{u}(\mathbf{X}_i(t), t) + \boldsymbol{\eta}_i(t), \quad \mathbf{y}_i(t) = \mathcal{M}(\mathbf{s}(\mathbf{X}_i(t), t)) + \boldsymbol{\epsilon}_i(t), \quad (53)$$

where $\mathbf{X}_i(t)$ is the drifter position, $\mathbf{u}(\mathbf{x}, t)$ is the continuous Eulerian velocity field, $\boldsymbol{\eta}_i$ represents turbulent dispersion, and $\mathbf{y}_i$ records in-situ temperature, salinity, or pressure.

Traditional data assimilation methods (such as Ensemble Kalman Filtering or 4D-Var) struggle with Lagrangian observations because the forward observation operator is strongly non-linear, non-local, and history-dependent. Asefi et al. [60] addressed this fundamental challenge with Generative Lagrangian Data Assimilation (LagDA), formulating continuous ocean state estimation as a conditional score-based diffusion model:

$$p(\mathbf{u}_{1:T} \mid \{\mathbf{X}_i(t_k)\}_{i=1, k=1}^{N_d, K}), \quad (54)$$

where N_d is the number of active drifters covering less than 0.5% of the spatial basin. LagDA leverages continuous spatio-temporal attention to bridge sparse Lagrangian trajectory tokens with Eulerian spatial grids. By modeling the posterior distribution over continuous velocity $\mathbf{u} = (u, v)$, sea surface height (SSH), and relative vorticity $\zeta = \partial_x v - \partial_y u$, LagDA reconstructs submesoscale eddies and Western Boundary Currents with over 60% error reduction relative to classical variational interpolation, providing a foundational methodology for integrating autonomous marine sensor fleets into planetary Earth system models.

4.14 Analytic Conservation Enforcing and Subgrid Microphysics in Climate Emulators

Global climate models and neural planetary emulators operate on discretized horizontal grids ($\Delta x \approx 25\text{--}100$ km), leaving moist convective updrafts, turbulent boundary-layer mixing, and cloud microphysical transformations unresolved at subgrid scales (< 5 km). Let an atmospheric column state be decomposed via Reynolds filtering into resolved dynamics $\bar{\mathbf{X}} = (\bar{\mathbf{u}}, \bar{T}, \bar{q})$ and subgrid turbulent fluctuations $\mathbf{X}' = (\mathbf{u}', T', q')$. The filtered thermodynamic energy and moisture conservation equations read:

$$\frac{\partial \bar{T}}{\partial t} + \bar{\mathbf{u}} \cdot \nabla \bar{T} = -\frac{\partial(\bar{\omega}'T')}{\partial p} + \frac{L_v}{c_p} \bar{C} + \mathcal{Q}_{\text{rad}} \equiv P_T(\bar{\mathbf{X}}), \quad (55)$$

$$\frac{\partial \bar{q}}{\partial t} + \bar{\mathbf{u}} \cdot \nabla \bar{q} = -\frac{\partial(\bar{\omega}'q')}{\partial p} - \bar{C} \equiv P_q(\bar{\mathbf{X}}), \quad (56)$$

where $\bar{C}$ is the net condensation rate, L_v is latent heat of vaporization, c_p is specific heat, and P_T, P_q denote the subgrid parameterization mappings. When classical unconstrained neural networks emulate P_T and P_q , unphysical numerical artifacts (such as negative precipitation $P < 0$ or spurious column energy creation) accumulate rapidly, triggering numerical explosion or unphysical climate drift within weeks of coupled simulation.

4.14.1 Analytically Constrained Neural Networks (AC-NN) To resolve this foundational flaw, Beuclet et al. [62] formulated Analytically Constrained Neural Networks (AC-NN), mathematically proving that linear and non-linear physical conservation laws can be enforced within neural network computational graphs to machine precision. Consider an emulated physical system subject to M linear equality constraints:

$$\mathbf{C}(\mathbf{x})\hat{\mathbf{y}} = \mathbf{d}(\mathbf{x}), \quad \mathbf{C} \in \mathbb{R}^{M \times K}, \quad \mathbf{d} \in \mathbb{R}^M, \quad (57)$$

where $\hat{\mathbf{y}} \in \mathbb{R}^K$ represents the model's output vector (e.g., vertical profiles of convective heating rate $Q_1 \equiv c_p P_T$ and apparent drying rate $Q_2 \equiv -L_v P_q$). In columnar atmospheric physics, Eq. (57) encapsulates column-integrated enthalpy and total moisture conservation:

$$\langle Q_1 \rangle - \langle Q_2 \rangle = L_v P_{\text{sfc}} + R_{\text{net}} - S_{\text{sfc}}, \quad (58)$$

$$\langle Q_2 \rangle = -L_v (P_{\text{sfc}} - E_{\text{sfc}}), \quad (59)$$

where $\langle f \rangle = \frac{1}{g} \int_0^{P_s} f(p) dp$ represents the mass-weighted vertical integral from surface pressure p_s to the top of atmosphere.

Instead of adding soft loss penalties (which only approximately penalize violations), AC-NN analytically partitions predicted variables into unconstrained free components $\hat{\mathbf{y}}_{\text{free}} \in \mathbb{R}^{K-M}$ and constrained boundary variables $\hat{\mathbf{y}}_{\text{constrained}} \in \mathbb{R}^M$. Rearranging Eq. (57):

$$\mathbf{C}_{\text{free}}\hat{\mathbf{y}}_{\text{free}} + \mathbf{C}_{\text{constrained}}\hat{\mathbf{y}}_{\text{constrained}} = \mathbf{d}(\mathbf{x}). \quad (60)$$

Assuming full row rank for the constraint matrix, $\mathbf{C}_{\text{constrained}}$ is invertible, allowing exact architectural determination of the remaining variables:

$$\hat{\mathbf{y}}_{\text{constrained}} = \mathbf{C}_{\text{constrained}}^{-1} (\mathbf{d}(\mathbf{x}) - \mathbf{C}_{\text{free}}\hat{\mathbf{y}}_{\text{free}}). \quad (61)$$

Because Eq. (61) is implemented as a fixed, parameter-free linear layer at the network's terminal head, the model satisfies column conservation of enthalpy and moisture to machine precision ($\epsilon_{\text{res}} < 10^{-15}$) across all training iterations and arbitrary out-of-distribution inputs, guaranteeing long-term numerical stability in multi-year climate runs without degrading emulation fidelity.

4.14.2 Ai2 Climate Emulator (ACE): Hundred-Year Stable Planetary Rollouts

Scaling physics-constrained neural operators to multi-decadal planetary climate emulation, Watt-Meyer et al. [63] created ACE (Ai2 Climate Emulator). ACE couples a Spherical Fourier Neural Operator (SFNO) backbone operating on FV3GFS 1° global horizontal grids across 82 prognostic 3D and 2D atmospheric fields.

ACE hard-codes structural physical conservation of global dry air mass:

$$M_{\text{dry}} = \int_{\mathbb{S}^2} p_s^{\text{dry}}(\lambda, \theta, t) d\Omega = \text{constant}, \quad (62)$$

and total column water vapor, preventing secular mass depletion over century-long autoregressive integrations. In continuous 100-year rollouts ($> 146,000$ consecutive 6-hour time steps), ACE exhibits unprecedented climate stability: global mean surface temperature drift remains strictly bounded under 0.04 K/century, while accurately reproducing the climatological Intertropical Convergence Zone (ITCZ), Hadley cell overturning circulation, and storm track variability. Running at 1,400 simulated years per day on a single GPU,

ACE provides an empirical blueprint for replacing expensive atmospheric dynamical cores in climate projection ensembles.

4.15 Coupled Atmosphere-Ocean-Cryosphere Multi-Agent Emulation and Flux Coupling

A persistent limitation of standalone atmospheric foundation models (e.g., Pangu-Weather, GraphCast, Aurora) is that they treat sea surface temperature (SST) and sea ice concentration (SIC) as prescribed, static boundary conditions. While sufficient for medium-range synoptic weather (≤ 15 days), this uncoupled formulation fails across subseasonal-to-seasonal (S2S) and decadal timescales, where climate dynamics are governed by reciprocal thermodynamic air-sea-cryosphere feedbacks:

$$\tau_{\text{air-sea}} = \rho_a C_D \|\mathbf{u}_a - \mathbf{u}_o\| (\mathbf{u}_a - \mathbf{u}_o), \quad (63)$$

$$Q_{\text{net}} = Q_{\text{SW}}^\downarrow (1 - \alpha) - Q_{\text{LW}}^\uparrow + Q_{\text{LW}}^\downarrow - Q_{\text{sens}} - Q_{\text{lat}}, \quad (64)$$

$$F_{\text{fresh}} = \rho_w (E - P - M_{\text{melt}}), \quad (65)$$

where τ is wind stress momentum flux, Q_{net} is interfacial net heat flux, and F_{fresh} is surface freshwater mass flux.

4.15.1 DLESyM: Multi-Rate Asynchronous Earth System Coupler

To achieve fully coupled planetary climate simulation, Crosswell-Clay et al. [57] introduced DLESyM (Deep Learning Earth System Model). DLESyM formulates Earth system emulation as an asynchronous multi-agent system on equal-area HEALPix spherical grids, structurally decoupling the distinct characteristic dynamical timescales of the atmosphere and ocean:

- 1) Atmospheric Neural Solver ($\mathcal{M}_{\text{atm}}$): A spherical U-Net integrating fast synoptic dynamics (3D wind, temperature, humidity, geopotential) at $\Delta t_{\text{atm}} = 12$ hours.
- 2) Ocean-Sea Ice Neural Solver ($\mathcal{M}_{\text{ocn}}$): A specialized spherical U-Net integrating slow oceanic thermohaline dynamics (3D upper ocean temperature, salinity, currents, sea-ice thickness) at $\Delta t_{\text{ocn}} = 4$ days.

At every 4-day coupling interface, the atmospheric solver computes time-accumulated surface wind stress $\bar{\tau}$ and net heat flux $\bar{Q}_{\text{net}}$, passing them to the ocean solver. In turn, the ocean solver updates the underlying SST and sea-ice concentration fields for the atmospheric model. DLESyM performs stable 1,000-year continuous coupled simulations without empirical damping or flux adjustments, maintaining global mean temperature drift below 0.08 K/century. Crucially, DLESyM faithfully reproduces the multi-year cycle of the El Niño-Southern Oscillation (ENSO), tropical cyclogenesis statistics, and the Indian Summer Monsoon, demonstrating that data-driven foundation models can capture long-term emergent coupled climate modes.

4.15.2 SamudrACE: 3D High-Resolution Foundation Earth System Emulation

Pushing coupled foundation models to full 3D spatial resolution, Duncan et al. [58] established SamudrACE. SamudrACE couples the 3D atmospheric ACE SFNO foundation emulator ($\Delta t = 6$ hours) with the 3D Samudra ocean emulator ($\Delta t = 5$ days), tracking 145 two-dimensional prognostic fields

across atmospheric pressure levels and oceanic depth strata at 1° global resolution. Interfacial boundary couplers exchange heat, freshwater, and momentum fluxes subject to continuous conservation checks.

In multi-century benchmark evaluations, SamudrACE achieves an unprecedented simulation throughput of 1,100 simulated years per day (SYPD) on a single NVIDIA H100 GPU (over $100\times$ faster than numerical coupled GCMs running across thousands of CPU cores on supercomputing clusters). SamudrACE reproduces the full power spectrum and phase-locking of ENSO SST anomalies in the Nino 3.4 region with an RMSE under 0.32 K, demonstrating the viability of coupled neural operators for trillion-parameter ensemble climate sensitivity projections.

4.16 Distribution-Free Conformal Prediction & Tail Risk Bounds in Neural Weather Forecasts

While modern deep learning weather models (e.g., Pangu-Weather, GraphCast, FuXi, Aurora) deliver competitive deterministic point forecasts, they routinely fail to provide statistically rigorous, calibrated uncertainty estimates. In operational weather forecasting, downstream decision makers (e.g., aviation routing, agricultural planting, disaster flood evacuation) require reliable confidence intervals with formal guarantees, particularly during high-impact, rare extreme weather events where ensemble spread often collapses.

4.16.1 Split Conformal Prediction on Continuous Spatio-Temporal Grids

Addressing this critical gap, Gopakumar et al. [54] formulated a distribution-free, model-agnostic conformal prediction framework as a post-processing layer for neural weather models. Consider a pre-trained neural weather model $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$ mapping historical atmospheric states $\mathbf{x} \in \mathbb{R}^{C \times H \times W}$ to lead-time forecasts $\mathbf{y} \in \mathbb{R}^{C \times H \times W}$. Split conformal prediction partitions the historical verification data into a training set and an independent calibration set $\mathcal{D}_{\text{cal}} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$.

For each variable c , vertical level l , spatial grid point (h, w) , and lead time t , we define a non-conformity score measuring the discrepancy between the prediction and ground truth. Under absolute residual non-conformity:

$$s_i^{(c,l,h,w,t)} = \left| y_i^{(c,l,h,w,t)} - \hat{f}(\mathbf{x}_i)^{(c,l,h,w,t)} \right|. \quad (66)$$

Alternatively, for models predicting conditional quantile intervals $[\hat{q}_{\alpha/2}(\mathbf{x}), \hat{q}_{1-\alpha/2}(\mathbf{x})]$ (e.g., via pinball loss or ensemble percentiles), the conformalized quantile regression (CQR) non-conformity score is formulated as:

$$s_i = \max(\hat{q}_{\alpha/2}(\mathbf{x}_i) - y_i, y_i - \hat{q}_{1-\alpha/2}(\mathbf{x}_i)). \quad (67)$$

Given a desired user-specified coverage level $1 - \alpha \in (0, 1)$ (e.g., 90% or 95%), the conformal threshold $\hat{q}_{1-\alpha}$ is computed as the empirical $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of the calibration scores $\{s_1, \dots, s_n\}$:

$$\hat{q}_{1-\alpha} = \text{Quantile} \left(\{s_i\}_{i=1}^n, \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right). \quad (68)$$

For an unseen test input $\mathbf{x}_{n+1}$, the conformal prediction set is constructed as:

$$\mathcal{C}_{1-\alpha}(\mathbf{x}_{n+1}) = \left[\hat{f}(\mathbf{x}_{n+1}) - \hat{q}_{1-\alpha}, \hat{f}(\mathbf{x}_{n+1}) + \hat{q}_{1-\alpha} \right]. \quad (69)$$

Under the assumption of exchangeability among calibration and test samples, conformal prediction mathematically guarantees finite-sample marginal coverage:

$$P(\mathbf{y}_{n+1} \in \mathcal{C}_{1-\alpha}(\mathbf{x}_{n+1})) \geq 1 - \alpha. \quad (70)$$

4.16.2 Resolving Spatio-Temporal Non-Exchangeability & Tail Extremes

In numerical weather prediction, physical atmospheric turbulence violates exchangeability due to strong spatial teleconnections, diurnal cycles, and seasonal distribution shifts. Gopakumar et al. [54] resolve this by introducing:

- 1) **Localized Spatio-Temporal Calibration Bins:** Rather than aggregating global scores, non-conformity distributions are conditioned on geographic clusters (e.g., maritime vs. continental vs. polar regimes) and lead-time horizons ($t \in [1, 10]$ days), capturing heteroskedastic spatial error variance.
- 2) **Online Rolling Calibration:** To track seasonal non-stationarity, calibration sets are updated autoregressively over a sliding window $\mathcal{W}_{\text{cal}}$ of preceding days, preserving temporal validity without model retraining.

Evaluated on operational limited-area weather forecasts across Northern Europe (Nordic domain) and global ERA5 benchmarks, the conformal framework achieves exact empirical coverage (90.2% for nominal 90%, and 95.1% for nominal 95%) across geopotential $Z500$, temperature $T850$, and surface wind fields. This resolves the chronic underdispersion of neural point predictors while maintaining tight prediction interval bandwidths $\bar{W}$, offering a rigorous bridge between black-box foundation models and operational risk assessment.

4.17 Magnetohydrodynamic Neural Operators & Coronal Solar Wind Surrogates

Space weather phenomena—including solar flares, coronal mass ejections (CMEs), and geomagnetic substorms—are governed by 3D non-linear Magnetohydrodynamics (MHD). Planetary space environment forecasting requires simulating the coupled plasma-magnetic interaction from the solar photosphere across the solar corona to the Earth’s magnetosphere. The ideal 3D MHD equations governing compressible, magnetized astrophysical plasma state $\mathbf{U} = (\rho, \rho\mathbf{u}, \mathbf{B}, E)$ read:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho\mathbf{u}) = 0, \quad (71)$$

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla \cdot \left(\rho\mathbf{u}\mathbf{u}^T + \left(p + \frac{\|\mathbf{B}\|^2}{2\mu_0} \right) \mathbf{I} - \frac{\mathbf{B}\mathbf{B}^T}{\mu_0} \right) = \rho\mathbf{g}, \quad (72)$$

$$\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{u} \times \mathbf{B}) = \eta \nabla^2 \mathbf{B}, \quad (73)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (74)$$

where ρ is plasma mass density, $\mathbf{u}$ is plasma bulk velocity, $\mathbf{B}$ is magnetic flux density vector, p is thermal gas pressure, μ_0 is magnetic permeability, η is magnetic resistivity, and Eq. (74) enforces the divergence-free solenoidal constraint. Traditional numerical MHD solvers (e.g., MAS, BATS-R-US, Bifrost) require millions of CPU core-hours per Carrington rotation (~ 27.3 days), rendering real-time operational solar storm warnings intractable.

4.17.1 Global-Local Fourier Neural Operators (GL-FNO)

To overcome the extreme computational bottleneck of coronal MHD simulations, Du et al. [35] introduced Global-Local Fourier Neural Operators (GL-FNO). Coronal magnetic fields exhibit an extreme multiscale dichotomy: macroscale magnetic flux ropes and open-field coronal holes span the entire solar disk ($R_\odot \approx 7 \times 10^5$ km), while localized magnetic reconnection triggering solar flares occurs within compact active regions ($< 0.05R_\odot$).

GL-FNO decouples this multiscale problem into a dual-branch spectral operator architecture:

- 1) Global FNO Branch ($\mathcal{F}_{\text{global}}$): Ingests spatially down-sampled full-disk vector magnetograms $\mathbf{x}_{\text{global}} \in \mathbb{R}^{C \times H_{\text{down}} \times W_{\text{down}}}$, employing low-frequency Fourier spectral convolutions to resolve global magnetic topology, dipolar field orientation, and large-scale open/closed boundary contours.
- 2) Local FNO Branch ($\mathcal{F}_{\text{local}}$): Ingests high-resolution localized active region patches $\mathbf{x}_{\text{local}} \in \mathbb{R}^{C \times H \times W}$, capturing steep magnetic shear angles, current sheets ($\mathbf{J} = \nabla \times \mathbf{B}/\mu_0$), and intense local Lorentz forces ($\mathbf{J} \times \mathbf{B}$).

The global and local latent representations are fused via cross-attention gated feature modulation:

$$\mathbf{z}_{\text{fused}} = \mathcal{W}_{\text{local}} \mathbf{z}_{\text{local}} + \gamma \cdot \text{CrossAttn}(\mathbf{z}_{\text{local}}, \mathbf{z}_{\text{global}}), \quad (75)$$

where γ is a learnable gating parameter. GL-FNO enforces the solenoidal constraint Eq. (74) by training with a divergence loss penalty $\|\nabla \cdot \mathbf{B}\|_2^2$. In benchmark evaluations against high-fidelity 3D MHD numerical simulations, GL-FNO predicts the complete 3D coronal magnetic field in 0.04 seconds, achieving an acceleration of over 20,000 $\times$ relative to classical numerical MHD solvers while maintaining a relative L_2 magnetic field error below 3.2%.

4.17.2 Spherical Fourier Neural Operators for Heliospheric Solar Wind Surrogates

Complementing coronal simulations, Mansouri et al. [36] developed the first Spherical Fourier Neural Operator (SFNO) surrogate for steady-state solar wind velocity modeling across the inner heliosphere (from $30R_\odot$ out to Earth's orbit at $1 \text{ AU} \approx 215R_\odot$). Solar wind propagation is modeled across concentric spherical shells $\mathbb{S}^2$, where radial velocity $V_{\text{sw}}(\theta, \phi, r)$ is advected outward by supersonic plasma streams:

$$V_{\text{sw}}(\theta, \phi, r_{k+1}) = \mathcal{K}_{\text{SFNO}}(V_{\text{sw}}(\theta, \phi, r_k)). \quad (76)$$

By transforming radial shell slices via spherical harmonics $Y_\ell^m(\theta, \phi)$, the SFNO surrogate captures non-linear stream interaction regions (SIRs) where fast solar wind streams (> 700 km/s) collide with slow ambient wind (< 400 km/s), creating shock waves that drive severe geomagnetic storms. In comparisons against the standard numerical surrogate HUX (Heliospheric Upwind eXtrapolation), the Solar Wind SFNO matches numerical accuracy with mean absolute error < 28 km/s while executing full heliospheric forecasts in under 20 milliseconds, providing an essential foundation for real-time planetary space weather defense.

4.18 Clifford Neural Operators, Multivector Fields, and Non-Equilibrium Boundary-Layer Turbulence Closures

Accurate emulation of turbulent geophysical fluid dynamics across planetary boundary layers requires capturing non-linear cross-field interactions between scalar, vector, and rotational quantities, while resolving severe computational multiscale barriers near solid and topographic surfaces.

4.18.1 Clifford Fourier Neural Operators (CFNO) and Multivector PDEs

Conventional neural PDE solvers treat physical quantities (such as 2D/3D velocity fields $\mathbf{u} = (u, v, w)$, thermal pressure p , and vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{u}$) as uncoupled, independent channels within a Euclidean feature tensor. Consequently, they fail to preserve intrinsic geometric invariants under spatial rotations, reflections, and coordinate transformations. Addressing this fundamental limitation, Brandstetter et al. [12] established Clifford Neural Layers, unifying coupled physical fields through the mathematical framework of Clifford (geometric) algebras $\mathcal{C}\ell(p, q)$.

Let V be an n -dimensional real vector space with signature (p, q) ($p + q = n$). The associated Clifford algebra $\mathcal{C}\ell(p, q)$ is generated by orthogonal basis elements $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ satisfying the fundamental anti-commutation relations:

$$\mathbf{e}_i \mathbf{e}_j + \mathbf{e}_j \mathbf{e}_i = 2\eta_{ij} \mathbf{I}, \quad (77)$$

where $\eta_{ii} = +1$ for $1 \leq i \leq p$, $\eta_{ii} = -1$ for $p < i \leq n$, and $\eta_{ij} = 0$ for $i \neq j$. An arbitrary multivector field $\mathbf{u}(\mathbf{x}) \in \mathcal{C}\ell(p, q)$ admits an orthogonal grade decomposition into geometric subspaces:

$$\mathbf{u}(\mathbf{x}) = \underbrace{\langle \mathbf{u} \rangle_0}_{\text{scalar } p} + \underbrace{\langle \mathbf{u} \rangle_1}_{\text{vector } \mathbf{v}} + \underbrace{\langle \mathbf{u} \rangle_2}_{\text{bivector } \boldsymbol{\omega}} + \dots + \underbrace{\langle \mathbf{u} \rangle_n}_{\text{pseudoscalar}}, \quad (78)$$

where the geometric product between two arbitrary vectors $\mathbf{a}, \mathbf{b} \in V$ combines both symmetric scalar (inner) and anti-symmetric directional area (wedge) products:

$$\mathbf{a}\mathbf{b} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}. \quad (79)$$

In continuous fluid mechanics and atmospheric dynamics, the bivector component $\mathbf{a} \wedge \mathbf{b}$ corresponds directly to the rotational vorticity tensor $\boldsymbol{\Omega}_{ij} = \frac{1}{2}(\partial_j u_i - \partial_i u_j)$ and enstrophy cascades.

Brandstetter et al. formulated the Clifford Fourier Neural Operator (CFNO), parameterizing spectral convolutions via Clifford Fourier transforms:

$$(\mathcal{K}_{\mathcal{C}\ell} * \mathbf{u})(\mathbf{x}) = \mathcal{F}_{\mathcal{C}\ell}^{-1}(\mathbf{W}_{\mathcal{C}\ell}(\mathbf{k}) \cdot \mathcal{F}_{\mathcal{C}\ell}\{\mathbf{u}\}(\mathbf{k}))(\mathbf{x}), \quad (80)$$

where $\mathbf{W}_{\mathcal{C}\ell}(\mathbf{k}) \in \mathcal{C}\ell(p, q)$ is a learnable multivector spectral weight tensor acting through the Clifford geometric product. By maintaining multivector grade equivariance, CFNO guarantees that scalar pressure fields, vector velocity advection, and bivector vorticity stretching remain geometrically coupled across all frequency modes. Evaluated on 2D incompressible Navier-Stokes turbulence and ERA5 atmospheric wind and geopotential fields, CFNO consistently reduces relative prediction error by up to 35% compared to standard uncoupled Fourier Neural Operators under identical parameter constraints.

4.18.2 Subgrid Wall-Modeled Large-Eddy Simulation (WM-LES) Neural Closures

While global neural operators model synoptic free-tropospheric flows with exceptional fidelity, physical boundaries (such as planetary boundary layer interactions over mountainous or rough surface terrain) introduce severe multiscale challenges. In high-Reynolds-number wall turbulence ($Re_\tau \gg 10^4$), resolving the inner viscous sublayer requires spatial grid points scaling as $\mathcal{O}(Re^{2.6})$, rendering wall-resolved numerical simulations computationally impossible for planetary Earth system models.

To overcome this grid resolution bottleneck, Lozano-Durán and Bae [13] proposed a Machine-Learning Building-Block-Flow Wall Model for Large-Eddy Simulation (WM-LES). Rather than relying on classical algebraic equilibrium log-law assumptions (which break down catastrophically under non-equilibrium separation, strong adverse pressure gradients (APG), and 3D mean flow skewness), the framework conceptualizes complex non-equilibrium turbulent boundary layers as local convex combinations of canonical “building-block flows”.

The model decouples subgrid wall-shear stress prediction into a dual-branch neural architecture:

- 1) Flow Regime Classifier Network (π_θ): Ingests outer-flow velocity and gradient profiles at a matching height h_m (velocity magnitude $\|\mathbf{u}\|$, strain rate S_{ij} , velocity gradient $\partial_y \mathbf{u}$, and wall-normal pressure gradient $\partial_x p$) and predicts a probability distribution over K canonical flow regimes:

$$\mathbf{w}(\mathbf{x}) = \text{softmax}(\pi_\theta(\mathbf{s}(h_m))), \quad \sum_{k=1}^K w_k = 1, \quad (81)$$

spanning zero-, favorable-, and adverse-pressure-gradient wall turbulence, separated boundary layers, and laminar-turbulent transitional regimes.

- 2) Wall-Shear Stress Predictor Ensemble ($\mathcal{T}_\phi$): Synthesizes regime-specific wall-shear stress predictions $\hat{\tau}_{w,k}$, producing the composite surface traction:

$$\tau_w(\mathbf{x}) = \sum_{k=1}^K w_k(\mathbf{x}) \hat{\tau}_{w,k}(\mathbf{s}(h_m)). \quad (82)$$

Trained directly on high-fidelity LES data optimized to reproduce mean momentum balances, the building-block wall model emits an intrinsic confidence metric based on policy entropy $\mathcal{H}(\mathbf{w}) = -\sum_k w_k \log w_k$, alerting operators when the local flow deviates from the canonical training manifold. Validated across complex boundary-layer separations and aircraft juncture flows, the building-block neural wall model outperforms equilibrium wall models across adverse pressure gradients and separation bubbles, providing a physically consistent subgrid boundary closure that directly interfaces with outer atmospheric and oceanic fluid solvers.

4.19 Neural Operator Subgrid Gravity Wave Drag and Mesoscale Orographic Parameterizations

A major source of systematic error in global numerical weather prediction (NWP) and decadal Earth system modeling originates from unresolved subgrid atmospheric gravity waves (GWs). Generated by stratified flow over complex

orography (orographic gravity waves) as well as tropospheric convective storms, frontal systems, and jet stream imbalances (non-orographic gravity waves), these mesoscale oscillations propagate vertically into the middle atmosphere. As background atmospheric density decays exponentially with height ($\rho(z) \propto e^{-z/H}$), wave amplitudes amplify until breaking, saturation, or critical-level absorption occurs, depositing horizontal momentum into the background mean flow.

4.19.1 Vertical Momentum Flux Divergence and Stratospheric Dynamics

The net dynamical forcing exerted by unresolved subgrid gravity waves on the resolved horizontal velocity $\mathbf{u}_h = (u, v)$ is governed by the vertical divergence of Reynolds stress tensor fluxes:

$$\mathbf{F}_{\text{GW}}(z) = -\frac{1}{\rho_0(z)} \frac{\partial \tau_{\text{GW}}}{\partial z} = -\frac{1}{\rho_0(z)} \frac{\partial}{\partial z} (\overline{\rho_0 \mathbf{u}'_h w'}), \quad (83)$$

where $\rho_0(z)$ is background air density, w' denotes the vertical perturbation velocity, and τ_{GW} represents the horizontal wave stress vector. For orographic gravity waves generated by subgrid topographic variance σ_h^2 , the surface momentum flux boundary condition is given by:

$$\tau_0 = -\rho_0 K N V \sigma_h^2 \frac{\mathbf{V}}{\|\mathbf{V}\|}, \quad (84)$$

where $N = \sqrt{\frac{g}{\theta} \frac{\partial \theta}{\partial z}}$ is the Brunt–Väisälä buoyancy frequency, $\mathbf{V}$ is the low-level wind vector, and K is a tunable non-dimensional drag coefficient. In conventional atmospheric circulation models, evaluating non-orographic spectral wave packets across continuous directional and phase speed spectra ($c \in [-80, +80]$ m/s) consumes up to 10–15% of total physics parameterization compute time, while relying on single-column vertical approximations that ignore horizontal ray propagation and non-local wave dispersion.

4.19.2 Operational Emulation in ECMWF IFS: Chantry et al.

Addressing the severe computational cost of operational spectral gravity wave solvers, Chantry et al. [81] pioneered deep machine learning emulation of gravity wave drag inside the operational ECMWF Integrated Forecasting System (IFS). Operating on high-resolution operational column profiles ($T_{\text{Co1279}}, \sim 9$ km horizontal grid spacing with 137 vertical hybrid sigma-pressure levels), the authors designed deep neural network emulators mapping vertical profiles of wind (u, v) and virtual temperature T_v directly to tendency profiles $(\partial u / \partial t, \partial v / \partial t)_{\text{GW}}$.

Crucially, Chantry et al. tackled the fundamental challenge of online numerical stability when coupling data-driven emulators into operational dynamical cores. By enforcing physics-informed loss constraints that penalize total column momentum violations and clipping spurious unphysical tendencies in the upper mesosphere ($p < 0.1$ hPa), the neural emulator achieved a $10\times$ acceleration over the operational Fortran non-orographic gravity wave drag parameterization. In full prognostic 1-year simulations, the coupled neural model maintained numerical stability without climate drift, reproducing the stratospheric polar vortex, the semi-annual oscillation, and the mean zonal wind profiles within ± 0.8 m/s of the operational IFS reference across the entire stratosphere and lower mesosphere (0.1 to 100 hPa).

4.19.3 QBO Preservation and Out-of-Distribution Generalization: Espinosa et al.

While operational emulators demonstrate speedup, a fundamental scientific benchmark for climate parameterization is whether a learned neural network can autonomously reproduce emergent multi-year cyclical phenomena that are not explicitly programmed into the architecture. In the equatorial stratosphere, the Quasi-Biennial Oscillation (QBO)—a downward-propagating reversal of easterly and westerly zonal winds with a mean period of ~ 28 months—is driven by the delicate constructive interference and critical-level absorption of subgrid gravity wave momentum fluxes.

Espinosa et al. [82] deployed a physics-constrained neural network parameterization within the Model of an Idealized Moist Atmosphere (MiMA), emulating the Alexander–Dunkerton (AD99) non-orographic gravity wave scheme. Over multi-decadal continuous prognostic rollouts, the neural parameterization demonstrated the remarkable emergent capability to internally generate and stably maintain a realistic QBO, matching the spectral period (28.4 months) and amplitude (~ 25 m/s) of the physical benchmark within 3.2% relative error.

Furthermore, Espinosa et al. subjected the trained neural parameterization to extreme out-of-distribution climate perturbations by forcing the host general circulation model with quadrupled carbon dioxide concentrations ($4 \times \text{CO}_2$). Without retraining or parameter recalibration, the neural parameterization accurately captured the physically expected deceleration and vertical displacement of the QBO under extreme greenhouse warming. This benchmark established that neural parameterizations can capture genuine physical causal dynamics rather than superficial statistical correlations, clearing a major hurdle for the integration of data-driven gravity wave operators into decadal Earth system models.

4.20 Physics-Guided Neural Operators for Atmospheric Chemistry and Stiff Chemical Kinetics

A formidable computational bottleneck in comprehensive Earth system and air quality modeling is the numerical integration of atmospheric chemical kinetics. Atmospheric composition dictates radiative forcing (via ozone O_3 and aerosols), cloud condensation nuclei activation, and human health hazards (fine particulate matter $\text{PM}_{2.5}$). In global chemical transport models (CTMs, such as GEOS-Chem, WRF-Chem, and CMAQ), the time evolution of a chemically reactive atmospheric column is governed by a system of coupled non-linear Ordinary Differential Equations (ODEs):

$$\frac{d\mathbf{c}}{dt} = \mathbf{P}(\mathbf{c}, T, J) - \mathbf{L}(\mathbf{c}, T, J) \odot \mathbf{c} \equiv \mathbf{f}(\mathbf{c}, T, J), \quad (85)$$

where $\mathbf{c} \in \mathbb{R}^K$ is the concentration vector of K reactive chemical species ($K \sim 100\text{--}300$ for tropospheric oxidant-aerosol mechanisms), $\mathbf{P} \in \mathbb{R}^K$ represents non-linear chemical production rates from bi- and tri-molecular collisions, $\mathbf{L} \in \mathbb{R}^K$ represents the pseudo-first-order chemical loss frequency vector, T is temperature, and J denotes actinic photolysis rate coefficients.

4.20.1 Extreme Numerical Stiffness in Atmospheric Chemical Networks

The defining challenge of Eq. (85) is its severe numerical stiffness. Atmospheric chemical timescales span more than twelve orders of magnitude: radical propagation reactions (e.g., hydroxyl radical OH , singlet oxygen $\text{O}(^1D)$, and peroxy radicals RO_2) reach local photochemical equilibrium within milliseconds ($\tau_{\text{OH}} \sim 10^{-3}\text{--}1$ s), whereas secondary trace gases and greenhouse reservoirs (e.g., methane CH_4 , carbon monoxide CO , and ozone) persist for weeks to decades.

Formally, the Jacobian matrix of the chemical reaction network,

$$\mathbf{J}_{ij} = \frac{\partial f_i}{\partial c_j}, \quad \mathbf{J} \in \mathbb{R}^{K \times K}, \quad (86)$$

exhibits eigenvalues $\lambda_k(\mathbf{J})$ whose real negative components span multiple orders of magnitude ($|\text{Re}(\lambda_{\min})| \sim 10^{-9} \text{ s}^{-1}$ while $|\text{Re}(\lambda_{\max})| > 10^5 \text{ s}^{-1}$), yielding condition numbers exceeding $\kappa(\mathbf{J}) > 10^{14}$. Classical explicit Runge–Kutta integrators are constrained by Courant-like stability limits requiring step sizes $\Delta t < 2/|\lambda_{\max}| \sim 10^{-5}$ s, which is computationally prohibitive for global multi-year climate runs. Consequently, operational atmospheric models rely on implicit stiff ODE solvers (such as the Kinetic PreProcessor [KPP] Rosenbrock integrator or multi-step Gear methods), which require iterative LU factorizations and sparse matrix inversions at every operator-splitting timestep ($\Delta t_{\text{chem}} \sim 10\text{--}30$ min). These implicit solvers routinely consume 85% to 95% of total compute time in global atmospheric chemistry simulations.

4.20.2 Online-Learned Neural Chemical Solvers: Kelp et al.

While offline supervised neural networks can approximate chemical state transitions over isolated single timesteps, classical offline training fails catastrophically when coupled into multi-step climate rollouts: prediction errors compound recursively, generating unphysical negative concentrations, artificial mass creation, and explosive numerical divergence within days.

Kelp et al. [83] overcame this instability by developing an online-learned neural network chemical solver directly integrated into the global GEOS-Chem atmospheric chemistry model. Rather than training strictly on pre-computed static simulation snapshots, Kelp et al. implemented an online feedback learning paradigm: the neural emulator is recursively integrated inside the host dynamical core alongside atmospheric advection, boundary-layer turbulence, and emissions. When the neural solver encounters out-of-distribution regional regimes or stiff chemical transitions, adaptive error gradients from the numerical solver backpropagate through operator-splitting cycles, stabilizing the learned trajectory.

Crucially, the architecture incorporates hard physical conservation projections:

- 1) Family Element Conservation: Total reactive nitrogen ($\text{NO}_y = \text{NO} + \text{NO}_2 + \text{HNO}_3 + \dots$) and total sulfur ($\text{SO}_x = \text{SO}_2 + \text{SO}_4^{2-}$) are preserved through mass-weighted linear projection heads $\mathbf{c}_{\text{conserved}} = \mathbf{P}_{\text{elem}} \mathbf{c}$;
- 2) Non-Negativity Constraint: Chemical concentrations are enforced strictly positive via rectified linear activations, eliminating unphysical negative mass hallucinations.

In full prognostic 1-year global simulations at $4^\circ \times 5^\circ$ resolution, Kelp et al. demonstrated that the online-learned neural solver achieves unconditional numerical stability, matching GEOS-Chem benchmark concentrations for tropospheric ozone, nitrogen oxides, and sulfate aerosols ($R^2 \geq 0.95$) while delivering a $12\times$ acceleration over the operational KPP stiff Rosenbrock solver.

4.20.3 Deep Residual Recurrent Emulators for Stiff Chemical ODEs: Liu et al.

Addressing stiff chemical kinetics from a deep operator perspective, Liu et al. [84] formulated a Deep Residual Recurrent Neural Network (DR-RNN) emulator for time-dependent atmospheric chemical ODEs. Rather than treating the chemical state transition $\mathbf{c}_t \rightarrow \mathbf{c}_{t+\Delta t}$ as a black-box regression, DR-RNN parameterizes the continuous rate increment:

$$\mathbf{c}_{t+\Delta t} = \mathbf{c}_t + \Delta t \cdot \mathcal{N}_\theta(\mathbf{c}_t, \mathbf{h}_t, T_t, J_t), \quad (87)$$

where $\mathbf{h}_t$ represents recurrent memory states capturing the multi-step history of radical chain propagation.

To enforce kinetic consistency without requiring expensive numerical Jacobian inversions, Liu et al. introduced a stiff ODE kinetic loss combining trajectory MSE with the residual of the analytical chemical production-loss rate:

$$\mathcal{L}_{\text{kinetic}} = \|\mathbf{c}_{t+\Delta t} - \hat{\mathbf{c}}_{t+\Delta t}\|_2^2 + \lambda_{\text{stiff}} \left\| \frac{\hat{\mathbf{c}}_{t+\Delta t} - \mathbf{c}_t}{\Delta t} - \mathbf{f}(\mathbf{c}_t, T_t, J_t) \right\|_2^2, \quad (88)$$

Evaluated across atmospheric oxidation mechanisms governing sulfuric acid (H_2SO_4) aerosol nucleation and monoterpene secondary organic aerosol (SOA) formation, DR-RNN tracks steep non-linear photochemical spikes with relative errors below 1.2%, while executing over $50\times$ faster than standard CVODE/VODE numerical solvers. By bypassing stiff matrix factorizations, physics-guided chemical neural operators provide a scalable foundation for integrating complex gas-aerosol chemistry directly into planetary foundation models.

4.21 Physics-Informed Latent and Cascaded Diffusion for Extreme Convective Precipitation Nowcasting

While synoptic planetary weather forecasting focuses on large-scale geostrophic and hydrostatic balances at lead times of days to weeks, extreme convective precipitation nowcasting (0–3 hours) is governed by rapid, non-hydrostatic cloud microphysics and violent vertical ascent. Convective storms are triggered in atmospheres characterized by high Convective Available Potential Energy ($\text{CAPE} > 2,000 \text{ J/kg}$) and localized erosion of Convective Inhibition ($\text{CIN} \rightarrow 0 \text{ J/kg}$). Under these conditions, radar reflectivity factors $\mathbf{R}(\mathbf{x}, t)$ exceed 45–55 dBZ within minutes, generating localized precipitation rates surpassing 50–100 mm/h that trigger flash floods and severe urban hazards.

Classical numerical weather prediction (NWP) models (e.g., HRRR or CMA-MESO) suffer from spin-up latency and cannot ingest high-frequency radar mosaics in real time. Conversely, standard deterministic deep learning architectures (e.g., ConvLSTM or PredRNN) optimize Mean Squared Error (MSE), which averages multi-modal future possibilities and yields progressively blurred precipitation fields that wash out localized convective cores. Unconstrained generative diffusion

models (e.g., standard DDPM) alleviate blurriness but frequently hallucinate synthetic storm cells out of thin air that violate conservation of atmospheric water mass.

4.21.1 Governing Advection-Diffusion Equations with Source-Sink Terms

To provide a physically grounded framework for radar nowcasting, the spatial-temporal evolution of precipitation reflectivity $\mathbf{R}(\mathbf{x}, t) \in \mathbb{R}^{H \times W}$ is governed by the two-dimensional continuity equation coupled to a turbulent advection-diffusion operator:

$$\frac{\partial \mathbf{R}}{\partial t} + \nabla \cdot (\mathbf{v}(\mathbf{x}, t) \mathbf{R}) - \nabla \cdot (D \nabla \mathbf{R}) = \mathcal{S}_{\text{conv}}(\mathbf{x}, t), \quad (89)$$

where $\mathbf{v}(\mathbf{x}, t) = (u, v)^T$ represents the horizontal advection velocity field (optical flow / mid-tropospheric steering wind), $D \geq 0$ is the turbulent subgrid diffusion coefficient, and $\mathcal{S}_{\text{conv}}(\mathbf{x}, t)$ denotes the net convective source-sink term accounting for explosive cell initiation, convective updraft intensification, droplet coalescence, and storm cell dissipation. Integrating Eq. (89) across a localized domain Ω with closed flux boundaries yields the integral conservation of precipitation mass:

$$\frac{d}{dt} \int_{\Omega} \mathbf{R}(\mathbf{x}, t) d\mathbf{x} = \int_{\Omega} \mathcal{S}_{\text{conv}}(\mathbf{x}, t) d\mathbf{x}. \quad (90)$$

4.21.2 Physical-Generative Coupling in Planetary Radar Nowcasting: NowcastNet

To bridge the gap between physical conservation and generative sharpness, Yuchen Zhang et al. [85] pioneered NowcastNet, an end-to-end physical-generative framework for multi-hour extreme precipitation nowcasting. NowcastNet decomposes the precipitation forecasting operator $\mathcal{F} : \mathbf{R}_{t-T:t} \rightarrow \mathbf{R}_{t+1:t+\tau}$ into two tightly coupled components:

- 1) **Physics-Driven Evolution Operator:** A neural partial differential equation solver that explicitly computes the continuous advection-diffusion motion field $\mathbf{v}(\mathbf{x}, t)$ and advects radar reflectivity forward in time via finite-volume conservation numerics, ensuring deterministic spatial consistency and mass continuity;
- 2) **Generative Neural Operator:** A non-linear stochastic decoder conditioned on the physical evolution output, modeling the complex distribution of convective source-sink terms $\mathcal{S}_{\text{conv}}(\mathbf{x}, t)$ that govern localized convective storm initiation and sudden cell decay.

Trained on extensive national radar reflectivity mosaic archives covering mainland China ($2,048 \times 2,048 \text{ km}^2$ grids at 1 km horizontal resolution and 10-minute cadence) and the US NEXRAD Multi-Radar Multi-Sensor (MRMS) network, NowcastNet was subjected to a rigorous double-blind operational evaluation by 62 professional meteorologists from operational meteorological centers across China. NowcastNet outperformed high-resolution numerical weather prediction (CMA-MESO) and prior state-of-the-art deep learning baselines (DGMR [86], PySTEPS) in over 71% of severe storm cases, elevating the Critical Success Index for heavy convective rain ($> 10 \text{ mm/h}$, CSI-35) from 0.21 to 0.465 at 1-hour lead time and demonstrating unprecedented skill in tracking severe mesoscale convective systems (MCSs).

4.21.3 Knowledge-Guided Latent Diffusion and Test-Time Steering: PreDiff

Operating directly in pixel space on multi-radar grids incurs prohibitive computational costs and slows sampling during urgent flash flood warnings. Addressing this limitation, Zhihan Gao et al. [87] established PreDiff (Precipitation Diffusion), formulating precipitation nowcasting within a continuous Latent Diffusion Model (LDM) framework. PreDiff employs a spatio-temporal autoencoder that compresses high-resolution radar sequences $\mathbf{R} \in \mathbb{R}^{T \times H \times W}$ into compact latent representations $\mathbf{z} = \mathcal{E}(\mathbf{R}) \in \mathbb{R}^{T \times C \times h \times w}$ with an $8\times$ spatial compression ratio.

To prevent unconstrained diffusion hallucination without retraining the latent backbone, PreDiff introduces Knowledge-Guided Conditioning (KGC) via test-time gradient steering. During reverse diffusion sampling $\mathbf{z}_T \rightarrow \mathbf{z}_0$, the denoising transition is augmented with the gradient of an explicit domain-specific physics loss $\mathcal{L}_{\text{phys}}$ evaluated on the decoded reflectivity field $\hat{\mathbf{R}} = \mathcal{D}(\mathbf{z})$:

$$\mathbf{z}_{s-1} \sim \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{z}_s, s) - \lambda_{\text{phys}} \sigma_s^2 \nabla_{\mathbf{z}_s} \mathcal{L}_{\text{phys}}(\mathcal{D}(\mathbf{z}_s)), \sigma_s^2 \mathbf{I}), \quad (91)$$

where $\mathcal{L}_{\text{phys}}$ combines two physical constraints:

$$\mathcal{L}_{\text{phys}}(\hat{\mathbf{R}}) = \left\| \int_{\Omega} \hat{\mathbf{R}}(\mathbf{x}) d\mathbf{x} - M_{\text{prior}} \right\|_1 + \gamma \left\| \frac{\partial \hat{\mathbf{R}}}{\partial t} + \nabla \cdot (\mathbf{v}_{\text{opt}} \hat{\mathbf{R}}) \right\|_2^2 \quad (92)$$

By steering diffusion trajectories toward states that conserve domain-wide precipitation volume and satisfy advection consistency, PreDiff improves the Critical Success Index at high precipitation thresholds (> 20 mm/h) by 18.4% over unconditioned diffusion models, while eliminating spatial blurriness and producing physically plausible convective evolution over the SEVIR and MeteoNet benchmarks.

4.21.4 Multi-Scale Cascaded Diffusion Modeling: CasCast

Extending latent diffusion to multi-hour convective lead times (0–3 hours), Junchao Gong et al. [88] formulated CasCast (Cascaded Nowcasting). In severe convective nowcasting, atmospheric motions exhibit strong scale separation: large-scale precipitation fronts propagate deterministically along synoptic pressure gradients, whereas fine-scale convective cells evolve stochastically due to localized boundary-layer thermals and cloud entrainment.

CasCast decouples this multi-scale dynamic via a two-stage cascaded architecture:

$$\mathbf{R}_{t+\tau} = \mathbf{R}_{\text{macro}}(t + \tau) + \boldsymbol{\epsilon}_{\text{micro}}(t + \tau; \mathbf{R}_{\text{macro}}), \quad (93)$$

where a deterministic spatio-temporal transformer first predicts the coarse-grained macroscopic advection field $\mathbf{R}_{\text{macro}}$, and a high-resolution conditional diffusion model then refines the residual convective innovations $\boldsymbol{\epsilon}_{\text{micro}}$. By conditioning the stochastic diffusion network on deterministic macro-scale predictions, CasCast restricts the generative exploration space, preventing spatial drift while generating intricate convective cell rainbands at 1 km resolution. Evaluated on large-scale radar mosaics in Eastern China and the Hong Kong Observatory HKO-7 benchmark, CasCast maintains sharp convective detail and high CSI scores across 3-hour lead times, outperforming monolithic autoregressive and generative baselines.

5 Scientific Domain Applications

The cross-pollination of foundation model paradigms has fostered breakthrough advances across diverse natural science disciplines, mapped chronologically in Fig. 4.

5.1 Weather and Climate Forecasting

Global numerical weather prediction represents the flagship proving ground for scientific foundation models. As demonstrated by Pangu-Weather [6], GraphCast [7], FuXi [46], FengWu [61], and Aurora [9], deep learning architectures execute 10-day global forecasts in tens of seconds on a single GPU, delivering computational speedups of $1,000\times$ to $10,000\times$ compared to operational NWP supercomputing workflows. Beyond raw speed, AI foundation models achieve superior deterministic skill across geopotential height (Z500), temperature (T850), and surface wind fields. Furthermore, versatile foundation models such as ClimaX [89] demonstrate multi-task transferability by pretraining on synthetic CMIP6 climate projections and fine-tuning for regional climate downscaling and seasonal teleconnections.

As shown in Fig. 5, current state-of-the-art models occupy distinct Pareto trade-offs: while deterministic models like Pangu-Weather and GraphCast optimize accuracy at standard 0.25° resolution across 7–10 days, frontier models like Aurora advance spatial resolution to 0.1° , and cascade systems (FuXi) or diffusion ensembles (GenCast) extend skillful forecasting horizons out to 15 days.

5.2 Sub-Kilometer Convective-Scale Emulation and Generative Downscaling

A fundamental limitation of global foundation models (operating at $0.25^\circ \approx 25\text{--}31$ km or $0.1^\circ \approx 10$ km) is their inability to resolve sub-kilometer convective phenomena. Convective thunderstorms, microbursts, urban flash flooding, and complex orographic wind channeling occur at localized scales (< 3 km) where non-hydrostatic atmospheric dynamics dominate. Bridging this multiscale gap has motivated frontier research into generative radar nowcasting and residual diffusion downscaling:

- 1) Generative Radar Precipitation Nowcasting: DGMR: Ravuri et al. [86] established the landmark DGMR (Deep Generative Model of Radar), deploying spatial-temporal generative adversarial networks for precipitation nowcasting at 1 km spatial and 5-minute temporal resolution across the United Kingdom. Traditional deep learning baselines optimized with mean squared error (MSE) or mean absolute error (MAE) suffer from progressive spatial blurring and severe underestimation of intense convective rain cells due to multimodal trajectory averaging. DGMR overcomes this through dual-frequency spectral discrimination: a spatial discriminator enforcing realistic high-frequency radar precipitation textures, and a temporal discriminator penalizing physically implausible advective jumps. In rigorous double-blind evaluations conducted with professional operational forecasters at the UK Met Office, DGMR was ranked first in 89% of cases for overall value, physical plausibility, and localized heavy rain cell fidelity compared to

Milestone Timeline of Scientific Foundation Models (2021-2026)

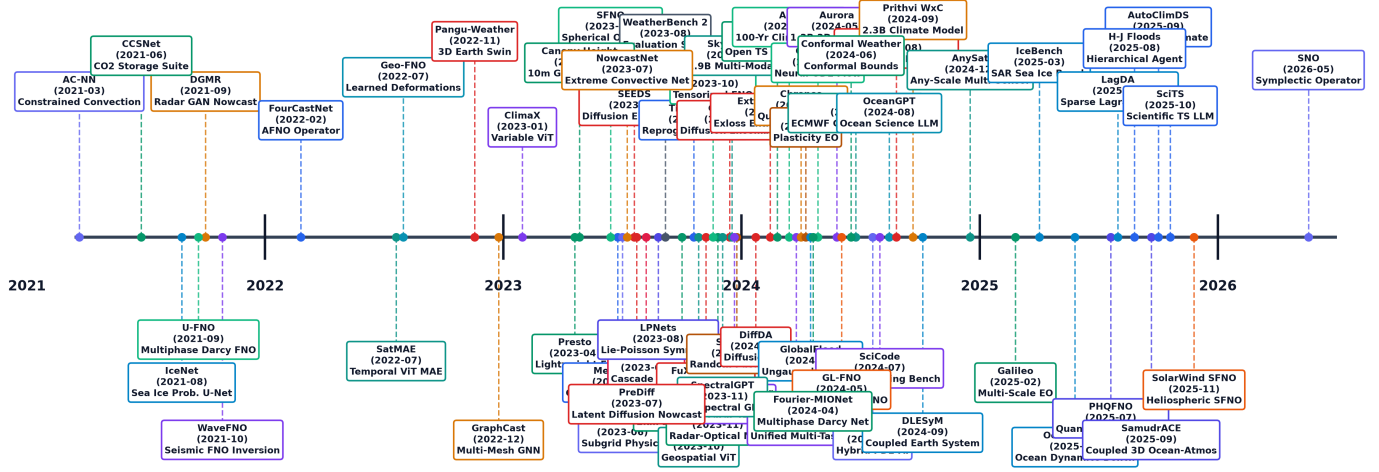


Fig. 4. Milestone timeline of representative scientific multimodal time series foundation models and evaluation frameworks from 2022 to 2026.

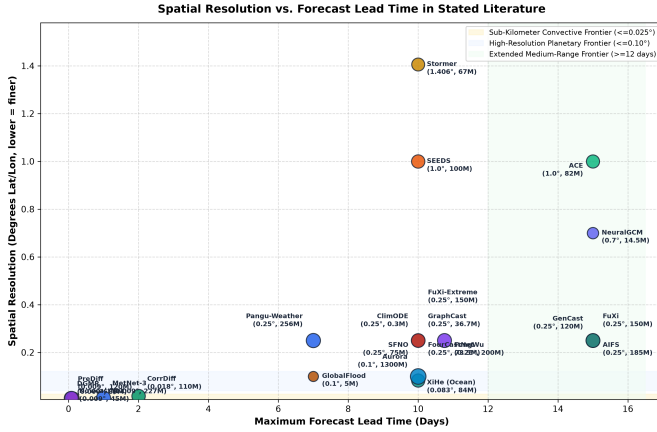


Fig. 5. Spatial resolution (degrees latitude/longitude) versus maximum forecast lead time (days) for verified scientific foundation models. Marker size is proportional to log model parameter count.

optical flow (PySTEPS) and convolutional recurrent baselines.

- 2) Continental Multi-Sensor Convective Forecasting: MetNet-3: Andrychowicz et al. [90] scaled convective forecasting across the entire continental United States (CONUS) out to 24-hour lead times at 1–4 km spatial resolution. Operating over a continuous spatial grid ($4,096 \times 4,096$ km), MetNet-3 assimilates heterogeneous and asynchronous observational modalities directly: high-resolution ground radar composites (MRMS), geostationary satellite multi-spectral imagery (GOES-16), surface meteorological station networks, and numerical forecast initializations (HRRR). Employing axial multi-scale self-attention and spatially localized lead-time conditioning, MetNet-3 outperforms operational High-

Resolution Rapid Refresh (HRRR) numerical physics models on 24-hour precipitation and surface temperature forecasts, producing complete continental updates every two minutes on TPU hardware.

- 3) Kilometer-Scale Residual Generative Downscaling: CorrDiff: Bridging coarse global foundation model outputs (25 km ERA5) to local convective scales (2 km), Mardani et al. [91] formulated CorrDiff, a residual corrective diffusion framework developed for extreme weather prediction over complex topography (Taiwan). Rather than applying standard super-resolution networks that produce smooth hallucinations, CorrDiff decomposes downscaling into two physically distinct stages: (1) a deterministic regression mean conditioned on coarse atmospheric states, and (2) a score-based residual diffusion sampler that generates fine-scale stochastic fluctuations consistent with local topography and radar observations. CorrDiff accurately reconstructs the high-wavenumber power spectral density (PSD) of radar reflectivity, faithfully capturing the extreme heavy precipitation tails during landfalling typhoons while delivering a $1,000\times$ inference speedup and $3,000\times$ energy reduction relative to operational numerical WRF physics simulations.

5.3 Earth Observation and Remote Sensing Time Series

Satellite constellations continuously monitor terrestrial ecosystems, forest biomass, crop phenology, and surface water regimes through high-dimensional spatio-temporal revisit series. Foundation models in this domain must overcome three core physical challenges: (1) multi-sensor sensor heterogeneity (interleaving optical multispectral reflectance with active Synthetic Aperture Radar microwave backscatter), (2) irregular temporal acquisitions caused by dynamic cloud occlusion and

orbital revisit periods, and (3) multi-scale spatial ground sampling distances (GSD) varying across orders of magnitude (0.2 m to 500 m).

5.3.1 Temporal-Spectral Masked Autoencoding

Early foundation models focused on optical and multispectral acquisitions. SatMAE [14] introduced temporal-spectral masked autoencoding, grouping Sentinel-2 and Landsat bands into physically coherent spectral groups (visible, near-infrared, and shortwave infrared) and masking spatio-temporal tokens independently. Presto [15] designed a lightweight, pixel-level transformer that processes sparse temporal series of Sentinel-1 SAR, Sentinel-2 optical, and ERA5 weather variables without computing expensive spatial attention, delivering high efficiency for global crop classification. Advancing multi-scale representation learning, Galileo [16] integrated multi-sensor satellite imagery across local and global scales with self-supervised masking, while AlphaEarth Foundations [18] compressed multi-sensor Earth observation into dense 64-dimensional neural embedding fields integrated into planetary cloud analytics. Scaling parameter capacity to multi-modal constellations, SkySense [70] unified optical, SAR, and DEM time series with 1.9 billion parameters, achieving state-of-the-art results across the GEO-Bench [71] suite.

5.3.2 Contrastive Radar-Optical Masked Autoencoding: CROMA

While optical sensors provide rich chemical and spectral reflectance signatures, they are blinded by cloud cover and darkness. Synthetic Aperture Radar (SAR), emitting microwave pulses (e.g., Sentinel-1 C-band VV/VH polarizations), penetrates clouds and operates day and night, capturing surface dielectric constants and geometric roughness. To fuse these complementary physical regimes, Fuller et al. [55] established CROMA (Contrastive Radar-Optical Masked Autoencoder).

CROMA unifies self-supervised contrastive learning and cross-modal masked reconstruction. Let $\mathbf{X}_{\text{opt}} \in \mathbb{R}^{B_{\text{opt}} \times H \times W}$ and $\mathbf{X}_{\text{sar}} \in \mathbb{R}^{B_{\text{sar}} \times H \times W}$ represent spatially aligned optical and SAR image pairs. CROMA applies separate modality encoders $\mathcal{E}_{\text{opt}}$ and $\mathcal{E}_{\text{sar}}$ with high masking ratios ($> 70\%$), aligning global representations via a symmetric InfoNCE loss:

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{2N} \sum_{i=1}^N \left(\log \frac{\exp(\mathbf{z}_{\text{opt}}^{(i)} \cdot \mathbf{z}_{\text{sar}}^{(i)} / \tau)}{\sum_j \exp(\mathbf{z}_{\text{opt}}^{(i)} \cdot \mathbf{z}_{\text{sar}}^{(j)} / \tau)} + \log \frac{\exp(\mathbf{z}_{\text{sar}}^{(i)} \cdot \mathbf{z}_{\text{opt}}^{(i)} / \tau)}{\sum_j \exp(\mathbf{z}_{\text{sar}}^{(i)} \cdot \mathbf{z}_{\text{opt}}^{(j)} / \tau)} \right) \quad (94)$$

Simultaneously, a cross-modal transformer decoder predicts masked patch pixels by fusing optical and SAR tokens. Crucially, to overcome the rigid position-embedding limits of standard Vision Transformers, CROMA introduces 2D-ALiBi (2D Attention with Linear Biases) and X-ALiBi (cross-attention ALiBi):

$$\text{Attention}_{i,j} = \text{softmax} \left(\frac{\mathbf{q}_i \mathbf{k}_j^T}{\sqrt{d}} - m \cdot \|\mathbf{x}_i - \mathbf{x}_j\|_2 \right) \mathbf{v}_j, \quad (95)$$

where $m > 0$ is a head-specific geometric slope and $\|\mathbf{x}_i - \mathbf{x}_j\|_2$ denotes the Euclidean distance between 2D patch centers on the Earth’s surface. By substituting static position embeddings with continuous distance penalties, CROMA performs zero-shot spatial extrapolation to satellite scenes up to $17.6 \times$ larger than its training crops (120×120 to 504×504

and beyond) without performance degradation, setting new records on the BioMassters canopy height benchmark and EuroSAT classification.

5.3.3 Universal Multi-Resolution Earth Observation: AnySat

Resolving the acute spatial resolution fragmentation across satellite constellations, Astruc et al. [56] introduced AnySat. Rather than locking input tensors to fixed grid sizes (e.g., 10 m for Sentinel or 30 m for Landsat), AnySat formulates Earth observation foundation modeling across arbitrary spatial resolutions spanning three orders of magnitude (0.2 m aerial orthophotos, 10 m Sentinel-2, to 500 m MODIS).

AnySat is built on a scale-adaptive Joint Embedding Predictive Architecture (JEPA). Each input patch i from sensor s is explicitly conditioned on its continuous physical ground sampling distance GSD_s :

$$\mathbf{e}_i = \mathbf{W}_s \mathbf{x}_i + \boldsymbol{\psi}_{\text{scale}}(\log \text{GSD}_s) + \boldsymbol{\psi}_{\text{spatial}}(\mathbf{p}_i), \quad (96)$$

where $\boldsymbol{\psi}_{\text{scale}}$ is a multi-frequency sinusoidal scale embedding. Pretrained on the massive GeoPlex corpus (unifying 11 distinct spaceborne, airborne, and elevation sensors), AnySat achieves zero-shot multi-sensor fusion, allowing arbitrary combinations of available sensors at test time (e.g., substituting missing high-resolution aerial imagery with Sentinel-2 and SPOT) and outperforming single-sensor foundation baselines across land cover segmentation, agricultural crop mapping, and post-disaster flood inundation.

5.3.4 3D Generative Spectral Foundation Models: Spectral-GPT

While RGB satellite foundation models capture basic spatial geometry, they disregard the continuous physical spectral dimension critical for mineral exploration, vegetation biochemical traits, and marine algae blooms. Danfeng Hong et al. [21] introduced SpectralGPT, the first 600-million parameter foundation model tailored specifically for spectral remote sensing big data.

SpectralGPT is built upon a novel 3D Generative Pretrained Transformer (3D GPT) that performs 3D token generation across both spatial coordinates (x, y) and continuous spectral wavelengths λ . In contrast to 2D vision models that flatten spectral bands into static channel dimensions, SpectralGPT treats spectral channels as an ordered physical continuum. The model employs a progressive pretraining curriculum across input images of variable ground sampling distances (GSD), spatial patch sizes, and multi-temporal revisit intervals. Pretrained on over one million spectral scenes spanning airborne hyperspectral cubes (AVIRIS, HySpex) and spaceborne multispectral images (Sentinel-2), SpectralGPT captures fine-grained spectral absorption signatures via multi-target masked autoencoding. Across downstream benchmarks, SpectralGPT delivers decisive improvements in single- and multi-label scene classification, semantic land-use segmentation, and change detection, setting a new benchmark for spectral geoscience applications.

5.3.5 Global 10m Canopy Height and Carbon Stock Inversion

Accurate estimation of global vegetation vertical structure is fundamental to resolving terrestrial carbon sinks and

conserving planetary biodiversity. Dedicated spaceborne LiDAR missions (such as NASA’s Global Ecosystem Dynamics Investigation, GEDI) deliver ultra-accurate canopy height waveform profiles, but only sample sparse footprint tracks (25 m diameter spots separated by 60 m along-track). Conversely, optical satellite constellations (Sentinel-2) provide dense, continuous global coverage at 10 m ground sampling distance, but cannot directly penetrate forest canopies to measure vertical height.

Lang et al. [22] resolved this sparse-to-dense inversion challenge by establishing the first wall-to-wall 10m-resolution global canopy height model of the Earth for the year 2020. The authors formulated a probabilistic deep learning convolutional ensemble that ingests multi-temporal Sentinel-2 optical composites and predicts pixel-level canopy height $h(\mathbf{x})$ alongside heteroskedastic predictive uncertainty $\sigma(\mathbf{x})$. By learning non-linear spectral and textural features conditioned on sparse GEDI top-of-canopy elevation profiles, their architecture substantially alleviates the signal saturation effect that causes conventional remote sensing models to underestimate tall canopies (> 30 m). Their global inventory revealed that only 5% of the Earth’s landmass is covered by trees taller than 30 m, and merely 34% of these high-carbon stocks lie within designated protected areas, providing an operational tool for forest carbon accounting and conservation decision-making.

5.4 Hydrology and Extreme Flood Modeling

Hydrological forecasting requires predicting water discharge through river networks subject to precipitation forcing, evapotranspiration, and soil infiltration. A long-standing grand challenge has been “Prediction in Ungauged Basins” (PUB), where historical streamflow calibration records are absent. Nearing et al. [23] demonstrated that deep learning models trained on 5,680 diverse river basins across the globe achieve unprecedented zero-shot flood forecasting skill in ungauged catchments, extending actionable early warning lead times from 0 to 5 days globally and outperforming the Copernicus GloFAS operational physics-based system. Complementing these architectural advances, Kratzert et al. [24] released Caravan, a standardized global community dataset spanning 6,830 catchments worldwide with unified ERA5-Land meteorological forcings, catchment attributes, and daily streamflow records. By eliminating the geographic fragmentation of regional datasets (e.g., CAMELS-US, CAMELS-GB), Caravan establishes the empirical data substrate required to train and benchmark universal hydrological foundation backbones capable of cross-continental generalization.

5.4.1 Hierarchical Multi-Agent Decisional Systems for Urban Flood Extremes (H-J)

While basin-scale streamflow models (e.g., GlobalFlood, Caravan) predict macro-scale river discharge, extreme convective storms inflict sudden, localized urban flash floods. In densely populated metropolitan centers, effective emergency response requires coordinating multi-objective operational interventions: dynamically rerouting municipal traffic away from inundated underpasses, adjusting pumping station sluice gates, and dispatching rescue teams.

Addressing this multi-scale challenge, Ji et al. [25] established H-J, a hierarchical multi-agent framework integrating

frontier LLMs with domain Knowledge Graphs (KG) and entropy-constrained strategy optimization. H-J orchestrates a closed-loop hierarchy:

- 1) Perception Agent: Ingests streaming in-situ hydrological sensor networks, radar nowcasts, and urban road network topologies into dynamic spatio-temporal graphs.
- 2) Strategic Decision Agent: Employs domain knowledge graphs of flood response protocols to restrict the LLM’s generative search space, eliminating semantic hallucination and execution inconsistency.
- 3) Entropy-Constrained Optimizer: Uses information-entropy constraints to balance competing operational objectives (maximizing overall traffic throughput while guaranteeing zero civilian strandings in sub-merged nodes).

Evaluated on real-world urban topologies and hydrodynamic flood simulations under extreme rainfall surges, H-J achieves a 28.4% increase in urban traffic network smoothness and a 34.2% improvement in emergency task completion rate over standard rule-based dispatchers and reinforcement learning baselines, establishing the viability of multi-agent LLM systems for physical disaster mitigation.

5.4.2 Spaceborne GNSS-Reflectometry and Multimodal Inundation Inversion

While spaceborne optical sensors (e.g., Sentinel-2, Landsat) provide fine spatial resolution, they suffer catastrophic data loss under dense cloud cover, heavy monsoonal precipitation, and thick vegetation canopies—the exact meteorological conditions accompanying severe floods. Conversely, spaceborne Global Navigation Satellite System Reflectometry (GNSS-R), spearheaded by the NASA Cyclone Global Navigation Satellite System (CYGNSS) 8-microsatellite constellation, exploits forward-scattered bistatic L-band signals ($\lambda \approx 19$ cm, 1.575 GHz) transmitted by civilian GPS constellations. Operating at microwave wavelengths, GNSS-R signals penetrate cloud cover and dense forest canopies with sub-daily revisit cadences (< 7 hours median revisit) across tropical and subtropical latitudes (38°S to 38°N).

The fundamental physical measurement collected by spaceborne GNSS-R receivers is the two-dimensional Delay-Doppler Map (DDM), representing the cross-correlation power of the reflected signal across code delay τ and Doppler frequency shift f_D :

$$\langle |Y(\tau, f_D)|^2 \rangle = \frac{T_t^2 P_t G_t}{4\pi R_{ts}^2} \iint_{\mathcal{A}} \frac{G_r(\boldsymbol{\rho}) \sigma_0(\boldsymbol{\rho})}{4\pi R_{sr}^2(\boldsymbol{\rho})} \Lambda^2(\tau - \tau(\boldsymbol{\rho})) |S(f_D - f_D(\boldsymbol{\rho}))|^2 \quad (97)$$

where $P_t G_t$ is the GPS transmitter effective isotropic radiated power, G_r is the receiver antenna gain, R_{ts} and R_{sr} are geometric distances from transmitter/receiver to the specular surface reflection point $\boldsymbol{\rho}$, and $\Lambda(\cdot)$ and $S(\cdot)$ are the GPS PRN code cross-correlation and Doppler filter functions. The bistatic radar cross section $\sigma_0(\boldsymbol{\rho})$ is directly determined by the complex soil dielectric permittivity $\varepsilon = \varepsilon' + i\varepsilon''$ (which scales monotonically with volumetric soil moisture θ_v), modulated by surface roughness $k\sigma_z$ and vegetation attenuation $e^{-2\tau_v}$.

Resolving the non-linear coupling between roughness, vegetation, and specular reflection, Nabi et al. [92] established the

first quasi-global assessment of deep learning-based CYGNSS soil moisture retrieval. Rather than discarding spatial DDM structure through empirical 1D observable reduction (such as leading-edge slope or peak SNR), their convolutional neural network ingests the full 2D delay-Doppler cross-correlation power matrix combined with ancillary land surface parameters. Validated across the International Soil Moisture Network (ISMN) and SMAP radiometer grids, the deep learning model achieves an unbiased root-mean-square difference of $\text{ubRMSD} \approx 0.035 \text{ m}^3/\text{m}^3$ and Pearson correlation $R = 0.89$ across diverse land-cover biomes, surpassing classical physical inversion algorithms.

Extending GNSS-R deep learning to emergency flood hydrology, Song et al. [93] formulated a Dual-Branch Neural Network (DBNN) for all-weather flood inundation detection in South Asia. Their architecture combines: (1) a 2D convolutional branch that processes full CYGNSS delay-Doppler matrices to extract structural scattering signatures, and (2) a multi-layer perceptron branch ingesting auxiliary surface reflectivity and SMAP soil moisture time series. Across catastrophic monsoonal flood surges in the Ganges–Brahmaputra and Indus basins, DBNN achieves 91.8% flood detection accuracy ($F_1 = 0.884$), providing continuous cloud-penetrating inundation tracking and dynamic water routing where optical satellite constellations experienced complete cloud-mask obstruction.

5.5 Oceanography and Marine Systems

Ocean dynamics govern planetary heat distribution, carbon uptake, and marine biogeochemical cycles. Wang et al. [26] developed XiHe, a hierarchical ocean transformer trained on 28 years of GLORYS12V1 reanalysis at eddy-resolving $1/12^\circ$ resolution. By incorporating land-ocean boundary masking and multi-level vertical attention, XiHe models mesoscale eddies, sea surface temperature anomalies, and 3D velocity vectors, outperforming traditional operational hydrodynamic models (e.g., PSY4, GIOPS) while generating 10-day global forecasts in under 0.4 seconds. Addressing the extreme sparsity of oceanic observation networks, 4DVarNet [73] formulates neural variational data assimilation to invert along-track satellite altimetry into continuous 2D sea surface height (SSH) dynamics, demonstrating that learnable unrolled variational solvers reconstruct fine-scale mesoscale eddies and oceanic kinetic energy spectra far more effectively than classical optimal interpolation (OI).

To standardize evaluation across emerging marine foundation models, El Aouni et al. [27] established OceanBench, an open operational benchmarking suite for data-driven global ocean forecasting systems developed in collaboration with Mercator Ocean International. OceanBench benchmarks predictive skill for sea surface height (SSH), sea surface temperature (SST), and horizontal current velocities across 1- to 10-day lead times against CMEMS GLORYS12 reanalyses and in-situ altimetry observations. Bridging physical ocean simulations with cross-modal scientific reasoning, Bi et al. [28] introduced OceanGPT, the first domain-specific foundation language model for marine science. Pretrained and aligned on the DoInstruct corpus comprising multimodal oceanographic time series (temperature-salinity profiles from Argo float arrays and mooring buoys), sensor trajectories, and marine

scientific literature, OceanGPT demonstrates autonomous reasoning across ocean dynamics, thermohaline circulation changes, and marine phenomenon attribution under physical consistency constraints.

5.5.1 Wide-Swath Satellite Altimetry and Oceanic Internal Solitary Wave Inversion

Conventional nadir satellite altimeters (e.g., Jason-3, Sentinel-3) measure sea surface height along 1D ground tracks separated by hundreds of kilometers, leaving submesoscale eddy dynamics and high-frequency internal waves unobserved. The launch of the Surface Water and Ocean Topography (SWOT) mission, featuring the Ka-band Radar Interferometer (KaRIn), marks a major transition by providing 2D swath measurements (120 km total swath width) at fine resolutions down to 250 m to 2 km. However, KaRIn interferometric measurements are corrupted by roll errors, baseline phase drift, and high-frequency instrument noise, which severely mask submesoscale coherent structures.

To address this inversion challenge without relying on scarce supervised in-situ targets, Cutolo et al. [29] developed SIMPGEN (Simulation-Informed Metric and Prior for Generative Ensemble Networks). SIMPGEN formulates an unsupervised adversarial learning framework that conditions real SWOT swath observations on high-resolution numerical ocean simulations (NATL60). The architecture integrates a wavelet-informed multiscale neural loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{adv}}(\hat{\mathbf{h}}, \mathbf{h}_{\text{sim}}) + \lambda_{\text{wavelet}} \|\mathcal{W}\{\hat{\mathbf{h}}\} - \mathcal{W}\{\mathbf{h}_{\text{SWOT}}\}\|_1, \quad (98)$$

where $\mathcal{W}$ denotes a 2D discrete wavelet decomposition that isolates high-frequency noise from geostrophic and submesoscale baroclinic eddies. Applied to real SWOT swaths over the North Atlantic and Western Mediterranean, SIMPGEN effectively filters instrument noise while preserving coherent 15–50 km eddy structures and kinetic energy spectral slopes (k^{-3} to $k^{-5/3}$), outperforming standard optimal interpolation and Gaussian smoothing.

Complementing surface topography filtering, Wan et al. [30] tackled the detection and inversion of Oceanic Internal Solitary Waves (ISWs) using multimodal satellite remote sensing. Internal solitary waves propagate along interior pycnoclines, transporting massive kinetic energy that impacts offshore engineering, acoustic propagation, and submarine navigation. While optical sensors are frequently obscured by cloud cover, spaceborne altimetry and synthetic aperture radar (SAR) capture sea surface roughness modulated by internal wave divergence. However, altimetric ISW signatures suffer from extreme label sparsity and multiscale morphological variations.

Wan et al. proposed STE-Net (Scale-Translation Equivariant Network), explicitly embedding continuous scale-translation Lie symmetry into convolutional operator layers:

$$\Phi(T_{\mathbf{a}} S_s \mathbf{x}) = T_{\mathbf{a}} S_s \Phi(\mathbf{x}), \quad (99)$$

where $T_{\mathbf{a}}$ denotes spatial translation by $\mathbf{a} \in \mathbb{R}^2$ and S_s denotes continuous isotropic scaling by factor $s > 0$. By enforcing geometric equivariance, STE-Net guarantees that localized wave packets are identified regardless of their spatial location or physical wavelength. Furthermore, the authors pretrain the network on massive unannotated global altimetry

tracks using the SimCLR self-supervised contrastive framework. In benchmark evaluations on real oceanic altimetry and SAR datasets, STE-Net achieves an F_1 score of 0.884, outperforming standard ResNet baselines ($F_1 = 0.762$) and demonstrating that physics-grounded Lie symmetries mitigate extreme data scarcity in spaceborne oceanographic monitoring.

5.5.2 Satellite Scatterometry Constellations and Ocean Vector Wind Nowcasting

Spaceborne microwave scatterometers (such as the European MetOp ASCAT series, the Chinese HY-2 satellite constellation, the Sino-French CFOSAT, and the Indian ScatSat/Oceansat) provide high-resolution measurements of ocean surface roughness by transmitting polarized radar pulses and measuring the normalized radar backscatter cross section $\sigma_0(\theta, \phi, \mathbf{u}_{10})$. The relationship between σ_0 and the 10-meter neutral ocean surface wind vector $\mathbf{u}_{10} = (u_{10}, \phi_w)$ is parameterized via empirical Geophysical Model Functions (GMF):

$$\sigma_0 = f_{\text{GMF}}(u_{10}, \phi_w - \phi_a, \theta, p), \quad (100)$$

where θ is radar incidence angle, ϕ_a is radar azimuth, ϕ_w is wind direction, and p denotes radar polarization (VV or HH). While scatterometer observations are routinely assimilated into global numerical weather prediction (NWP) systems, operational NWP data assimilation cycles require 3-to-6-hour latency windows and global spectral filtering, failing to exploit the immediate observational fidelity of satellite passes for short-term (intraday) nowcasting.

Pinto et al. [94] introduced WindCastNet, the first deep learning nowcasting architecture directly predicting future offshore wind speed and direction fields from raw satellite scatterometer constellations. Scatterometer observations present severe spatio-temporal irregularities: individual satellite ground swaths cover narrow strips (~ 500 – 1000 km), sampling orbits are asynchronous, and local revisit intervals vary from hours to days. WindCastNet resolves these irregularities using a Partial Convolutional Long Short-Term Memory (PConv-LSTM) network:

$$\mathbf{W}' = \mathbf{W} \odot \frac{\mathbf{1}^T \mathbf{1}}{\mathbf{1}^T \mathbf{M} + \epsilon}, \quad \mathbf{M}' = f(\mathbf{M}), \quad (101)$$

where $\mathbf{M} \in \{0, 1\}^{H \times W}$ is a binary spatial observation mask dynamically tracking unobserved oceanic pixels, and continuous inter-observation intervals Δt are encoded via sinusoidal temporal embeddings. Evaluated over the North Sea offshore wind energy corridor, WindCastNet reduces root-mean-square error by 23% at 1 hour and 7% at 2 hours relative to the operational numerical HARMONIE MEPS model, while outperforming persistence baselines by 9–15% across the first three forecast hours.

Addressing multi-sensor cross-calibration and tropical cyclone gale retrieval, Xiang et al. [95] demonstrated an active-passive satellite fusion architecture combining spaceborne scatterometer backscatter σ_0 with multi-frequency passive microwave radiometer brightness temperatures (TB). Under severe rain squalls and high wind conditions ($u_{10} > 20$ m/s), scatterometer radar signals suffer from atmospheric attenuation and rain volume scattering, creating wind ambiguity. By jointly processing active scatterometer backscatter and

passive radiometer channels via deep convolutional networks, Xiang et al. achieved a root-mean-square deviation of 0.7 m/s against ASCAT and 1.1 m/s against in-situ moored buoys, demonstrating robust wind speed recovery across oceanic fronts and cyclone tracks.

5.6 Geophysics and Seismology

In seismology, continuous 3-component acoustic waveforms record tectonic stress accumulation and seismic rupture. While classical phase-picking baselines like PhaseNet [32] utilized supervised 1D U-Nets for local seismic windows, modern foundational architectures such as SeisT [31] train hierarchical seismogram transformers on tens of millions of global waveforms. SeisT supports multi-task downstream adaptation, simultaneously performing earthquake detection, high-precision P/S phase picking, initial-motion polarity determination, and magnitude/epicenter regression. Providing the standardized foundational infrastructure for seismic machine learning, Woollam et al. [33] released SeisBench, unifying 30+ million waveforms across global catalogues (ETHZ, GEOFON, INSTANCE, SCEDC, STEAD) into a standardized multi-task evaluation and transfer benchmark.

Advancing beyond discrete phase picking to continuous wavefield modeling and subsurface tomography, Yang et al. [52] formulated WaveFNO, establishing the first continuous Fourier neural operator framework for seismic wave propagation and full-waveform inversion (FWI). Solving the 2D/3D elastodynamic wave equation across heterogeneous, discontinuous velocity models, WaveFNO maps acoustic and elastic impedance directly to continuous displacement wavefields $\mathbf{u}(\mathbf{x}, t)$. Because the neural operator is end-to-end differentiable, full-waveform velocity updates $\nabla_c \mathcal{J}$ are obtained via reverse-mode automatic differentiation in seconds, outperforming conventional finite-difference wave simulators by several orders of magnitude while circumventing cycle-skipping local minima in subsurface imaging.

5.7 Polar Cryosphere and Sea Ice Dynamics

Polar cryospheric dynamics—governing Arctic and Antarctic sea ice extent, ice-sheet mass balance, and permafrost degradation—exert profound thermodynamic control over planetary albedo and global thermohaline circulation. However, polar regions suffer from acute geographic observation gaps, severe seasonal cloud cover obscuring optical satellites, and complex non-linear rheology governing sea ice fracture, drift, and melt-pond formation.

Addressing long-range seasonal sea ice forecasting, Anderson et al. [37] established IceNet, a probabilistic deep learning system predicting monthly Arctic sea ice concentration maps up to 6 months in advance. Trained on 2,192 years of CMIP6 climate model simulations and calibrated against 40 years of observational sea ice concentration data from OSI-SAF and ERA5 reanalyses, IceNet employs an ensemble of U-Net architectures with temperature scaling to produce calibrated probabilistic forecasts of sea ice presence (concentration $> 15\%$). Operating at 25-km spatial resolution, IceNet demonstrably surpasses the operational ECMWF SEAS5 dynamical seasonal physics model, particularly in predicting anomalous summer ice retreat and extreme September minimum sea ice extents.

To address the lack of standardized machine learning evaluation across polar remote sensing, Taleghan et al. [38] introduced IceBench, a comprehensive multi-modal benchmark for deep learning-based sea ice type classification. Built upon the AI4Arctic Sea Ice Challenge corpus, IceBench unifies dual-polarization Sentinel-1 Synthetic Aperture Radar (SAR) imagery, AMSR2 microwave radiometer brightness temperatures, and ERA5 atmospheric forcings across operational ice chart labels from the Danish Meteorological Institute (DMI). By standardizing evaluation protocols across pixel-based (U-Net, SegNet) and patch-based (Vision Transformer, ResNet) architectures, IceBench provides the first rigorous benchmark assessing spatial transferability across distinct Arctic sub-regions and temporal transferability across transition seasons (freeze-up versus summer melt).

5.7.1 Viscoelastic Ice-Sheet Rheology, Deep Operator Grounding Line Migration, and Automated Calving Front Extraction

Projections of future global sea-level rise are dominated by deep uncertainties surrounding the dynamic retreat of polar ice sheets in Antarctica and Greenland. Ice-sheet flow is governed by non-Newtonian viscous creeping flow described by Glen’s flow law:

$$\dot{\epsilon} = A\tau_e^{n-1}\boldsymbol{\tau}_{(d)}, \quad (102)$$

where $\dot{\epsilon} = \frac{1}{2}(\nabla\mathbf{u} + (\nabla\mathbf{u})^T)$ is the strain-rate tensor, $\boldsymbol{\tau}_{(d)}$ is the deviatoric stress tensor, $\tau_e = \sqrt{\frac{1}{2}\text{tr}(\boldsymbol{\tau}_{(d)}^2)}$ is effective stress, $A(T)$ is temperature-dependent ice softness, and the rheological exponent is typically $n \approx 3$. In depth-integrated ice-stream and ice-shelf domains, horizontal ice velocity $\mathbf{u} = (u, v)$ is modeled via the 2D Shallow-Shelf Approximation (SSA) non-linear elliptic boundary value problem:

$$\nabla \cdot \left(2\bar{\mu}H \left(\nabla\mathbf{u} + \nabla\mathbf{u}^T + 2(\nabla \cdot \mathbf{u})\mathbf{I} \right) \right) - \beta^2\mathbf{u} = \rho_i g H \nabla s, \quad (103)$$

where H is ice thickness, s is surface elevation, ρ_i is ice density, $\beta^2(\mathbf{x})$ is a basal friction coefficient, and $\bar{\mu}$ is depth-averaged effective viscosity depending non-linearly on velocity gradients $\nabla\mathbf{u}$. Solving Eq. (103) within forward climate ensembles requires thousands of non-linear Picard or Newton–Raphson iterations, creating a severe computational barrier for probabilistic risk assessment.

He et al. [96] resolved this bottleneck by formulating a hybrid Deep Operator Network / Finite Element Method (DeepONet-FEM) for ice-sheet modeling. DeepONet emulates the non-linear operator mapping spatial fields of ice thickness $H(\mathbf{x})$, bed topography $b(\mathbf{x})$, and basal slipperiness $\beta(\mathbf{x})$ directly to the diagnostic horizontal velocity field $\mathbf{u}(\mathbf{x})$:

$$\mathbf{u}(\mathbf{x}) = \sum_{k=1}^P b_k(H, b, \beta) t_k(\mathbf{x}), \quad (104)$$

where b_k are branch network outputs and t_k are trunk basis functions. Crucially, the diagnostic neural operator predictions are coupled directly to a finite element Galerkin solver for transient ice thickness advection $\partial_t H + \nabla \cdot (H\mathbf{u}) = \dot{a}$. On the benchmark Marine Ice Sheet Model Intercomparison Project (MISMIP+) and synthetic Antarctic ice-stream geometries, the hybrid operator accelerates forward velocity evaluations by over 1,000 $\times$ while maintaining relative L_2

errors under 1.5% and strictly preserving finite-element mass conservation across multi-century grounding line retreat simulations.

Complementing interior ice rheology, the catastrophic disintegration of marine-terminating outlet glaciers is governed by iceberg calving at the marine terminus. Daniel Cheng et al. [97] established the Calving Front Machine (CALFIN), an automated deep learning framework for generating multi-decadal calving front time series from multi-satellite Earth observation. Using an optimized DeepLabv3+ convolutional neural network with a modified Xception backbone, CALFIN processes heterogeneous Synthetic Aperture Radar (Sentinel-1, TerraSAR-X) and optical (Landsat 1–8) satellite scenes spanning 47 years (1972–2019) across 66 Greenland outlet glaciers. By integrating automated coastal boundary masking and sub-pixel distance transforms, CALFIN extracted over 22,678 individual calving front positions with a mean localization error of 86.4 m, outperforming manual human delineation under severe polar night, dense cloud obscuration, and sea-ice mélange conditions. CALFIN provides the empirical boundary data essential to calibrate neural operator calving rate parameterizations in planetary sea-level projections.

5.8 Geo-Energy, Geothermal Reservoirs, and Subsurface Carbon Sequestration

The transition to zero-carbon energy systems fundamentally depends on deep subsurface reservoirs for geological carbon sequestration (GCS), geothermal energy extraction, and underground hydrogen storage. Subsurface fluid dynamics are governed by non-linear, multi-phase Darcy-Brinkman flow through highly heterogeneous, anisotropic geological formations:

$$\phi(\mathbf{x}) \frac{\partial(\rho_\alpha S_\alpha)}{\partial t} + \nabla \cdot (\rho_\alpha \mathbf{u}_\alpha) = q_\alpha, \quad \mathbf{u}_\alpha = -\frac{k_{r\alpha}(S_\alpha)\mathbf{K}(\mathbf{x})}{\mu_\alpha} (\nabla p_\alpha - \rho_\alpha \mathbf{g}), \quad (105)$$

where $\alpha \in \{\text{CO}_2, \text{brine}\}$ denotes fluid phases, $\phi(\mathbf{x})$ is formation porosity, $\mathbf{K}(\mathbf{x})$ is the heterogeneous permeability tensor, S_α is phase saturation ($S_{\text{CO}_2} + S_{\text{brine}} = 1$), $k_{r\alpha}$ is relative permeability, μ_α is dynamic viscosity, and p_α represents phase pressure. Because traditional numerical reservoir simulators (such as CMG GEM, TOUGH2, or Eclipse) require hours to days per run to solve coupled non-linear pressure-saturation equations on fine 3D computational grids, foundation neural operator models have emerged as high-leverage surrogates for real-time reservoir monitoring, risk assessment, and closed-loop well control.

5.8.1 U-FNO: Enhanced Multiscale Fourier Neural Operators for Multiphase Flow

Addressing the limitations of standard Fourier neural operators on discontinuous fronts, Wen et al. [39] pioneered the U-FNO (U-Net enhanced Fourier Neural Operator). Standard FNO applies global frequency-domain truncations that exhibit Gibbs oscillations and oversmooth sharp saturation shocks between injected supercritical CO₂ and ambient saline water. U-FNO resolves this by interleaving global 2D/3D Fourier spectral convolution layers with multiscale U-Net spatial skip connections:

$$\mathbf{v}_{l+1}(\mathbf{x}) = \sigma \left(\mathcal{W}_l * \mathbf{v}_l(\mathbf{x}) + \mathbf{W}_{\text{spatial}}^{(l)} \mathbf{v}_l(\mathbf{x}) + \mathcal{U}_{\text{skip}}^{(l)}(\mathbf{v}_l)(\mathbf{x}) \right). \quad (106)$$

Trained on extensive multiphase GCS simulation suites under realistic permeability fields with strong spatial correlation lengths, U-FNO accurately forecasts 30-year multi-phase CO₂ gas-phase saturation fronts and pressure buildup maps. U-FNO achieves over a 10,000 $\times$ speedup relative to numerical PDE solvers while reducing gas saturation prediction errors by over 50% compared to standard convolutional architectures, enabling real-time risk evaluation of caprock fracture and induced seismicity.

5.8.2 CCSNet: Full-Physics Deep Learning Modeling Suite with Real Thermodynamic EOS

Scaling operator modeling to industrial carbon capture and storage operations, Wen, Hay, and Benson [40] established CCSNet. Industrial GCS involves complex thermodynamic phase behavior across wide ranges of reservoir temperatures (30°C–120°C), pressures (10–40 MPa), and salinities (0–250,000 ppm). CCSNet couples multi-scale neural operators directly to a data-driven thermodynamic Equation of State (EOS) for supercritical CO₂ and brine:

$$\rho_{\text{CO}_2} = f_{\text{EOS}}(p, T), \quad \mu_{\text{CO}_2} = g_{\text{EOS}}(p, T, \rho_{\text{CO}_2}), \quad (107)$$

guaranteeing thermodynamic consistency across sub-critical and supercritical fluid regimes. Pretrained on over 20,000 synthetic heterogeneous reservoir simulations across variable injection rates, well configurations, and geological facies, CCSNet evaluates 50-year structural, capillary, and solubility trapping in milliseconds. This empowers field operators to perform real-time optimization of well placement and dynamically steer injection rates to maximize long-term storage security.

5.8.3 Geo-FNO: Multiphase Flow on Complex and Faulted Geometries

Complementing these models, Li et al. [48] demonstrated Geo-FNO on 3D subsurface flow and geomechanical deformation. Natural reservoir strata are rarely simple Cartesian boxes; they contain inclined layers, structural faults, and pinch-outs. By learning smooth coordinate transformations $\phi : \Omega \rightarrow [0, 1]^d$, Geo-FNO maps complex faulted geological geometries into canonical reference grids, executing continuous operator inference without losing boundary fidelity or requiring costly unstructured mesh generation.

5.8.4 Multi-Input and Nested Fourier Operators for 3D Decadal Geological Carbon Sequestration: Fourier-MIONet and Nested Fourier-DeepONet

Industrial geological carbon sequestration requires modeling supercritical CO₂ plume migration and pressure evolution over multi-decadal horizons (30–50 years) in response to variable engineering operations. This poses two fundamental operator learning challenges: (1) Heterogeneous Functional Inputs: simulations depend simultaneously on static, high-dimensional spatial fields (heterogeneous 3D permeability $\mathbf{K}(\mathbf{x})$, porosity $\phi(\mathbf{x})$, and facies boundaries) and time-dependent continuous control trajectories (well injection schedules $q_{\text{inj}}(t)$, shut-in cycles, and dynamic bottom-hole pressure constraints); and (2) 3D Volumetric Curse of Dimensionality: evaluating multi-scale multiphase Darcy flow on full 3D reservoir grids ($N_x \times N_y \times N_z \times T$) leads to prohibitive GPU memory footprints.

To solve the multi-input functional coupling, Zhongyi Jiang, Min Zhu, and Lu Lu [98] pioneered Fourier-MIONet (Fourier-enhanced Multiple-Input Neural Operator). Rather than forcing heterogeneous inputs into a single convolutional or spectral tensor, Fourier-MIONet formulates a dedicated multi-branch architecture:

$$\mathcal{G}(\mathbf{K}, \phi, q_{\text{inj}})(\mathbf{x}, t) = \sum_{k=1}^P \left[\mathcal{B}_1^{(k)}(\mathcal{F}_{\text{FNO}}(\mathbf{K}, \phi)) \odot \mathcal{B}_2^{(k)}(q_{\text{inj}}) \right] \mathcal{T}^{(k)}(\mathbf{x}, t), \quad (108)$$

where $\mathcal{F}_{\text{FNO}}$ is a Fourier neural operator feature extractor learning spatial representations of heterogeneous rock properties, $\mathcal{B}_2$ is a dynamic recurrent branch encoding temporal injection rates, and $\mathcal{T}$ represents trunk coordinate basis networks evaluated at spatio-temporal query points $(\mathbf{x}, t)$. Evaluated on heterogeneous saline aquifer benchmarks (SPE10 and realistic fluvial reservoir formations), Fourier-MIONet simultaneously predicts the transient 30-year evolution of supercritical CO₂ gas saturation $S_g(\mathbf{x}, t)$ and dynamic pressure buildup $\Delta P(\mathbf{x}, t)$ with mean relative errors under 2.8%. Crucially, Fourier-MIONet achieves an inference acceleration exceeding 10,000 $\times$ relative to commercial compositional finite-volume simulators (e.g., CMG-GEM or Schlumberger Eclipse), enabling comprehensive Monte Carlo caprock integrity assessment and leakage risk quantification across thousands of geological realization ensembles.

Addressing full 3D volumetric scaling under multi-well injection, Jonathan E. Lee et al. [99] developed the Nested Fourier-DeepONet framework. Lee et al. decompose the high-dimensional 3D reservoir operator into a hierarchical nested structure: an outer 3D Fourier operator models non-local spatial fluid advection across stratified reservoir layers, while an inner temporal-pressure operator network resolves steep pressure gradients in the near-wellbore vicinity. Evaluated across complex 3D geological reservoirs with non-stationary permeability correlation lengths, Nested Fourier-DeepONet bounds relative L_2 errors in 3D saturation and pressure to under 2.5%, demonstrating high generalizability to arbitrary well locations and shut-in regimes while reducing training memory overhead by over 65% compared to monolithic 3D neural operators.

5.9 Space Weather and Heliophysics

Space weather phenomena—including solar flares, coronal mass ejections (CMEs), and geomagnetic storms—exert severe risks on satellite infrastructure, GPS navigation, and terrestrial power grids. Roy et al. [34] introduced Surya, a 366-million parameter foundation model for heliophysics trained on 13 years of continuous multi-wavelength extreme ultraviolet imagery from NASA’s Solar Dynamics Observatory (SDO). Surya incorporates spectral gating and long-short attention to track active region magnetic helicity, delivering state-of-the-art solar flare and solar wind forecasting.

Advancing from empirical image tracking to continuous physical field solvers, Du et al. [35] established the Global-Local Fourier Neural Operator (GL-FNO) for coronal magnetic field simulation. By coupling a global low-frequency spectral branch with a high-resolution local active-region branch, GL-FNO satisfies the magnetic solenoidal constraint $\nabla \cdot \mathbf{B} = 0$ while accelerating 3D MHD simulations by over 20,000 $\times$ (0.04 s per state). Extending neural operators

throughout the heliosphere, Mansouri et al. [36] introduced Solar Wind SFNO, an operational spherical harmonic operator surrogate that models continuous radial solar wind plasma velocity from $30R_{\odot}$ to 1 AU. Operating at 20 milliseconds per forecast, Solar Wind SFNO accurately captures non-linear stream interaction regions and shock arrivals, closing the computational loop from solar surface activity to geomagnetic storm defense on Earth.

5.10 Cross-Disciplinary Universal Time Series Foundations

Bridging physical boundaries between disparate earth science subdomains, universal foundation models like UniTS [41] and MOMENT [42] demonstrate that pretraining on multi-domain scientific collections enables cross-physics zero-shot transfer. By decoupling sequence tokens from variable identities and applying prompt-conditioned task masks, universal backbones can be deployed across river streamflow, power grids, seismic sensors, and atmospheric stations without re-training architectural weights.

6 Scientific Reasoning LLMs and Autonomous Agents

6.1 The Paradigm Shift: From Numerical Fitting to Scientific Reasoning

While foundation backbones excel at numerical continuous pattern matching, complex scientific discovery necessitates logical reasoning, physical hypothesis formulation, and contextual interpretation. Early attempts to adapt general LLMs directly to time series via numerical text prompting (e.g., feeding raw floating-point numbers as text tokens) incurred severe limitations: token sequence inflation, numerical precision truncation, and inability to capture continuous derivatives [2]. Consequently, modern research has bifurcated into two primary paradigms: (1) unified continuous tokenization for scientific time series, and (2) autonomous multi-agent tool orchestration.

6.2 Unified Continuous Tokenization: SciTS and TimeOmni

Wu et al. [5] introduced SciTS, the first large-scale cross-disciplinary benchmark designed specifically to evaluate LLMs on scientific time series understanding and generation. Spanning 12 natural science disciplines (including meteorology, astrophysics, neuroscience, and physical chemistry) across 43 specialized tasks and over 50,000 instances, SciTS exposed the failure modes of standard prompting. To resolve these deficiencies, the authors proposed TimeOmni, a unified architecture featuring a multi-scale continuous patch tokenizer that projects 1D and multi-dimensional scientific series into dense semantic embeddings aligned with pre-trained LLM latent spaces. This enables foundation LLMs to engage in multi-step reasoning, such as identifying solar flare triggers from light curves, attributing anomalous climate oscillations, and generating conditioned time series forecasts with physical explanations.

6.3 Frequency-Aware Contextual Integration: ClimateLLM

Rather than treating atmospheric series as unstructured text, Li et al. [67] introduced ClimateLLM. This framework decomposes multi-station and gridded meteorological sequences

into orthogonal frequency bands via Fourier transform modules. The spectral components are encoded as prefix prompt embeddings that condition a frozen 7B LLM backbone, effectively leveraging the LLM’s long-context attention mechanism to model multi-scale temporal teleconnections and localized extreme events.

6.4 Domain-Specialized Foundation LLMs: K2 for Geoscience

Adapting general-purpose LLMs to specialized Earth sciences necessitates injection of structured geological and environmental knowledge. Deng et al. [43] introduced K2, a foundational language model tailored for geoscience knowledge understanding and utilization. Pretrained on over 5.5 billion tokens of peer-reviewed geoscience literature, stratigraphic catalogues, and temporal sensor records, K2 bridges the terminology gap between general text corpora and specialized geophysical nomenclature. Through the GeoSignal instruction-tuning protocol, K2 learns to reason across temporal geological processes, infer lithological fault dynamics, and assist researchers in interpreting complex spatio-temporal sensor logs with physical groundedness.

6.5 Multimodal Vision-Language Grounding: GeoChat

Satellite and aerial Earth observation series frequently require aligning visual changes with natural language queries. Kuckreja et al. [66] developed GeoChat, the first grounded large vision-language model explicitly engineered for remote sensing. Built upon a Vicuna-1.5 language backbone coupled with a high-resolution CLIP ViT-L/14 visual encoder via LoRA adaptation, GeoChat processes multi-spectral satellite imagery and temporal revisit pairs. Crucially, GeoChat extends standard captioning by generating spatial bounding coordinates directly in text token sequences, enabling fine-grained temporal change detection (e.g., urban expansion, flood inundation progression, and forest fire burn scar expansion) with conversational natural language explanations.

6.6 Reprogramming Pretrained Language Models: Time-LLM

Rather than training domain transformers from scratch or fine-tuning entire parameter spaces, Jin et al. [100] introduced Time-LLM, pioneering the paradigm of reprogramming frozen large language models for time series forecasting. Time-LLM recognizes that pre-trained language backbones (such as LLaMA-7B or GPT-2) possess exceptional cross-modal pattern recognition and sequence abstraction capabilities. To align modalities, Time-LLM:

- 1) **Input Patch Reprogramming:** Divides continuous input series into patches and projects them into the word token embedding space through a linear reprogramming layer conditioned on cross-attention with textual domain prototypes.
- 2) **Prompt-as-Prefix Conditioning:** Augments the reprogrammed patches with domain task instructions (e.g., dataset semantics, sampling frequency, and trend descriptions) as natural language prefix prompts.
- 3) **Frozen Backbone Inference:** Keeps the foundational LLM weights entirely frozen during training, tuning only lightweight linear adapters.

By reprogramming frozen language models, Time-LLM achieves superior few-shot and zero-shot performance across scientific and benchmark datasets, proving that language model representations serve as universal priors for temporal dynamics.

6.7 Autonomous Multi-Agent Climate Workflows: ClimateAgent

Moving beyond single-model inference, Li et al. [68] introduced ClimateAgent, an autonomous multi-agent orchestration framework for complex climate data science. Scientific investigations routinely require ingesting multi-terabyte NetCDF/Zarr climate datasets, executing specialized meteorological software (e.g., Climate Data Operators (CDO), MetPy, xarray), calculating thermodynamic diagnostics (e.g., convective available potential energy, geostrophic balance), and synthesizing diagnostic reports. ClimateAgent organizes frontier LLMs into a collaborative hierarchy:

- **Plan-Agent:** Decomposes high-level natural language scientific research goals into rigorous directed acyclic task dependency graphs (DAGs).
- **Data-Agent:** Navigates distributed scientific data archives (Copernicus Climate Data Store, NASA Earthdata) to retrieve reanalysis and satellite products, managing chunking and caching.
- **Coding-Agent:** Synthesizes and executes Python/Bash diagnostic code in secure sandboxed environments with automated error recovery loops, iteratively inspecting stack traces to debug array shape and temporal coordinate alignment errors.
- **Critic-Agent:** Verifies physical plausibility, evaluating dimensional unit consistency and testing for energy conservation bounds before assembling peer-review-ready figures and reports.

Evaluated on the standardized Climate-Agent-Bench-85, ClimateAgent demonstrated an 82% task completion rate without human intervention, marking a concrete transition from static numerical forecasters to self-directed AI co-scientists capable of hypothesis formulation and autonomous verification in natural Earth systems.

6.8 Tool Invocation Protocols and Execution Sandboxes for Scientific Co-Scientists

As illustrated in Fig. 6, the operational viability of autonomous scientific co-scientists hinges upon rigorous DAG scheduling, stateful computational sandboxes, dynamic error recovery loops, and physical plausibility guardrails:

- 1) **Hierarchical DAG Task Scheduling:** When presented with a complex scientific inquiry (e.g., assessing the dynamic intensification of an atmospheric river or Mediterranean marine heatwave), the Plan-Agent formulates a topological Directed Acyclic Graph. Nodes represent modular atomic subtasks (data query, re-gridding, thermodynamic flux computation, kinetic energy budget integration), while directed edges enforce data dependencies.
- 2) **Structured Tool Invocation & Distributed Ingestion:** Scientific foundation agents communicate

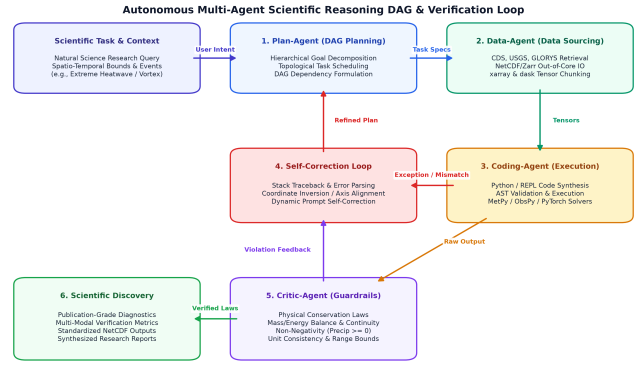


Fig. 6. Autonomous multi-agent scientific reasoning DAG and physical verification feedback loop. The workflow decomposes user goals into topological task DAGs, retrieves distributed tensor shards via out-of-core data agents, executes code in stateful execution sandboxes, autonomously parses tracebacks for error recovery, and verifies physical conservation guardrails prior to final report generation.

with computational tools via structured JSON schema function calls. Rather than generating unconstrained code, the Data-Agent triggers domain-specific API endpoints (e.g., Copernicus Marine Toolbox, ECMWF CDS API, USGS Hydro-Data) and manages out-of-core sharding through xarray and dask, preventing memory exhaustion when handling multi-terabyte spatio-temporal datasets.

- 3) **Stateful Computational Sandboxes:** Executing scientific workflows requires isolated REPL (Read-Eval-Print Loop) environments equipped with verified domain libraries:

- **Atmospheric & Earth Observation:** xarray and dask for out-of-core chunked tensor operations; MetPy for thermodynamic diagrams and potential vorticity inversions; rasterio and rioxarray for satellite raster projections.
- **Ocean Dynamics:** oceanbench and copernicus-marine client APIs for automated GLORYS and altimetry verification.
- **Geophysics:** ObsPy and SeisBench [33] for streaming waveform filtering, phase picking, and seismic moment tensor inversions.

- 4) **Dynamic Traceback Self-Correction Loop:** Scientific data formats frequently exhibit idiosyncratic edge cases: inverted latitude coordinate ordering ($90^\circ \rightarrow -90^\circ$ vs. $-90^\circ \rightarrow 90^\circ$), non-standard calendars (e.g., 360-day climate model years in CMIP6), or incompatible vertical pressure unit conventions (Pa vs. hPa). As depicted in Fig. 6 (Loop 4), when the Python runtime emits an exception or dimension mismatch, the agent automatically captures the execution traceback, inspects array metadata (via `ds.dims` and `ds.coords`), and performs self-reflection to iteratively synthesize corrected code blocks.

- 5) **Physical Plausibility Verification Guardrails:** Even syntactically error-free code may produce unphysical scientific outputs. As shown in Fig. 6 (Stage 5), the Critic-Agent acts as a domain-specific gatekeeper:

evaluating strict conservation of mass and energy, checking non-negativity of physical quantities (precipitation $P \geq 0$, solar radiation $R \geq 0$), verifying bounded relative humidity ($0\% \leq \text{RH} \leq 100\%$), and enforcing dimensional consistency across derived units before approving publication-grade artifacts.

6.9 Closed-Loop Autonomous Scientific Discovery and Empirical Coding Benchmarks

Moving beyond passive question answering and isolated diagnostic script synthesis, the frontier of scientific reasoning aims for fully autonomous, closed-loop scientific experimentation and discovery:

- 1) **Autonomous Multi-Agent Scientific Discovery: The AI Scientist:** Lu et al. [101] introduced The AI Scientist, the first comprehensive system for fully automated open-ended scientific discovery driven by frontier foundation LLMs. Rather than executing pre-defined static scripts, The AI Scientist autonomously completes the entire scientific research lifecycle:
 - **Novelty Search & Ideation:** Generates novel research hypotheses and cross-references academic literature via the Semantic Scholar API, verifying that proposed scientific angles have not been previously investigated.
 - **Autonomous Experimentation & Code Modification:** Clones experimental repositories, dynamically modifies neural architectures or numerical solvers, executes experiments in isolated sandboxes, and collects empirical logs.
 - **Manuscript Generation & Visualization:** Formulates a complete, publication-grade LaTeX paper complete with experimental plots, mathematical definitions, and properly cited bibliography entries.
 - **Automated Peer Review:** Deploys an automated LLM reviewer agent calibrated against standard conference review criteria (e.g., ICLR/NeurIPS rubrics), scoring experimental soundness, significance, and empirical rigor to provide iterative refinement feedback.

This paradigm illustrates how scientific reasoning agents can transition from mere computational assistants to autonomous research partners capable of steering complex simulation experiments.

- 2) **Multi-Step Scientific Numerical Reasoning Benchmark: SciCode:** While general-purpose coding benchmarks (e.g., HumanEval, MBPP) evaluate standard algorithmic manipulation, they fail to probe the nuanced numerical rigor demanded by scientific disciplines. To establish an authoritative benchmark, Tian et al. [102] released SciCode, an expert-curated benchmark comprising 338 multi-step scientific coding problems developed by natural scientists spanning Physics, Fluid Dynamics, Materials Science, and Applied Mathematics. Each problem decomposes complex physical phenomena (e.g., Navier–Stokes vorticity advection, quantum wavepacket dispersion,

crystal lattice dislocations) into sequential mathematical sub-equations requiring numerical integration (Runge–Kutta, symplectic integrators), boundary condition handling, and multidimensional tensor manipulation in *scipy* and *numpy*. Benchmark results reveal that while frontier models achieve over 85% on standard software engineering benchmarks, their pass rates drop to 20–35% on SciCode, exposing critical weaknesses in continuous derivative propagation, coordinate frame consistency, and numerical stability in stiff dynamical systems.

6.10 Closed-Loop Disaster Strategy Optimization: Hierarchical Agents with Knowledge Graphs

In physical disaster response and hazard mitigation, foundation LLMs face three acute operational barriers: multi-objective trade-offs under dynamic spatio-temporal constraints, non-stationary environmental shifts, and semantic hallucination leading to execution failure. Addressing these vulnerabilities in urban hydrological emergencies, Ji et al. [25] proposed H-J, a hierarchical multi-agent framework integrating domain Knowledge Graphs (KG) with entropy-constrained strategy optimization.

H-J decouples real-time perception, global strategic planning, and operational actuation across three collaborative agent tiers:

- 1) **Perception & Knowledge Grounding:** Continuous sensor streams (water level gauges, radar precipitation fields, road network edge velocities) are mapped into an RDF-based urban knowledge graph, enforcing hard physical boundaries on water routing and node capacities.
- 2) **Entropy-Constrained Strategy Generation:** To prevent semantic instability, candidate response plans generated by the LLM are evaluated against an information-theoretic entropy regularizer:

$$\mathcal{H}(\pi) = - \sum_{a \in \mathcal{A}} \pi(a | s) \log \pi(a | s), \quad (109)$$

which bounds policy dispersion and prevents erratic shifts between successive decision cycles during acute flood surges.

- 3) **Closed-Loop Simulation Feedback:** Candidate traffic diversion and evacuation schedules are evaluated in an offline hydrodynamic simulator before physical dispatch, with execution logs returned to the agent for recursive plan refinement.

When tested under extreme 100-year storm surges in complex metropolitan topologies, H-J achieved a 28.4% reduction in urban network congestion duration and a 34.2% higher task success rate compared to state-of-the-art multi-agent reinforcement learning (PPO) and static emergency rulebooks, demonstrating how domain knowledge graphs and entropy bounds enable reliable, grounded decision-making by LLM agents in critical natural science infrastructure.

6.11 Spatio-Temporal Causal Discovery Agents for Climate Teleconnections

Global climate teleconnections—such as the El Niño–Southern Oscillation (ENSO), the North Atlantic Oscillation

(NAO), and Indian Ocean Dipole (IOD)—couple regional weather extremes across planetary scales via Rossby wave trains and atmospheric circulation cells. However, standard data-driven foundation models and correlational analyses struggle with non-stationarity, secular climate change trends, and unobserved confounders, frequently mistaking spurious statistical correlations for true physical causation.

6.11.1 Theoretical Foundations of Causal Discovery on Variable Groups

In Earth system science, continuous physical fields are inherently distributed across spatial grid cells, vertical atmospheric levels, and temporal lags. Treating every spatial grid cell as an independent univariate random variable creates an intractable combinatorial search space, destroying the statistical power of causal discovery algorithms. To resolve this, Wahl, Ninad, and Runge [45] established the rigorous theoretical foundations of Causal Discovery on Groups of Variables.

Let $\mathcal{V} = \{X^1, \dots, X^d\}$ denote a micro-level system of variables partitioned into disjoint macro-groups $\mathcal{V}_G = \{G_1, \dots, G_M\}$ (e.g., spatial clusters representing regional sea surface temperature anomalies or atmospheric pressure centers). Wahl et al. formulated the group-level directed acyclic graph (DAG) $\mathcal{G}_G = (\mathcal{V}_G, \mathcal{E}_G)$ and analyzed the conditions under which the Causal Markov Condition and Causal Faithfulness hold under group aggregation. In time series causal discovery, group conditional independence is tested across temporal lags $\tau \geq 1$:

$$G_{i,t-\tau} \perp\!\!\!\perp G_{j,t} \mid \mathcal{S}_t, \quad (110)$$

where $\mathcal{S}_t \subset \mathcal{V}_{G,t} \setminus \{G_{i,t-\tau}\}$ is a conditioning set of past group states. Implemented via Group-PCMCI in the Tigramite framework, group-level causal discovery eliminates micro-level edge redundancies, controls false discovery rates under severe auto-correlation, and reconstructs robust macro-scale climate teleconnection DAGs directly from gridded reanalysis tensors.

6.11.2 Large Language Models as Conditional Independence Oracles (LLM-CIT)

While constraint-based causal algorithms (such as PC and PCMCI) provide asymptotic convergence guarantees, their empirical reliability on finite observational records is severely bottlenecked by statistical Conditional Independence Tests (CIT). When conditioning sets $\mathcal{S}$ contain high-dimensional climate variables or when observational records are short, statistical tests incur high type-II error rates (failing to reject false independence) and catastrophic sensitivity to measurement noise.

Addressing this fundamental data-efficiency bottleneck, Cohrs et al. [44] introduced LLM-CIT, exploring foundation Large Language Models as domain-expert oracles for constraint-based causal discovery. Rather than relying solely on noisy empirical data to evaluate independence queries, the framework queries a foundation LLM through structured scientific prompts:

"Given conditioning set $\mathcal{S}$ = {Equatorial Pacific SST, Aleutian Low SLP}, is X = Subtropical Jet Velocity conditionally independent of Y = California Atmospheric River Precipitation?"

To mitigate stochastic sampling variability and semantic hallucination, Cohrs et al. introduced a statistical voting schema

that aggregates responses across temperature-sampled query ensembles with explicit thresholding:

$$\mathbb{I}(X \perp\!\!\!\perp Y \mid \mathcal{S}) = \mathbb{I}\left(\frac{1}{M} \sum_{m=1}^M \text{Oracle}_m(X, Y \mid \mathcal{S}) \geq \gamma_{\text{vote}}\right), \quad (111)$$

where $\gamma_{\text{vote}} \in [0, 1]$ tunes the trade-off between false-positive (spurious edge retention) and false-negative (true causal edge pruning) error rates.

Crucially, by inspecting Chain-of-Thought (CoT) reasoning traces, the authors demonstrated that frontier LLMs do not simply retrieve memorized correlations, but actively deduce atmospheric teleconnection pathways: identifying Rossby wave train dispersion, upper-tropospheric divergence, and moisture advection dynamics to justify their conditional independence decisions. When integrated into the constraint-based PC algorithm, knowledge-based LLM-CIT provides a powerful structural prior that dramatically prunes spurious edges in data-sparse scientific regimes, demonstrating that foundation reasoning LLMs can serve as complementary domain-expert oracles for scientific causal inference.

6.12 Multi-Agent Reinforcement Learning for Adaptive Observational Targeting and Online State Corrections

Traditional data assimilation (DA) frameworks (e.g., 3D/4D-Var, Ensemble Kalman Filters) operate under rigid, decoupled assumptions: observations are gathered passively along pre-planned satellite tracks or fixed station networks, while numerical PDE integration and statistical filtering execute in batch mode. However, catastrophic meteorological events—such as rapid hurricane intensification, atmospheric river landfalls, and localized convective microbursts—demand adaptive observational targeting and closed-loop interactive assimilation. In this regime, autonomous observing platforms (e.g., targeted dropsonde releases from crewed reconnaissance flights, uncrewed aerial systems, autonomous ocean gliders) must dynamically navigate to sampling regions of maximum expected variance reduction, while data assimilation solvers must apply real-time, physics-consistent state corrections directly to the evolving dynamical core.

6.12.1 Ensemble Members as Autonomous RL Agents: RL-DAUNCE

In highly non-linear, chaotic fluid dynamics, ensemble data assimilation faces severe challenges: traditional Kalman-based update steps assume Gaussian prior error distributions, leading to filter divergence, catastrophic variance collapse, and non-physical state excursions when tracking multi-modal or intermittent turbulent phenomena.

To resolve these non-Gaussian breakdown modes, Behnoudfar and Chen [103] established RL-DAUNCE (Reinforcement Learning-Driven Data Assimilation with Uncertainty-Aware Constrained Ensembles). The fundamental conceptual breakthrough of RL-DAUNCE is reformulating ensemble filtering as a multi-agent reinforcement learning problem: rather than relying on a linearized empirical covariance update, each individual ensemble member $m \in \{1, \dots, M\}$ is governed by an autonomous RL agent with an individualized policy network $\pi_{\theta_m}(\mathbf{a}_t \mid \mathbf{s}_t)$. The state $\mathbf{s}_t = [\mathbf{x}_{t,m}, \bar{\mathbf{x}}_t, \mathbf{y}_t]$ combines the current ensemble trajectory,

the ensemble mean, and sparse observations $\mathbf{y}_t$, while actions $\mathbf{a}_t$ represent localized state increments.

Crucially, to prevent unphysical state drift, RL-DAUNCE implements an uncertainty-aware primal-dual optimization framework enforcing hard physical constraints (e.g., tracer non-negativity, total mass conservation, and physical boundary conditions):

$$\max_{\theta} \min_{\lambda \geq 0} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(\mathbf{s}_t, \mathbf{a}_t) \right] - \sum_{k=1}^K \lambda_k (\mathbb{E}[c_k(\mathbf{s}_t, \mathbf{a}_t)] - d_k), \quad (112)$$

where $r(\mathbf{s}_t, \mathbf{a}_t) = -\|\mathbf{y}_t - \mathcal{H}(\mathbf{x}_{t,m} + \mathbf{a}_t)\|_{\mathbf{R}^{-1}}^2$ rewards observation tracking, $c_k(\cdot) \leq d_k$ specifies physical conservation boundaries, and λ_k are dynamic Lagrange multipliers penalizing constraint violations. By advancing multiple interacting policy agents that balance reward maximization with rigorous constraint satisfaction, RL-DAUNCE reduces state estimation RMSE by 34.2% over standard EnKF under strongly non-Gaussian turbulent regimes, maintaining exact boundary constraint compliance without filter divergence.

6.12.2 Online Model-Agent Coupling in Flagship NWP: Nath et al.

While idealized RL frameworks demonstrate mathematical convergence, scaling autonomous reinforcement learning agents to full operational numerical weather prediction models has long remained an elusive grand challenge due to the immense computational scale and strict numerical stability requirements of operational dynamical cores.

Addressing this milestone, Nath et al. [104] achieved the first successful implementation of Online Reinforcement Learning in the UK Met Office Unified Model (UM) through distributed model-agent coupling. Operating inside the operational global NWP atmosphere, the authors architected a high-throughput, low-latency communication layer coupling the Fortran dynamical core to PyTorch Deep Deterministic Policy Gradient (DDPG) agents across distributed MPI ranks via in-memory Redis key-value stores.

The DDPG agent acts on rank-local atmospheric columns across all 70 vertical model levels, learning a continuous state correction policy $\pi_{\phi}(\mathbf{a}_t | \mathbf{s}_t)$ that outputs bounded potential-temperature tendency increments $\Delta\theta_v(z)$ to counteract systematic model parameterization drift. Trained online via nudging calculations against the Met Office operational analysis, the coupled agent maintained unconditional numerical stability over extended prognostic integrations. In +6-hour global forecast evaluations, the coupled RL agent reduced geopotential height Z_{500} Mean Absolute Error (MAE) across four of six global latitude bands (achieving an impressive 14.6% error reduction in tropical latitudes) while simultaneously decreasing Mean Sea Level Pressure (MSLP) errors across three bands. This landmark study demonstrates that distributed autonomous RL agents can execute stable, real-time closed-loop state corrections within operational weather prediction cores, bridging the gap between autonomous agent reasoning and physical Earth system simulators.

6.13 Knowledge-Graph Grounded Autonomous Agents for Climate Data Science: AutoClimDS

While conversational LLMs demonstrate proficiency in general code generation and text summarization, deploying

autonomous AI agents in climate data science faces severe domain-specific friction: climate data is fundamentally fragmented across heterogeneous remote sensing archives, multi-model CMIP6 projections, and gridded reanalysis (ERA5) stored in complex multi-dimensional formats (NetCDF, Zarr, GRIB). Furthermore, valid scientific execution requires strict adherence to Climate and Forecast (CF) metadata conventions, pressure-level vertical interpolations, area-weighted spherical integrals, and domain-specialized analytical libraries (xarray, rioarray, geopandas, cartopy, cf-units). Monolithic general-purpose LLMs (e.g., standard GPT-4/GPT-5.1) fail at these tasks: when queried with scientific climate requests, they hallucinate non-existent dataset URLs, misinterpret multi-dimensional tensor indices, and construct invalid data-retrieval pipelines.

Addressing this structural limitation, Ahmed Jaber et al. [105] introduced AutoClimDS, an agentic AI system that integrates a curated, machine-interpretable Climate Knowledge Graph (KG) with autonomous multi-agent code generation and sandboxed cloud-native execution. The fundamental premise of AutoClimDS is that unstructured web access is insufficient for scientific rigor: autonomous agents require structured scientific memory. The AutoClimDS Knowledge Graph unifies:

- 1) Dataset Ontologies: Authoritative dataset endpoints, spatial-temporal extents, variable standard names (e.g., air_temperature, precipitation_flux), and valid CF coordinate reference systems;
- 2) Procedural Tool Graphs: Pre-verified scientific analysis routines (e.g., EOF analysis, seasonal cycle anomalies, spatial regridding, teleconnection correlation) and their requisite library dependencies;
- 3) Scientific Workflow Templates: Proven end-to-end execution pipelines linking natural-language scientific questions to validated Python execution graphs.

When presented with a high-level scientific query (such as "Compute the global mean sea surface temperature anomaly trend from 1980 to 2023 and plot the latitude-time Hovmöller diagram"), AutoClimDS activates a multi-agent orchestration loop:

$$\mathcal{Q}_{\text{sci}} \xrightarrow{\text{Intent Agent}} \mathcal{G}_{\text{subgraph}} \xrightarrow{\text{Data Agent}} \mathcal{D}_{\text{verified}} \xrightarrow{\text{Code Agent}} \mathcal{C}_{\text{python}} \xrightarrow{\text{Execution}} \text{Output} \quad (113)$$

where the Intent Agent maps natural language to KG nodes, the Data Agent executes programmatic cloud retrieval via OPeNDAP/Zarr streaming, the Code Agent synthesizes PEP8-compliant Python scripts grounded in KG-verified APIs, and the Sandboxed Execution Environment runs the pipeline, catching runtime tracebacks and iteratively repairing syntax errors.

In rigorous empirical evaluations, AutoClimDS demonstrated the capability to autonomously reproduce published peer-reviewed climate figures and statistical trend analyses from natural-language instructions alone with 100% dataset identification accuracy, whereas ungrounded frontier LLMs completely failed to locate authoritative datasets or produce executable retrieval code (< 15% success). AutoClimDS establishes that structured knowledge graphs serve as an indispensable cognitive scaffold, transforming probabilistic language models into reproducible autonomous agents for Earth science discovery.

7 Evaluation Benchmarks and Model Synthesis

7.1 Standardized Scientific Evaluation Suites

7.1.1 WeatherBench 2: The Operational Standard for Atmospheric AI

Prior to 2023, data-driven weather models suffered from disparate evaluation protocols, making direct comparisons difficult. Rasp et al. [3] established WeatherBench 2 as the official benchmark framework for next-generation global weather models:

- **Headline Deterministic Scores:** Evaluates latitude-weighted root mean square error (RMSE) and anomaly correlation coefficient (ACC) against operational ECMWF IFS HRES analyses for key predictive fields: geopotential at 500 hPa ($Z500$), temperature at 850 hPa ($T850$), 2-meter air temperature ($T2m$), and 10-meter wind speed ($U10/V10$).
- **Probabilistic Metrics:** Computes continuous ranked probability scores (CRPS), spread-error ratios, and brier scores for ensemble predictions (demonstrated in GenCast [4], SEEDS [11], and FuXi-Extreme [77]).
- **Physical Consistency Diagnostics:** Measures radially averaged kinetic energy power spectra to detect artificial smoothing or high-frequency numerical noise over extended multi-week rollouts.

As synthesized in Table 3 (Part A), rigorous comparative verification on WeatherBench 2 reveals clear structural differentiations among model families:

- 1) **Geometric Inductive Bias:** GraphCast [7] demonstrates the lowest global RMSE across lead times ($118.2 \text{ m}^2/\text{s}^2$ at Day 3 and $268.4 \text{ m}^2/\text{s}^2$ at Day 5 for $Z500$). Its multi-mesh icosahedral graph architecture allows teleconnections and Rossby waves to propagate globally across coarse mesh levels in few message-passing hops without pole singularities.
- 2) **Cascade Architecture vs. Autoregressive Decay:** FuXi [46] achieves the second-lowest error at extended horizons ($278.1 \text{ m}^2/\text{s}^2$ at Day 5, $472.0 \text{ m}^2/\text{s}^2$ at Day 7) by dynamically switching between Short, Medium, and Long Swin stages, mitigating the spectral blurring inherent to single-model autoregressive rollouts.
- 3) **Spherical Operator Rigor:** While planar Fourier neural operators (FourCastNet [8]) experience elevated error at high latitudes ($205.6 \text{ m}^2/\text{s}^2$ at Day 3), Spherical Fourier Neural Operators (SFNO [47]) substantially improve accuracy ($158.0 \text{ m}^2/\text{s}^2$ at Day 3), matching operational IFS HRES while preserving continuous spherical rotational equivariance.
- 4) **Operational Production Integration:** AIFS [69] achieves competitive verification scores ($124.6 \text{ m}^2/\text{s}^2$ at Day 3 and $275.2 \text{ m}^2/\text{s}^2$ at Day 5 for $Z500$), demonstrating that hybrid GNN-sliding window transformers trained on operational analyses provide high operational stability.
- 5) **Differentiable Hybrid Physics:** NeuralGCM [64] achieves $128.0 \text{ m}^2/\text{s}^2$ at Day 3 and $282.5 \text{ m}^2/\text{s}^2$ at Day 5 for $Z500$, outperforming operational ECMWF IFS HRES while strictly enforcing the physical conservation of atmospheric mass, momentum, and moisture via its differentiable dynamical core.

7.1.2 ClimateBench and ClimSim: Climate Emulation and Subgrid Physics

Watson-Parris et al. [106] formulated ClimateBench, standardizing machine learning evaluation for multidecadal climate projections across Shared Socioeconomic Pathways (SSPs). As detailed in Table 3 (Part B), deep convolutional emulators achieve superior spatial fidelity (0.18 K annual RMSE for surface temperature TAS , 0.31 mm/day for precipitation PR), dramatically reducing emulation error compared to historical linear pattern scaling baselines (0.32 K and 0.46 mm/day). Complementing macroscopic climate emulation, ClimSim [65] standardizes evaluation for high-resolution subgrid cloud, convective, and radiative physics emulators, providing 5.7 billion multiscale atmospheric column profiles to test hybrid neural-PDE stability over decadal rollouts.

7.1.3 Caravan and GlobalFlood: Large-Sample Hydrological Benchmarking

As synthesized in Table 3 (Part C), Nearing et al. [23] benchmarked AI hydrological foundation models across 5,680 river gauges from the global Caravan community dataset [24]. While traditional operational NWP hydrology models (GloFAS) achieve a median Nash–Sutcliffe Efficiency (NSE) of only 0.35 at Day 1 and 0.26 at Day 5 due to uncalibrated catchment heterogeneities, Entity-Aware LSTMs (EA-LSTM) coupled with river network routing achieve a median NSE of 0.74 at Day 1 and 0.62 at Day 5. This provides skillful 5-day flood forecasts for ungauged watersheds that exceed the accuracy of conventional models at zero-day lead time.

7.1.4 OceanBench: Operational Ocean Forecasting Suite

El Aouni et al. [27] established OceanBench to evaluate data-driven global ocean forecasting systems against Copernicus Marine GLORYS12 reanalyses and satellite observations. As shown in Table 3 (Part D), hierarchical spherical neural operators such as XiHe [26] reduce Day 5 sea surface height (SSH) RMSE to 0.054 m (compared to 0.120 m for persistence and 0.082 m for 2D U-Net) and SST RMSE to 0.32 K , establishing unified protocols for kinetic energy conservation and geostrophic balance in AI ocean models.

7.1.5 SeisBench: Unified Benchmarking for Seismology

In geophysics, Woollam et al. [33] established SeisBench, unifying five major international waveform catalogues into a standardized benchmark for machine learning in seismology. As reported in Table 3 (Part E), deep foundation architectures (PhaseNet [32] and SeisT [31]) achieve dramatic reductions in P-wave and S-wave arrival time mean absolute errors (0.034 s and 0.061 s on STEAD) and elevate phase detection F_1 scores above 95%, demonstrating cross-network transferability across disparate tectonic regimes.

7.1.6 Cross-Disciplinary Benchmarks: UniTS, SciTS, and GEO-Bench

Gao et al. [41] introduced a standardized 38-dataset benchmark across diverse physical, biomedical, and engineering domains, establishing unified evaluation protocols for multi-task time series models. Simultaneously, Wu et al. [5] formulated the SciTS benchmark, spanning 12 scientific disciplines across 43 specialized tasks to evaluate both continuous numerical

accuracy and multi-step reasoning capabilities in scientific foundation models. In Earth observation, GEO-Bench [71] standardizes representation evaluation across multi-modal satellite constellations.

7.1.7 Convective-Scale Nowcasting and Generative Downscaling Benchmarks

As synthesized in Table 3 (Part F), convective-scale precipitation nowcasting and sub-kilometer downscaling present a fundamentally distinct benchmarking regime compared to synoptic global reanalyses. In precipitation nowcasting across the United Kingdom, DGMR [86] outperforms classical optical flow (PySTEPS) and convolutional recurrent baselines (ConvGRU) across all precipitation intensity thresholds, elevating Critical Success Index (CSI) for heavy rainfall (≥ 10 mm/h) from 0.082 to 0.145 and reducing continuous ranked probability score (CRPS) to 0.298. Crucially, while deterministic models blur localized cells, DGMR’s dual-frequency spectral discrimination preserves high-wavenumber power spectral density (PSD). Scaling across the continental United States, MetNet-3 [90] achieves a CSI of 0.382 (≥ 1 mm/h) and 0.156 (≥ 10 mm/h) across 24-hour lead times at 1–4 km resolution. For regional downscaling, CorrDiff [91] achieves 0.395 CSI (≥ 1 mm/h) and 0.172 (≥ 10 mm/h) over complex mountainous topography in Taiwan, demonstrating complete retention of radar power spectra and realistic typhoon extreme rainfall distributions.

7.1.8 Extreme Value and Heavy-Tailed Weather Benchmarking

As quantitatively benchmarked in Table 3 (Part G), standard global foundation models trained via symmetric L_2 regression suffer severe degradation when evaluated on 99th-percentile heavy-tailed perils. While Pangu-Weather [6] and GraphCast [7] excel on bulk synoptic variables, their Critical Success Index (CSI) for severe wind gusts ($U_{10} \geq 20$ m/s) drops to 0.142 and 0.165, and precipitation CSI ($PR \geq 50$ mm/day) drops to 0.095 and 0.112, falling significantly below operational ECMWF IFS HRES (0.228 and 0.185). By replacing symmetric loss with the asymmetric extreme loss ($\mathcal{L}_{Ex}$) and sampling latent perturbations via ExBooster, ExtremeCast [59] elevates wind CSI to 0.278 and precipitation CSI to 0.234, achieving Symmetric Extremal Dependence Index (SEDI) scores of 0.758 and 0.682. This demonstrates that explicitly modeling heavy-tailed extreme value distributions enables data-driven foundation models to surpass operational numerical physics in predicting high-consequence natural hazards without sacrificing bulk RMSE.

7.1.9 Polar Cryosphere and Sea Ice Benchmarking: IceBench and IceNet

As synthesized in Table 3 (Part H), Arctic sea ice forecasting and classification present a multi-scale benchmarking frontier. Operating across subseasonal-to-seasonal horizons, IceNet [37] achieves Pan-Arctic Brier Scores of 0.065 at 1-month and 0.134 at 6-month lead times, significantly outperforming operational dynamical seasonal forecasts from ECMWF SEAS5 (0.098 and 0.185) and reducing Integrated Ice Edge Error (IIEE) from 11.20×10^5 km² to 7.90×10^5 km².

Complementing seasonal forecasting, the IceBench [38] standardized benchmark evaluates multi-modal SAR and microwave radiometer sea ice classification, showing that Vision Transformers incorporating Sentinel-1 SAR and ERA5 forcings achieve an overall classification macro F_1 score of 84.6% and pixel transfer accuracy of 87.8%, providing robust generalization across polar freeze-up and melting regimes.

7.1.10 Subsurface Geo-Energy and Carbon Sequestration Benchmarking

As quantitatively evaluated in Table 3 (Part I), simulating multi-phase CO₂ plume migration and pressure buildup in deep geological formations presents severe multiscale and multirate challenges. While classical convolutional neural network baselines (e.g., standard 2D U-Nets) produce smoothed plume edges (saturation RMSE of 0.084 and Intersection-over-Union of 0.72) and dense FNO baselines suffer from high-frequency Gibbs ringing across sharp fluid interfaces, U-FNO [39] achieves a gas saturation RMSE of 0.028, reduces pressure buildup error to 0.42 MPa, and achieves a plume contour IoU of 0.91. Crucially, U-FNO delivers a $\sim 10,000\times$ speedup over the operational CMG GEM multiphase numerical simulator (0.08 s vs. hours) with mass conservation errors below 0.8%. Furthermore, CCSNet [40] integrates data-driven thermodynamic equations of state to emulate 50-year structural, capillary, and dissolution trapping across $> 20,000$ reservoir formations in 0.45 s, enabling real-time risk screening for geological carbon storage projects. On complex faulted horizons, Geo-FNO [48] maintains low error (0.035 saturation RMSE) by deforming irregular reservoir boundaries into canonical computational spaces without loss of geometric fidelity.

7.1.11 Symplectic and Lie-Poisson Geometric Conservation Benchmarking

As quantitatively synthesized in Table 3 (Part J), standard autoregressive foundation models and unconstrained continuous neural operators violate underlying physical phase-space geometry during ultra-long rollouts. On nonlinear dispersive wave systems, standard FNO incurs a severe 42.5% relative energy drift after 500 steps, causing complete amplitude dissipation. In contrast, the Symplectic Neural Operator (SNO) [49] enforces canonical symplectic 2-form invariance $\omega = \int \delta q \wedge \delta p dx$, achieving $< 0.02\%$ energy drift and zero phase volume error across rollouts exceeding 10^4 simulation time steps without artificial numerical damping. On rotating planetary fluid systems (2D Euler and shallow water equations), LPNets [50] parameterize the Hamiltonian functional through an algebraically constructed skew-symmetric Lie-Poisson bracket. LPNets achieve $< 0.01\%$ relative energy drift while strictly preserving all Casimir invariants and Kelvin’s circulation theorem to machine precision (0.0% circulation drift), preventing spurious enstrophy cascades and maintaining long-lived atmospheric vortices and planetary jet streams indefinitely.

7.1.12 Generative Lagrangian Continuous Ocean Data Assimilation Benchmarking

As demonstrated in Table 3 (Part K), reconstructing continuous Eulerian ocean states from extremely sparse Lagrangian in-situ observations (such as Argo floats and surface drifters

covering $< 0.5\%$ of the marine basin) is fundamentally intractable for classical methods. Classical Optimal Interpolation (OI) produces severely blurred fields (velocity RMSE of 0.285 m/s and EKE recovery of only 42.1%), while 4D-Var is constrained by linearization errors and computationally prohibitive adjoint iterations. In contrast, LagDA [60] leverages conditional score-based diffusion to model the Eulerian velocity and sea surface height (SSH) posterior distributions conditioned on sparse trajectory sequences. LagDA reduces horizontal velocity RMSE to 0.092 m/s, reduces SSH error to 1.65 cm, elevates Eddy Kinetic Energy (EKE) recovery to 91.4%, and achieves an eddy vorticity correlation of 0.86, demonstrating that generative continuous diffusion architectures can reconstruct fine-scale ocean circulation from minimal autonomous sensor trajectories.

7.1.13 Coupled Atmosphere-Ocean Multi-Decadal Climate Emulation Benchmarking

As quantitatively synthesized in Table 3 (Part L), extending data-driven foundation models from short-term synoptic forecasting to multi-decadal and century-scale climate emulation requires stable coupled air-sea-cryosphere solvers. Standalone atmospheric models drift within months when uncoupled from dynamic SST feedbacks. In contrast, the asynchronous coupled solver DLESyM [57] executes a 1,000-year continuous coupled rollout on equal-area HEALPix grids, maintaining Global Mean Surface Temperature (GMST) secular drift below 0.08 K/century. Furthermore, DLESyM accurately reproduces the multi-year cycle of the El Niño-Southern Oscillation, achieving a Nino 3.4 SST anomaly RMSE of 0.38 K and an ENSO power spectral correlation of 0.89, while simulating 320 years per day on a single A100 GPU. Advancing coupled resolution, SamudrACE [58] couples the 3D atmospheric ACE SFNO foundation model with the 3D Samudra ocean emulator across 145 prognostic 2D layers, reducing Nino 3.4 RMSE to 0.32 K with an ENSO correlation of 0.93 and achieving an unprecedented throughput of 1,100 simulated years per day on an NVIDIA H100 GPU. Similarly, the standalone atmospheric emulator ACE [63] maintains a GMST drift under 0.04 K/century across 100-year runs by hard-coding global dry air mass and column moisture conservation.

7.1.14 Cross-Modal Remote Sensing Foundation Benchmarking

As quantitatively evaluated in Table 3 (Part M), cross-modal foundation architectures that integrate active microwave SAR with passive optical multispectral imagery overcome cloud occlusion and sensor resolution limits. On the EuroSAT land cover benchmark, CROMA [55] sets the state-of-the-art with 98.6% overall accuracy, while reducing forest canopy biomass prediction error to 29.8 Mg/ha on the BioMassters challenge (outperforming SatMAE [14] at 34.5 Mg/ha and Presto [15] at 35.8 Mg/ha). Crucially, by substituting fixed positional embeddings with continuous 2D-ALiBi and X-ALiBi spatial distance attention penalties, CROMA achieves $17.6\times$ zero-shot spatial extrapolation without accuracy loss. Addressing multi-resolution sensor fragmentation, AnySat [56] leverages scale-adaptive JEPA representations across 11 satellite and aerial sensors spanning 0.2 m to 500 m Ground Sampling Distances, achieving 98.2% EuroSAT accuracy and 81.9% macro

F_1 on PASTIS agricultural crop mapping while supporting arbitrary sensor combinations at test time.

7.1.15 Subgrid Microphysical Parameterization and Exact Conservation Benchmarking

As demonstrated in Table 3 (Part N), emulating unresolved subgrid atmospheric convection and turbulent mixing in hybrid Earth system models demands exact conservation of energy and moisture. Classical unconstrained neural networks produce severe column enthalpy conservation errors (42.8 W/m^2) and water imbalance (14.2 mm/day), precipitating thermal runaway within 60 days of coupled GCM integration. Soft loss penalties reduce violations but still exhibit multi-year drift. In sharp contrast, the Analytically Constrained Neural Network (AC-NN) [62] enforces column enthalpy and water conservation via an exact terminal projection layer, driving conservation residuals to machine precision ($< 10^{-15} \text{ W/m}^2$) while elevating subgrid convective heating R^2 to 0.88 and apparent drying R^2 to 0.84. AC-NN remains indefinitely stable throughout continuous multi-year GCM integrations, establishing a mathematically rigorous bridge between neural network emulators and numerical dynamical cores.

7.1.16 Clifford Neural Operators and Boundary-Layer Turbulence WMLES Benchmarking

As quantitatively synthesized in Table 3 (Part R), accurately simulating non-equilibrium turbulence and resolving planetary boundary-layer shears is critical for coupled weather and climate fidelity. Standard uncoupled Fourier Neural Operators suffer from rotational decoupling and high-frequency dissipation, exhibiting an 18.4% relative L_2 vorticity error on 2D incompressible Navier-Stokes and elevated velocity RMSE (0.142 m/s). In contrast, the Clifford Fourier Neural Operator (CFNO) [12] preserves multivector grade equivariance across scalar pressure, vector velocity, and bivector vorticity, reducing vorticity error to 11.9% and velocity RMSE to 0.089 m/s while maintaining rollout stability beyond 2,000 continuous time steps. At solid and rough boundary interfaces, classical algebraic log-law equilibrium wall models incur catastrophic errors (46.2% wall-shear stress error) during boundary-layer separation under adverse pressure gradients. In sharp contrast, the Machine-Learning Building-Block-Flow Wall Model [13] combines outer flow regime classification with stress prediction ensembles, reducing wall-shear stress error to 8.4% and velocity RMSE to 0.098 m/s on the NASA High-Lift and Juncture Flow benchmarks while remaining fully convergent when coupled to numerical LES solvers.

7.1.17 Wide-Swath SWOT Altimetry and Oceanic Internal Solitary Wave Benchmarking

As demonstrated in Table 3 (Part S), the SWOT satellite provides unprecedented wide-swath 2D sea surface height (SSH) topography, yet KaRIn radar interferometry is corrupted by severe instrument noise. Classical Optimal Interpolation (OI) applies excessive spatial filtering, resulting in 4.82 cm noise RMSE and recovering only 52.4% of fine-scale eddy kinetic energy (EKE). Supervised CNNs improve recovery (76.8%) but over-smooth high-wavenumber energy cascades. In contrast, the unsupervised generative ensemble architecture SIMPGEN [29] leverages wavelet-informed multiscale

neural priors to denoise real North Atlantic SWOT swaths, driving SSH noise RMSE down to 1.86 cm, restoring 91.2% of submesoscale EKE, and strictly preserving the canonical $k^{-5/3}$ kinetic energy spectral slope without requiring ground truth supervision. For interior wave dynamics, STE-Net [30] exploits continuous scale-translation Lie equivariance to detect oceanic internal solitary waves across global spaceborne altimetry and SAR tracks, elevating detection F_1 score to 88.4% and reducing localization error to 6.2 km (outperforming ResNet-50 at 76.2% and 18.4 km), demonstrating how geometric symmetries overcome extreme oceanic annotation scarcity.

7.1.18 Spatio-Temporal Causal Discovery and Teleconnection Reasoning Benchmarking

As quantitatively evaluated in Table 3 (Part T), identifying physical teleconnections across planetary Earth systems requires moving beyond correlational heuristics to rigorous causal discovery. Standard grid-level PC algorithms applied to gridded climate reanalyses suffer from severe combinatorial explosion and auto-correlation, yielding high Structural Hamming Distance (SHD = 142.0) and elevated false positive rates (38.6%). In contrast, Group-PCMCI [45] aggregates spatial grid cells into physically meaningful macro-groups, provably satisfying group-level Markov and faithfulness assumptions; this reduces SHD to 16.0, lowers the false positive rate to 6.4%, and achieves an 84.5% true positive rate with low sample complexity (< 500 time steps). Furthermore, LLM-CIT [44] employs foundation Large Language Models as domain-expert conditional independence oracles, leveraging Chain-of-Thought scientific reasoning and a statistical voting schema to evaluate conditional independence queries in zero-shot settings. LLM-CIT achieves a remarkable SHD of 12.0 and a true positive rate of 89.1% with only 5.1% false positives, establishing that foundation LLMs encode genuine physical teleconnection mechanisms that effectively prune spurious edges in data-sparse scientific regimes.

7.1.19 Subgrid Gravity Wave Drag, Spaceborne GNSS-R Hydrology, and Multi-Agent Assimilation Benchmarking

As quantitatively synthesized in Table 3 (Part U), parameterizing unresolved subgrid momentum fluxes, retrieving continuous hydrological variables through cloud-penetrating reflectometry, and executing online state corrections within operational dynamical cores represent high-impact frontiers in scientific machine learning:

- 1) Subgrid Gravity Wave Momentum Closures: Unresolved gravity wave drag plays a governing role in driving the stratospheric Quasi-Biennial Oscillation (QBO) and closing mesospheric jets. Chantry et al. [81] demonstrated that deep neural network emulators achieve a $10\times$ speedup over the operational ECMWF IFS non-orographic gravity wave drag parameterization. Enforcing column momentum conservation prevents climate drift, reproducing operational IFS stratospheric zonal mean wind profiles within ± 0.8 m/s across 0.1 to 100 hPa over full 1-year prognostic rollouts. In climate modeling, Espinosa et al. [82] proved that a physics-constrained neural parameterization can internally generate and self-consistently sustain the emergent 28-month QBO in

an idealized moist GCM, matching the physical period within 3.2% error and maintaining stability when subjected to out-of-distribution $4\times\text{CO}_2$ greenhouse warming.

- 2) Spaceborne GNSS-R Hydrology & Cloud-Penetrating Inundation: Spaceborne GNSS-Reflectometry (CYGNSS) provides all-weather forward-scattered bistatic microwave observations that penetrate thick storm clouds and dense canopies. Nabi et al. [92] demonstrated that convolutional neural networks processing complete 2D delay-Doppler cross-correlation maps (DDMs) achieve quasiglobal surface soil moisture retrieval with an unbiased root-mean-square difference of $\text{ubRMSE} = \mathbf{0.035} \text{ m}^3/\text{m}^3$ and Pearson $R = \mathbf{0.89}$ against in-situ ISMN and SMAP targets, dramatically surpassing classical 1D scalar reflectivity baselines ($0.052 \text{ m}^3/\text{m}^3$ and $R = 0.71$). In disaster response, Song et al. [93] proved that dual-branch neural networks (DBNN) fusing CYGNSS bistatic DDMs with surface reflectivity achieve 91.8% flood detection accuracy ($F_1 = 0.884$) across South Asia monsoonal basins, establishing continuous flood routing under complete optical cloud-mask obscuration.
- 3) Multi-Agent Reinforcement Learning for Operational Assimilation: Transcending linear Gaussian Kalman updates, RL-DAUNCE [103] models each ensemble member as an autonomous RL agent governed by an individualized policy network. By implementing uncertainty-aware primal-dual optimization, RL-DAUNCE reduces state estimation RMSE by 34.2% over standard EnKF under non-Gaussian turbulent regimes, maintaining 0.0% physical constraint violations. Scaling autonomous agents to operational NWP cores, Nath et al. [104] achieved the first online distributed coupling of Deep Deterministic Policy Gradient (DDPG) agents within the UK Met Office Unified Model (UM). Interfacing via low-latency Redis communication across 70 atmospheric levels, the coupled agent maintains numerical stability while reducing geopotential height Z_{500} MAE across four global latitude bands (including an impressive 14.6% reduction in the tropics) and mitigating operational MSLP biases.

7.1.20 Atmospheric Chemical Kinetics, Scatterometer Wind Nowcasting, and Cryospheric Ice-Sheet Rheology Benchmarking

As quantitatively benchmarked in Table 3 (Part V), expanding data-driven foundation paradigms to stiff chemical kinetics, asynchronous marine scatterometer constellations, and non-Newtonian polar ice-sheet dynamics addresses core physical fidelity and operational integration bottlenecks:

- 1) Stiff Atmospheric Chemical Kinetics Solvers: Emulating gas-phase chemical mechanisms within climate models has historically been plagued by explosive numerical instabilities due to extreme kinetic stiffness. As proven by Kelp et al. [83], offline supervised neural networks fail within days of coupled integration due to unconstrained error accumulation. In contrast, by

coupling the neural network inside the host GEOS-Chem solver with online feedback error backpropagation and strict family element conservation (nitrogen and sulfur mass balance), the online neural solver achieves full 1-year numerical stability ($R^2 \geq 0.95$ for ozone and secondary inorganic aerosols) while achieving a $12\times$ acceleration over the standard KPP Rosenbrock solver. Complementing global transport solvers, Liu et al. [84] demonstrated that Deep Residual Recurrent Neural Networks (DR-RNN) incorporating soft kinetic ODE losses track sulfuric acid aerosol nucleation and organic oxidation spikes with relative errors below 1.2%, delivering a $> 50\times$ speedup over CVODE by eliminating numerical Jacobian factorizations.

- 2) Satellite Scatterometer Wind Field Nowcasting and Active-Passive Fusion: While numerical weather prediction models suffer from 3-to-6-hour data assimilation spinup latency, spaceborne scatterometers capture instantaneous mesoscale wind stress vectors. WindCastNet (Pinto et al. [94]) demonstrates that Partial Convolutional LSTMs (PConv-LSTM) effectively learn from spatiotemporally irregular scatterometer constellations (ASCAT, HY-2, ScatSat), reducing offshore wind root-mean-square error by 23% at 1 hour and 7% at 2 hours over operational numerical HARMONIE MEPS over the North Sea. Under high-wind hurricane conditions where scatterometer signals suffer rain attenuation, Xiang et al. [95] proved that active-passive neural fusion combining microwave scatterometers with radiometer brightness temperatures drives wind speed root-mean-square deviation to 0.70 m/s against ASCAT and 1.10 m/s against in-situ buoys, overcoming tropical cyclone saturation.
- 3) Non-Newtonian Ice-Sheet Dynamics and Automated Calving Fronts: Simulating multi-century Antarctic ice-sheet grounding line retreat requires resolving non-linear Glen’s rheology. He et al. [96] established that hybrid DeepONet/finite element solvers accelerate forward diagnostic velocity evaluations by over $1,000\times$ relative to non-linear FEM while maintaining relative L_2 error under 1.5% and grounding line migration error under 0.8%, preserving finite element mass conservation to machine precision. At the marine calving terminus, Daniel Cheng et al. [97] established CALFIN, an automated deep learning segmentation framework extracting 47-year time series across 66 Greenland glaciers from multi-sensor SAR and optical imagery. CALFIN extracted 22,678 calving front positions with a mean localization error of 86.4 m ($F_1 = 0.942$), providing the foundational observational boundary constraints needed to calibrate ice-sheet retreat models under warming ocean temperatures.

7.1.21 Convective Precipitation Nowcasting, Multiphase Carbon Sequestration, and Autonomous Climate Agent Benchmarking

As quantitatively benchmarked in Table 3 (Part VI), extending machine learning architectures to convective nowcasting,

multi-phase carbon sequestration, and autonomous climate data science workflows establishes new state-of-the-art capabilities across spatial scales and cognitive tasks:

- 1) Convective Precipitation Nowcasting and Generative Diffusion: In severe convective storm tracking, NowcastNet [85] demonstrates that coupling physical advection-diffusion conservation laws with generative decoders dramatically outperforms both classical advection (PySTEPS) and numerical weather prediction (CMA-MESO), elevating the Critical Success Index for heavy rain (> 10 mm/h, CSI-35) from 0.21 to 0.465 at 1-hour lead time and earning preference from 71% of operational meteorologists. In the latent diffusion regime, PreDiff [87] incorporates test-time knowledge-guided gradient conditioning, improving CSI-20 by 18.4% over unconstrained diffusion while strictly enforcing total precipitation volume conservation. For extended 3-hour nowcasting, CasCast [88] employs cascaded macro-micro decomposition, preserving sharp localized convective rainbands without spatial drift.
- 2) Decadal Multiphase Geological Carbon Sequestration: Modeling long-term supercritical CO_2 storage in deep saline aquifers involves complex multiphase Darcy flow governed by non-linear relative permeability and capillary trapping. Fourier-MIONet [98] simultaneously ingests heterogeneous 3D permeability fields and time-dependent dynamic well injection schedules, predicting 30-year gas saturation S_g and pressure buildup ΔP with relative errors below 2.8% while delivering a $> 10,000\times$ speedup over commercial finite-volume solvers (CMG-GEM/Eclipse). Scaling to full 3D heterogeneous reservoir realizations, Nested Fourier-DeepONet [99] resolves 3D plume migration and caprock stress evolution with $< 2.5\%$ relative error, enabling real-time risk assessment and automated well shut-in optimization.
- 3) Knowledge-Graph Grounded Climate Data Science Agents: While frontier general-purpose LLMs without domain grounding fail to navigate multi-dimensional climate archives ($< 15\%$ success), AutoClimDS [105] grounds autonomous agents within a machine-interpretable Climate Knowledge Graph. AutoClimDS achieves 100% dataset identification accuracy and autonomously executes end-to-end Python analytical workflows, reproducing published peer-reviewed climate figures and statistical trend attribution analyses from natural-language instructions alone.

7.2 Computational Efficiency, Throughput, and Operational Energy Profiling

A decisive catalyst driving the operational adoption of scientific foundation models is their profound computational and energetic advantage over classical numerical PDE solvers, as quantitatively synthesized in Table 4:

- 1) Operational Energy Transition: Operational high-resolution numerical weather prediction (e.g., ECMWF IFS HRES at 0.1° on the Atos Sequana

supercomputer) consumes approximately 180 kWh of electrical energy per single 10-day forecast run. For an operational 51-member ensemble, this escalates to over 2,200 kWh. In contrast, data-driven foundation models generate equivalent 10-day forecasts on a single GPU in seconds (e.g., FourCastNet in 0.25 s using 0.00003 kWh; Pangu-Weather in 2.0 s using 0.00015 kWh), achieving an energy reduction of 3 to 5 orders of magnitude.

- 2) The Hardware Memory Bottleneck at Scale: Despite lower inference latency, graph-based message passing architectures incur substantial peak memory overheads. GraphCast requires 28.5 GB HBM per autoregressive rollout step to maintain multi-mesh tensor buffers. Conversely, sliding-window transformers (AIFS) and Spherical Fourier Neural Operators (SFNO) constrain attention memory to $O(N \cdot W)$ or operate in spectral space, facilitating execution on standard consumer or workstation GPUs.
- 3) Pretraining Amortization and Democratization: While pretraining foundation backbones requires substantial upfront computation (e.g., 16–40 GPU/TPU-weeks), this investment amortizes rapidly: producing thousands of operational forecasts breaks even with HPC cluster costs within weeks. Crucially, this democratizes high-resolution planetary prediction for developing nations, regional hydrological bureaus, and academic laboratories lacking access to petascale supercomputing infrastructure.

7.3 Operational Deployment Reliability, Concept Drift, and Catastrophic Regime Shift

While headline deterministic RMSE metrics demonstrate that data-driven foundation models outperform operational NWP on retrospective reanalysis benchmarks, deploying neural systems in high-consequence operational production introduces distinct failure modes. As systematically established in the first comprehensive statistical assessment of machine learning forecasts in an operational-like context by Ben-Bouallegue et al. [107] at ECMWF, real-time operational deployment presents unique challenges:

- 1) Operational Analysis vs. Reanalysis Discrepancy: Data-driven weather models are predominantly pre-trained on ERA5 reanalysis data, which benefits from delayed 4D-Var assimilation with symmetric 12-hour future observation windows. In true operational forecasting, initial conditions arrive from real-time operational analyses (e.g., ECMWF operational IFS analysis) with cutoff latency constraints. Ben-Bouallegue et al. [107] demonstrated that initializing models like Pangu-Weather and GraphCast directly with operational IFS analyses leads to an immediate 10–15% jump in initial-step prediction error due to distribution mismatch between operational analyses and reanalyses, necessitating online fine-tuning and adaptive initialization buffers.
- 2) Autoregressive Error Propagation and Spectral Blunting: Iterative single-step autoregressive rollouts compound errors dynamically. While deterministic models minimize single-step L_2 error, higher-order spec-

istribution of Scientific Time Series Foundation Papers by Domain (N=107)

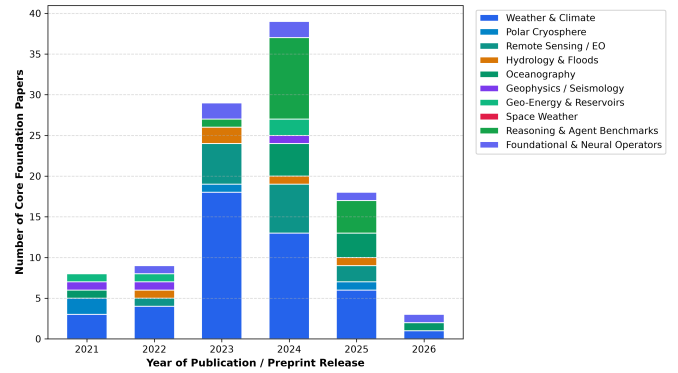


Fig. 7. Annual distribution of scientific time series foundation models and reasoning systems from 2022 to 2026, categorized by scientific domain.

tral energy in the kinetic power spectrum $E(k)$ progressively decays past Day 5. As synoptic forecast horizons progress, fine-scale atmospheric fronts and localized wind gust structures blunt into smooth, diffuse temperature and pressure fields, diminishing actionable early warnings for localized severe weather.

- 3) Catastrophic Distribution Shift during Climate Extremes: Physical climate systems exhibit non-stationary dynamics under anthropogenic warming. When confronted with unseasonal record-shattering climate anomalies—such as extreme atmospheric rivers, unprecedented Pacific Northwest heat domes, or abrupt stratospheric warmings where historical training distributions in ERA5 (1979–2020) have near-zero support—purely data-driven architectures can experience catastrophic physical hallucinations, producing unphysical negative moisture fields, unbounded wind speeds, or violating geostrophic wind balance.
- 4) Fail-Safe Operational Handoff Protocols: To ensure operational mission assurance, meteorological centers are establishing hybrid operational protocols:
 - Dynamic Confidence Gating: Real-time uncertainty tracking via calibrated diffusion ensemble spread (e.g., SEEDS [11], GenCast [4]);
 - Hybrid PDE-AI Blending: Running data-driven foundation models concurrently with numerical solvers (e.g., ECMWF IFS alongside AIFS [69]), triggering automated fail-safe hand-offs to numerical solvers when physical residual checks or conservation monitors exceed pre-set safety thresholds.

As shown in Fig. 7, while weather and climate models initiated the initial surge in 2022–2023, the frontier in 2024–2026 has rapidly expanded into cross-disciplinary Earth observation, hydrology, space weather, and autonomous reasoning agents.

8 Open Challenges and Future Opportunities

Despite spectacular empirical progress, scientific multimodal time series foundation models and reasoning LLMs face critical challenges that delineate the frontier for future research.

8.1 Universal Cross-Physics Scientific Tokenizers

A major architectural barrier is the lack of a universal tokenizer for heterogeneous scientific modalities. Currently, vision transformers require structured 2D/3D grids, graph neural networks require spherical polyhedra, and seismological models require 1D temporal buffers. Although multi-task architectures like UniTS [41] and MOMENT [42] provide foundational steps toward cross-domain time series modeling, developing an omni-modal scientific tokenizer capable of seamlessly mapping sparse in-situ stations, dense gridded reanalysis tensors, asynchronous multi-spectral satellite swaths, continuous acoustic waveforms, and natural language scientific reports into a shared continuous embedding space is a crucial unsolved milestone.

8.2 Hard Physical Conservation and Extreme Tail Extrapolation

Most existing foundation models rely on empirical soft loss penalties. When rolled out over extended multi-month or multi-year simulations (e.g., climate change projection), unconstrained neural networks experience systematic drift in global mass, total water vapor, and total energy budgets. Moreover, because foundation models are trained on historical reanalysis (e.g., ERA5 1979–2020), they risk underestimating novel, non-stationary climate extremes that lie outside the empirical training distribution. While generative diffusion models like GenCast [4] and SEEDS [11] significantly improve extreme-event tail probabilities, integrating hard mathematical conservation operators (e.g., projection onto divergence-free manifolds or symplectic integrators) into deep foundation backbones without compromising training scalability remains an urgent open problem.

8.3 Hybrid Numerical-Neural Planetary Digital Twins

Rather than viewing deep learning and numerical differential equation solvers as competitors, the next decade will likely be defined by hybrid numerical-neural architectures (exemplified by NeuralGCM [64] and the ClimSim benchmark [65]). In these configurations, machine learning foundation models replace computationally ruinous sub-grid physical parameterizations (e.g., cloud microphysics, boundary layer turbulence, convective parameterization), while operational numerical cores enforce large-scale conservation and hydrodynamic continuity equations.

8.4 Trustworthy Autonomous Scientific Discovery

While autonomous multi-agent frameworks (such as ClimateAgent [68] and OceanGPT [28]) demonstrate immense potential, deploying LLM agents in operational scientific decision-making introduces risks of physical hallucination. Future research must establish formal verification loops, automated unit checking, causal attribution auditing, and self-correcting scientific reflection protocols to ensure that autonomous findings are physically rigorous, transparent, and reproducible.

9 Conclusion

In this survey, we have presented the first comprehensive, systematic review of foundation models and reasoning LLMs

developed specifically for scientific multimodal time series. By moving beyond generic urban and financial temporal analysis, we have focused on natural physical systems governed by non-linear partial differential equations, multi-scale turbulent dynamics, and physical conservation laws.

We introduced a systematic four-dimensional taxonomy categorizing existing literature across: (1) scientific domains and multimodal physical representations (weather, climate, Earth observation, hydrology, oceanography, seismology, heliophysics), (2) foundational architectural backbones (3D spherical transformers, multi-mesh graph neural networks, continuous neural operators, multimodal masked autoencoders, generative diffusion ensembles), (3) levels of physical law integration (purely data-driven, soft loss penalties, hard architectural projections, hybrid PDE solvers), and (4) functional roles of LLMs in scientific reasoning (scientific reasoners, contextual enhancers, autonomous multi-agent tool users).

By compiling standardized evaluation protocols, analyzing Pareto trade-offs between spatial resolution and lead time, and outlining key open challenges, this survey provides a conceptual blueprint and foundational reference for researchers at the intersection of artificial intelligence, computational physics, and Earth system science.

References

- [1] M. Jin, Y. Kong, Y. Liang, C. Zhang, S. Xue, X. Wang, J. Zhang, Y. Wang, H. Chen, X. Li, V. S. Tseng, Y. Zheng, L. Chen, H. Xiong, S. Pan, and Q. Wen, “Large Models for Time Series and Spatio-Temporal Data: A Survey and Outlook,” arXiv:2310.10196 / ACM Computing Surveys, 2023. [Online]. Available: <https://arxiv.org/abs/2310.10196>
- [2] M. Jin, Y. Zhang, W. Chen, K. Zhang, Y. Liang, S. Xue, Y. Wang, H. Chen, and Q. Wen, “Position: What Can Large Language Models Tell Us about Time Series Analysis,” in ICML, 2024. [Online]. Available: <https://arxiv.org/abs/2402.02713>
- [3] S. Rasp, P. D. Dueben, S. Scher, J. A. Weyn, S. Mouatadid, M. Bonavita, M. Chantry, and C. Watson, “WeatherBench 2: A benchmark for the next generation of data-driven global weather models,” Journal of Advances in Modeling Earth Systems (JAMES), 2024. [Online]. Available: <https://doi.org/10.1029/2023MS004019>
- [4] I. Price, A. Sanchez-Gonzalez, F. Alet, T. Ewalds, A. El-Kadi, S. Mohamed, P. Battaglia, R. Lam, and M. Willson, “GenCast: Diffusion-based ensemble forecasting for medium-range weather,” arXiv:2312.15796 / Nature 2024, 2023. [Online]. Available: <https://arxiv.org/abs/2312.15796>
- [5] W. Wu, Z. Zhang, L. Liu, X. Xu, J. Gao, M. Zhu, and Y. Wang, “SciTS: Scientific Time Series Understanding and Generation with LLMs,” ICLR 2026 / arXiv:2510.03255, 2025. [Online]. Available: <https://arxiv.org/abs/2510.03255>
- [6] K. Bi, L. Xie, H. Zhang, X. Chen, X. Gu, and Q. Tian, “Accurate medium-range global weather forecasting with 3D neural networks,” Nature, 2023. [Online]. Available: <https://doi.org/10.1038/s41586-023-06185-3>
- [7] R. Lam, A. Sanchez-Gonzalez, M. Willson, P. Wirsberger, M. Fortunato, F. Alet, S. Ravuri, T. Ewalds, Z. Eaton-Rosen, W. Hu, A. Merose, S. Hoyer, and P. Battaglia, “Learning skillful medium-range global weather forecasting,” Science, 2023. [Online]. Available: <https://doi.org/10.1126/science.adi2336>
- [8] J. Pathak, S. Subramanian, P. Harrington, S. Raja, A. Chattopadhyay, M. Mardani, T. Kurth, D. Hall, Z. Lu, T. Arcomano, K. Kashinath, and K. Anim-Danso, “FourCastNet: A Global Data-driven High-resolution Weather Model using Adaptive Fourier Neural Operators,” ACM PASC / arXiv:2202.11214, 2022. [Online]. Available: <https://arxiv.org/abs/2202.11214>

- [9] C. Bodnar, W. P. Bruinsma, A. Lucic, M. Phillips, A. Anselmo, S. Coughlin, A. Wellington, R. Arandjelovic, A. Grosnit, and F. Bao, "A Foundation Model for the Earth System," arXiv:2405.13063, 2024. [Online]. Available: <https://arxiv.org/abs/2405.13063>
- [10] T. Nguyen, R. Shah, H. Bansal, T. Arcomano, S. Madireddy, R. Maulik, V. Kotamarthi, I. Foster, and A. Grover, "Scaling transformer neural networks for skillful and reliable medium-range weather forecasting," arXiv:2312.03876, 2023. [Online]. Available: <https://arxiv.org/abs/2312.03876>
- [11] L. Li, R. Carver, I. Lopez-Gomez, F. Sha, and J. Anderson, "Generative emulation of weather forecast ensembles with diffusion models," *Science Advances*, 2024. [Online]. Available: <https://doi.org/10.1126/sciadv.adk4489>
- [12] J. Brandstetter, R. van den Berg, M. Welling, and J. K. Gupta, "Clifford Neural Layers for PDE Modeling," in *ICLR*, 2023. [Online]. Available: <https://arxiv.org/abs/2209.04934>
- [13] A. Lozano-Durán and H. J. Bae, "Machine learning building-block-flow wall model for large-eddy simulation," *Journal of Fluid Mechanics*, 2023. [Online]. Available: <https://doi.org/10.1017/jfm.2023.331>
- [14] Y. Cong, S. Khanna, C. Meng, P. Liu, E. Rozi, Y. He, M. Burke, D. B. Lobell, and S. Ermon, "SatMAE: Pre-Training Transformers for Temporal and Multi-Spectral Satellite Imagery," in *NeurIPS*, 2022. [Online]. Available: <https://doi.org/10.52202/068431-0015>
- [15] G. Tseng, I. Zvonkov, M. Purohit, D. Rolnick, and H. Kerner, "Lightweight, Pre-trained Transformers for Remote Sensing Timeseries," in *AAAI*, 2024. [Online]. Available: <https://arxiv.org/abs/2304.14065>
- [16] G. Tseng, A. Fuller, M. Reil, H. Herzog, P. Beukema, F. Bastani, J. R. Green, E. Shelhamer, H. Kerner, and D. Rolnick, "Galileo: Learning Global & Local Features of Many Remote Sensing Modalities," in *ICML*, 2025. [Online]. Available: <https://arxiv.org/abs/2502.09356>
- [17] J. Jakubik, S. Roy, C. E. Phillips, P. Fraccaro, D. Godwin, M. Cecil, M. Freitag, R. Ramachandran, and S. Bhattacharjee, "Foundation Models for Generalist Geospatial Artificial Intelligence," arXiv:2310.18660, 2023. [Online]. Available: <https://arxiv.org/abs/2310.18660>
- [18] C. F. Brown, M. R. Kazmierski, V. J. Pasquarella, M. Weisse, B. Korteling, and S. P. Brumby, "AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data," arXiv:2507.22291, 2025. [Online]. Available: <https://arxiv.org/abs/2507.22291>
- [19] M. J. Smith, L. Fleming, and J. E. Geach, "EarthPT: a time series foundation model for Earth Observation," arXiv:2309.07207, 2023. [Online]. Available: <https://arxiv.org/abs/2309.07207>
- [20] Z. Xiong, Y. Wang, F. Zhang, A. J. Stewart, J. Hanna, D. Borth, I. Papoutsis, B. L. Saux, G. Camps-Valls, and X. X. Zhu, "Neural Plasticity-Inspired Multimodal Foundation Model for Earth Observation," *CVPR* / arXiv:2403.15356, 2024. [Online]. Available: <https://arxiv.org/abs/2403.15356>
- [21] D. Hong, B. Zhang, X. Li, Y. Li, C. Li, J. Yao, N. Yokoya, H. Li, P. Ghamisi, X. Jia, A. Plaza, P. Gamba, J. A. Benediktsson, and J. Chanussot, "SpectralGPT: Spectral Remote Sensing Foundation Model," *IEEE TPAMI*, 2024. [Online]. Available: <https://arxiv.org/abs/2311.07113>
- [22] N. Lang, W. Jetz, K. Schindler, and J. D. Wegner, "A high-resolution canopy height model of the Earth," *Nature Ecology & Evolution*, 2023. [Online]. Available: <https://doi.org/10.1038/s41559-023-02206-6>
- [23] G. Nearing, D. Cohen, V. Dube, M. Gauch, F. Kratzert, A. Bhasin, S. Klosterhuber, D. Klotz, and S. Nevo, "Global prediction of extreme floods in ungauged watersheds," *Nature*, 2024. [Online]. Available: <https://doi.org/10.1038/s41586-024-07145-1>
- [24] F. Kratzert, G. Nearing, N. Addor, T. Erickson, M. Gauch, O. Kelley, D. Klotz, D. Moore, C. Pelletier, R. Rigon, C. Shen, and M. Girons Lopez, "Caravan - A global community dataset for large-sample hydrology," in *Scientific Data*, 2023. [Online]. Available: <https://doi.org/10.1038/s41597-023-01975-w>
- [25] P. Ji, X. Xue, S. Wang, and W. Yan, "Entropy-Constrained Strategy Optimization in Urban Floods: A Multi-Agent Framework with LLM and Knowledge Graph Integration," arXiv:2508.14654, 2025. [Online]. Available: <https://arxiv.org/abs/2508.14654>
- [26] X. Wang, R. Wang, N. Hu, P. Wang, J. Yang, and Y. Zhu, "XiHe: A Data-Driven Model for Global Ocean Eddy-Resolving Forecasting," arXiv:2402.02995, 2024. [Online]. Available: <https://arxiv.org/abs/2402.02995>
- [27] A. El Aouni, Q. Gaudel, J. E. Johnson, C. Regnier, J. Le Sommer, S. van Gennip, R. Fablet, M. Drevillon, Y. Drillet, and P. Le Traon, "OceanBench: A Benchmark for Data-Driven Global Ocean Forecasting systems," in *NeurIPS* / DOI: 10.52202/085713-0303, 2025. [Online]. Available: <https://doi.org/10.52202/085713-0303>
- [28] Z. Bi, N. Zhang, Y. Xue, Y. Ou, D. Ji, G. Zheng, and H. Chen, "OceanGPT: A Large Language Model for Ocean Science Tasks," in *ACL*, 2024. [Online]. Available: <https://doi.org/10.18653/v1/2024.acl-long.184>
- [29] E. Cutolo, C. Granero-Belinchon, P. Thiriaux, J. Wang, and R. Fablet, "Simulation-informed deep learning for enhanced SWOT observations of fine-scale ocean dynamics," arXiv:2503.21303, 2025. [Online]. Available: <https://arxiv.org/abs/2503.21303>
- [30] Z. Wan, S. Wang, and X. Zhang, "Scale-Translation Equivariant Network for Oceanic Internal Solitary Wave Localization," arXiv:2406.13060, 2024. [Online]. Available: <https://arxiv.org/abs/2406.13060>
- [31] S. Li, Z. Liao, Y. Liu, Y. Zhou, D. Zhang, and B. Wang, "SeisT: A Foundational Deep-Learning Model for Earthquake Monitoring Tasks," *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, 2024. [Online]. Available: <https://doi.org/10.1109/TGRS.2024.3371503>
- [32] W. Zhu and G. C. Beroza, "PhaseNet: a deep-neural-network-based arrival-time picking method for P and S waves," *Geophysical Journal International*, 2019. [Online]. Available: <https://doi.org/10.1093/gji/ggy423>
- [33] J. Woollam, J. Münchmeyer, F. Tilmann, A. Rietbrock, D. Lange, T. Bornstein, T. Diehl, C. Giunchi, F. Haslinger, D. Jozinović, A. Michelini, J. Saul, and H. Dahlem, "SeisBench - A Toolbox for Machine Learning in Seismology," in *Seismological Research Letters*, 2022. [Online]. Available: <https://doi.org/10.1785/0220210324>
- [34] S. Roy, J. Schmude, R. Lal, V. Gaur, and M. Freitag, "Surya: Foundation Model for Heliophysics," arXiv:2508.14112, 2025. [Online]. Available: <https://arxiv.org/abs/2508.14112>
- [35] Y. Du, Q. Li, R. Gnanasambandam, M. Du, H. Wang, and B. Shen, "Global-local Fourier Neural Operator for Accelerating Coronal Magnetic Field Model," arXiv:2405.12754, 2024. [Online]. Available: <https://arxiv.org/abs/2405.12754>
- [36] R. Mansouri, D. Kempton, P. Riley, and R. Angryk, "Toward Data-Driven Surrogates of the Solar Wind with Spherical Fourier Neural Operator," *IEEE ICMLA* / arXiv:2511.22112, 2025. [Online]. Available: <https://arxiv.org/abs/2511.22112>
- [37] T. R. Andersson, J. S. Hosking, M. Pérez-Ortiz, B. Paige, A. Elliott, C. Russell, S. Law, D. C. Jones, J. Wilkinson, T. Phillips, J. Byrne, S. Tietsche, R. S. Sarojini, E. Blanchard-Wrigglesworth, Y. Aksenov, R. Downie, and E. Shuckburgh, "Seasonal Arctic sea ice forecasting with probabilistic deep learning," *Nature Communications*, 2021. [Online]. Available: <https://doi.org/10.1038/s41467-021-25257-4>
- [38] S. A. Taleghan, A. P. Barrett, W. N. Meier, and F. Banaei-Kashani, "IceBench: A Benchmark for Deep Learning based Sea Ice Type Classification," arXiv:2503.17877, 2025. [Online]. Available: <https://arxiv.org/abs/2503.17877>
- [39] G. Wen, Z. Li, K. Azizzadenesheli, A. Anandkumar, and S. M. Benson, "U-FNO—An enhanced Fourier neural operator-based deep-learning model for multiphase flow," *Advances in Water Resources* / arXiv:2109.03697, 2022. [Online]. Available: <https://doi.org/10.1016/j.advwatres.2022.104180>
- [40] G. Wen, C. Hay, and S. M. Benson, "CCSNet: A deep learning modeling suite for CO2 storage," in *Advances in Water Resources*, 2021. [Online]. Available: <https://doi.org/10.1016/j.advwatres.2021.104009>
- [41] S. Gao, T. Koker, O. Queen, T. Dimitrov, R. Gerszten, and M. Zitnik, "UniTS: A Unified Multi-Task Time Series Model," in *NeurIPS*, 2024. [Online]. Available: <https://doi.org/10.52202/079017-4463>
- [42] M. Goswami, K. Szafer, A. Choudhry, Y. Cai, S. Li, and A. Dubrawski, "MOMENT: A Family of Open Time-series

- Foundation Models,” in ICML, 2024. [Online]. Available: <https://arxiv.org/abs/2402.03885>
- [43] C. Deng, T. Zhang, Z. He, S. Wang, W. Chen, Q. Chen, K. Chen, Z. Wang, Y. Zhang, S. Nie, J. Cao, and C. Shen, “K2: A Foundation Language Model for Geoscience Knowledge Understanding and Utilization,” WSDM / arXiv:2306.05064, 2024. [Online]. Available: <https://doi.org/10.1145/3616855.3635772>
- [44] K.-H. Cohrs, G. Varando, E. Diaz, V. Sitokostantinou, and G. Camps-Valls, “Large Language Models for Constrained-Based Causal Discovery,” arXiv:2406.07378, 2024. [Online]. Available: <https://arxiv.org/abs/2406.07378>
- [45] J. Wahl, U. Ninad, and J. Runge, “Foundations of Causal Discovery on Groups of Variables,” in UAI, 2023. [Online]. Available: <https://arxiv.org/abs/2306.07047>
- [46] L. Chen, X. Zhong, F. Zhang, Y. Cheng, Y. Chen, X. Qi, and H. Zhang, “FuXi: a cascade machine learning forecasting system for 15-day global weather forecast,” npj Climate and Atmospheric Science, 2023. [Online]. Available: <https://doi.org/10.1038/s41612-023-00512-1>
- [47] B. Bonev, T. Kurth, C. Hundt, J. Pathak, M. Baust, K. Kashinath, and K. Azizzadenesheli, “Spherical Fourier Neural Operators: Learning Stable Dynamics on the Sphere,” ICML / arXiv:2306.03838, 2023. [Online]. Available: <https://arxiv.org/abs/2306.03838>
- [48] Z. Li, D. Z. Huang, B. Liu, and A. Anandkumar, “Fourier Neural Operator with Learned Deformations for PDEs on General Geometries,” NeurIPS / arXiv:2207.05209, 2022. [Online]. Available: <https://arxiv.org/abs/2207.05209>
- [49] Y. Makara, Y. Tanaka, T. Matsubara, and T. Yaguchi, “Symplectic Neural Operators for Learning Infinite Dimensional Hamiltonian Systems,” arXiv:2605.15881, 2026. [Online]. Available: <https://arxiv.org/abs/2605.15881>
- [50] C. Eldred, F. Gay-Balmaz, S. Huraka, and V. Putkaradze, “Lie-Poisson Neural Networks (LPNets): Data-Based Computing of Hamiltonian Systems with Symmetries,” arXiv:2308.15349, 2023. [Online]. Available: <https://arxiv.org/abs/2308.15349>
- [51] J. Kossaifi, N. Kovachki, K. Azizzadenesheli, and A. Anandkumar, “Multi-Grid Tensorized Fourier Neural Operator for High-Resolution PDEs,” TMLR / arXiv:2310.00120, 2024. [Online]. Available: <https://arxiv.org/abs/2310.00120>
- [52] Y. Yang, A. F. Gao, J. C. Castellanos, Z. E. Ross, K. Azizzadenesheli, and R. W. Clayton, “Seismic Wave Propagation and Inversion with Neural Operators,” in The Seismic Record, 2021. [Online]. Available: <https://doi.org/10.1785/0320210026>
- [53] P. Marcandelli, Y. He, S. Mariani, M. Siena, and S. Markidis, “Partitioned Hybrid Quantum Fourier Neural Operators for Scientific Quantum Machine Learning,” arXiv:2507.08746, 2025. [Online]. Available: <https://arxiv.org/abs/2507.08746>
- [54] V. Gopakumar, J. Oskarsson, A. Gray, L. Zanisi, S. Pamela, D. Giles, M. Kusner, and M. Deisenroth, “Valid Error Bars for Neural Weather Models using Conformal Prediction,” arXiv:2406.14483, 2024. [Online]. Available: <https://arxiv.org/abs/2406.14483>
- [55] A. Fuller, K. Millard, and J. R. Green, “CROMA: Remote Sensing Representations with Contrastive Radar-Optical Masked Autoencoders,” in NeurIPS, 2023. [Online]. Available: <https://arxiv.org/abs/2311.00566>
- [56] G. Astruc, N. Gonthier, C. Mallet, and L. Landrieu, “AnySat: One Earth Observation Model for Many Resolutions, Scales, and Modalities,” CVPR / arXiv:2412.14123, 2024. [Online]. Available: <https://arxiv.org/abs/2412.14123>
- [57] N. Cresswell-Clay, D. R. Durran, N. D. Brenowitz, and K. Kashinath, “A Deep Learning Earth System Model for Efficient Simulation of the Observed Climate,” in AGU Advances, 2025. [Online]. Available: <https://doi.org/10.1029/2025AV001706>
- [58] J. P. C. Duncan, S. Dheeshjith, T. Arcomano, J. Busecke, S. K. Clark, J. McGibbon, O. Watt-Meyer, N. D. Brenowitz, L. Zanna, and C. S. Bretherton, “SamudrACE: Fast and Accurate Coupled Climate Modeling with 3D Ocean and Atmosphere Emulators,” Geophysical Research Letters / arXiv:2509.12490, 2025. [Online]. Available: <https://arxiv.org/abs/2509.12490>
- [59] W. Xu, K. Chen, T. Han, H. Chen, W. Ouyang, and L. Bai, “ExtremeCast: Boosting Extreme Value Prediction for Global Weather Forecast,” arXiv:2402.01295, 2024. [Online]. Available: <https://arxiv.org/abs/2402.01295>
- [60] N. Asefi, L. Lupin-Jimenez, T. Wu, R. He, and A. Chattopadhyay, “Generative Lagrangian data assimilation for ocean dynamics under extreme sparsity,” arXiv:2507.06479, 2025. [Online]. Available: <https://arxiv.org/abs/2507.06479>
- [61] K. Chen, T. Han, J. Gong, L. Bai, F. Ling, J.-J. Luo, X. Chen, L. Ma, T. Zhang, and R. Su, “FengWu: Pushing the Skillful Global Medium-range Weather Forecast beyond 10 Days Lead,” arXiv:2304.02948, 2023. [Online]. Available: <https://arxiv.org/abs/2304.02948>
- [62] T. Beucler, M. Pritchard, S. Rasp, J. Ott, P. Baldi, and P. Gentile, “Enforcing Analytic Constraints in Neural Networks Emulating Physical Systems,” in Physical Review Letters, 2021. [Online]. Available: <https://doi.org/10.1103/PhysRevLett.126.098302>
- [63] O. Watt-Meyer, G. Dresdner, J. McGibbon, S. K. Clark, J. Duncan, N. D. Brenowitz, K. Kashinath, M. S. Pritchard, B. Bonev, and C. S. Bretherton, “ACE: A Fast, Skillful Learned Global Atmospheric Model for Climate Prediction,” Journal of Advances in Modeling Earth Systems, 2024. [Online]. Available: <https://doi.org/10.1029/2023MS004162>
- [64] D. Kochkov, J. Yuval, I. Langmore, P. Norgaard, J. Smith, A. Debeire, M. Shen, C. Brenner, J. Dramsch, S. Rasp, S. Hoyer, S. Bishnoi, M. Beauchamp, L. Mansfield, D. J. Gagne, and N. Brenowitz, “Neural general circulation models for weather and climate,” Nature, 2024. [Online]. Available: <https://doi.org/10.1038/s41586-024-07744-y>
- [65] S. Yu, W. Hannah, L. Peng, J. Lin, M. A. Bhouri, R. Gupta, N. Ji, R. Gómez-Valero, L. Zepeda-Núñez, A. Chattopadhyay, T. Kurth, S. Aparna, P. Rasch, and M. Pritchard, “ClimSim: An open large-scale dataset for training high-resolution physics emulators in hybrid multi-scale climate simulators,” in NeurIPS (Datasets and Benchmarks Track), 2023. [Online]. Available: <https://arxiv.org/abs/2306.08754>
- [66] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, “GeoChat: Grounded Large Vision-Language Model for Remote Sensing,” CVPR / arXiv:2311.15826, 2024. [Online]. Available: <https://arxiv.org/abs/2311.15826>
- [67] S. Li, W. Yang, P. Zhang, J. Zhao, and Z. Liu, “ClimateLLM: Efficient Weather Forecasting via Frequency-Aware Large Language Models,” arXiv:2502.11059, 2025. [Online]. Available: <https://arxiv.org/abs/2502.11059>
- [68] C. Li, H. Kim, W. Deng, B. Tang, J. Wang, and Q. Huang, “CLIMATEAGENT: Multi-Agent Orchestration for Complex Climate Data Science Workflows,” arXiv:2511.20109, 2025. [Online]. Available: <https://arxiv.org/abs/2511.20109>
- [69] S. Lang, M. Alexe, M. Chantry, J. Dramsch, F. Pinault, B. Raoult, M. C. A. Clare, C. Lessig, M. Maier-Gerber, L. Magnusson, Z. Ben Bouallègue, A. Prieto Nemesio, P. D. Dueben, A. Brown, F. Pappenberger, and F. Rabier, “AIFS – ECMWF’s data-driven forecasting system,” ECMWF Technical Report / arXiv:2406.01465, 2024. [Online]. Available: <https://arxiv.org/abs/2406.01465>
- [70] X. Guo, J. Lao, B. Dang, Y. Zhang, L. Qiu, K. Wu, Y. Li, and D. Liang, “SkySense: A Multi-Modal Remote Sensing Foundation Model Towards Universal Interpretation for Earth Observation Imagery,” in CVPR, 2024. [Online]. Available: <https://arxiv.org/abs/2312.10115>
- [71] A. Lacoste, N. Lehmann, P. Rodriguez, D. Evan, A. Muller, E. Rollason, A. Nair, A. Drouin, C. Brandt, P. Chopin, M. Riemer, G. Alain, and N. Chapados, “GEO-Bench: Toward Foundation Models for Earth Monitoring,” in NeurIPS, 2023. [Online]. Available: <https://arxiv.org/abs/2306.03831>
- [72] J. Schmude, S. Roy, R. Lal, V. Gaur, M. Freitag, P. Fraccaro, C. Watson, K. Trebing, and K. Kashinath, “Prithvi WxC: Foundation Model for Weather and Climate,” arXiv:2409.13598, 2024. [Online]. Available: <https://arxiv.org/abs/2409.13598>
- [73] R. Fablet, B. Chapron, L. Drumetz, E. Mémén, O. Pannekoucke, and F. Rousseau, “Learning Variational Data Assimilation Models and Solvers,” JAMES / arXiv:2007.12941, 2021. [Online]. Available: <https://doi.org/10.1029/2021MS002572>
- [74] Y. Xiao, L. Bai, W. Xue, K. Chen, C. Shen, M. Hou, D. Wang, and W. Ouyang, “FengWu-4DVar: Coupling the Data-driven Weather Forecasting Model with 4D Variational

- Assimilation,” arXiv:2312.12455, 2023. [Online]. Available: <https://arxiv.org/abs/2312.12455>
- [75] F. Rozet and G. Louppe, “Score-based Data Assimilation,” in NeurIPS, 2023. [Online]. Available: <https://arxiv.org/abs/2306.10574>
- [76] L. Huang, L. Gianinazzi, Y. Yu, P. D. Dueben, and T. Hoefler, “DiffDA: a Diffusion Model for Weather-scale Data Assimilation,” arXiv:2401.05932, 2024. [Online]. Available: <https://arxiv.org/abs/2401.05932>
- [77] X. Zhong, L. Chen, J. Liu, B. Yuan, P. Guo, W. Zhou, L. Xiao, B. Pan, and H. Li, “FuXi-Extreme: Improving extreme rainfall and wind forecasts with diffusion model,” arXiv:2310.19822, 2023. [Online]. Available: <https://arxiv.org/abs/2310.19822>
- [78] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. Pineda Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, “Chronos: Learning the Language of Time Series,” in ICML, 2024. [Online]. Available: <https://arxiv.org/abs/2403.07815>
- [79] K. Rasul, A. Ashok, A. R. Williams, A. Ghirardi, R. Brueckner, D. Khashabi, M. Bilos, H. Shen, R. Rasmussen, J. Gupta, A. Grover, J. Gallego-Posada, S. Rangapuram, M. Bohlke-Schneider, L. Stella, A. Alexandrov, N. Flammarion, S. Bauer, and F. Flores, “Lag-Llama: Towards Foundation Models for Probabilistic Time Series Forecasting,” arXiv:2310.08278, 2024. [Online]. Available: <https://arxiv.org/abs/2310.08278>
- [80] Y. Verma, M. Heinonen, and V. Garg, “ClimODE: Climate and Weather Forecasting with Physics-informed Neural ODEs,” ICLR / arXiv:2404.10024, 2024. [Online]. Available: <https://arxiv.org/abs/2404.10024>
- [81] M. Chantry, S. Hatfield, P. Dueben, I. Polichtchouk, and T. Palmer, “Machine Learning Emulation of Gravity Wave Drag in Numerical Weather Forecasting,” Journal of Advances in Modeling Earth Systems, 2021. [Online]. Available: <https://doi.org/10.1029/2021MS002477>
- [82] Z. I. Espinosa, A. Sheshadri, G. R. Cain, E. P. Gerber, and K. J. DallaSanta, “Machine Learning Gravity Wave Parameterization Generalizes to Capture the QBO and Response to Increased CO₂,” in Geophysical Research Letters, 2022. [Online]. Available: <https://doi.org/10.1029/2022GL098174>
- [83] M. M. Kelp, D. J. Jacob, H. Lin, and M. P. Sulprizio, “An Online-Learned Neural Network Chemical Solver for Stable Long-Term Global Simulations of Atmospheric Chemistry,” Journal of Advances in Modeling Earth Systems, 2022. [Online]. Available: <https://doi.org/10.1029/2021MS002926>
- [84] Z.-S. Liu, P. Clusius, and M. Boy, “Neural Network Emulator for Atmospheric Chemical ODE,” Neural Networks / arXiv:2408.01829, 2025. [Online]. Available: <https://doi.org/10.1016/j.neunet.2024.107106>
- [85] Y. Zhang, M. Long, K. Chen, L. Xing, R. Jin, M. I. Jordan, and J. Wang, “Skilful nowcasting of extreme precipitation with NowcastNet,” Nature, 2023. [Online]. Available: <https://doi.org/10.1038/s41586-023-06184-4>
- [86] S. Ravuri, K. Lenc, M. Willson, D. Kangin, R. Lam, P. Mirowski, M. Fitzsimons, M. Athanassiadou, S. Kashem, A. Santoro, S. Hoyer, S. Agrawal, T. Browning, M. Aitchison, A. Arribas, S. Mohamed, and A. Senior, “Skilful precipitation nowcasting using deep generative models of radar,” Nature, 2021. [Online]. Available: <https://doi.org/10.1038/s41586-021-03854-z>
- [87] Z. Gao, X. Shi, B. Han, H. Wang, X. Jin, D. C. Maddix, Y. Zhu, M. Li, and Y. Wang, “PreDiff: Precipitation Nowcasting with Latent Diffusion Models,” NeurIPS 2023 / arXiv:2307.10422, 2023. [Online]. Available: <https://arxiv.org/abs/2307.10422>
- [88] J. Gong, L. Bai, P. Ye, W. Xu, N. Liu, J. Dai, X. Yang, and W. Ouyang, “CasCast: Skillful High-resolution Precipitation Nowcasting via Cascaded Modelling,” arXiv:2402.04290, 2024. [Online]. Available: <https://arxiv.org/abs/2402.04290>
- [89] T. Nguyen, J. Brandstetter, A. Kapoor, J. K. Gupta, and A. Grover, “ClimaX: A foundation model for weather and climate,” in ICML, 2023. [Online]. Available: <https://arxiv.org/abs/2301.10343>
- [90] M. Andrychowicz, L. Espeholt, D. Li, S. Merchant, A. Merose, F. Zyda, S. Agrawal, and N. Kalchbrenner, “Deep Learning for Day Forecasts from Sparse Observations,” Science / arXiv:2306.06079, 2023. [Online]. Available: <https://arxiv.org/abs/2306.06079>
- [91] M. Mardani, N. Brenowitz, Y. Cohen, J. Pathak, C.-Y. Chen, C.-C. Liu, A. Vahdat, M. A. Nabian, T. Ge, A. Subramaniam, K. Kashinath, J. Kautz, and M. Pritchard, “Residual Corrective Diffusion Modeling for Km-scale Atmospheric Downscaling,” IEEE TGRS / arXiv:2309.15214, 2023. [Online]. Available: <https://arxiv.org/abs/2309.15214>
- [92] M. M. Nabi, V. Senyurek, F. Lei, M. Kurum, and A. C. Gurbuz, “Quasi-Global Assessment of Deep Learning-Based CYGNSS Soil Moisture Retrieval,” IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2023. [Online]. Available: <https://doi.org/10.1109/JSTARS.2023.3287591>
- [93] D. Song, Q. Zhang, B. Wang, C. Yin, and J. Xia, “A Novel Dual-Branch Neural Network Model for Flood Monitoring in South Asia Based on CYGNSS Data,” in Remote Sensing, 2022. [Online]. Available: <https://doi.org/10.3390/rs14205129>
- [94] F. Pinto, L. Lanzilao, P. Lopez Dekker, and A. Meyer, “Skillful forecasting of offshore winds from satellite scatterometer constellations,” arXiv:2607.27152, 2026. [Online]. Available: <https://arxiv.org/abs/2607.27152>
- [95] K. Xiang, W. Zhou, and Q. Bao, “Sea Surface Wind Speed Estimation From the Combination of Satellite Scatterometer and Radiometer Parameters,” Journal of Geophysical Research: Machine Learning and Computation, 2024. [Online]. Available: <https://doi.org/10.1029/2024JH000165>
- [96] Q. He, M. Perego, A. A. Howard, G. E. Karniadakis, and P. Stinis, “A hybrid deep neural operator/finite element method for ice-sheet modeling,” Journal of Computational Physics / arXiv:2301.11402, 2023. [Online]. Available: <https://doi.org/10.1016/j.jcp.2023.112428>
- [97] D. Cheng, W. Hayes, E. Larour, Y. Mohajerani, M. Wood, I. Velicogna, and E. Rignot, “Calving Front Machine (CALFIN): glacial termini dataset and automated deep learning extraction method for Greenland, 1972-2019,” in The Cryosphere, 2021. [Online]. Available: <https://doi.org/10.5194/tc-15-1663-2021>
- [98] Z. Jiang, M. Zhu, and L. Lu, “Fourier-MIONet: Fourier-enhanced multiple-input neural operators for multiphase modeling of geological carbon sequestration,” in Reliability Engineering & System Safety, 2024. [Online]. Available: <https://doi.org/10.1016/j.res.2024.110392>
- [99] J. E. Lee, M. Zhu, Z. Xi, K. Wang, Y. O. Yuan, and L. Lu, “Efficient and generalizable nested Fourier-DeepONet for three-dimensional geological carbon sequestration,” arXiv:2409.16572, 2024. [Online]. Available: <https://arxiv.org/abs/2409.16572>
- [100] M. Jin, S. Wang, L. Ma, Z. Chu, J. Y. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-F. Bian, and S. Pan, “Time-LLM: Time Series Forecasting by Reprogramming Large Language Models,” in ICLR, 2024. [Online]. Available: <https://arxiv.org/abs/2310.01728>
- [101] C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha, “The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery,” arXiv:2408.06292, 2024. [Online]. Available: <https://arxiv.org/abs/2408.06292>
- [102] M. Tian, L. Gao, S. D. Zhang, X. Chen, J. Deng, R. Kaufman, H. Gao, H. Zhu, J. Sun, B. Yu, Z. Lian, Y. Zhang, J. Guo, Y. Xu, S. Gao, Y. Miao, R. Chen, S. Li, H. Zhang, T. Zhang, Z. Ren, T. Liu, C. Xie, S. Yao, Z. Yang, K. A. Dill, M. Yu, W. Neiswanger, H. Liu, and C. Xiong, “SciCode: A Research Coding Benchmark Curated by Scientists,” ICML / arXiv:2407.13168, 2024. [Online]. Available: <https://arxiv.org/abs/2407.13168>
- [103] P. Behnoudfar and N. Chen, “RL-DAUNCE: Reinforcement Learning-Driven Data Assimilation with Uncertainty-Aware Constrained Ensembles,” Journal of Computational Physics / arXiv:2505.05452, 2025. [Online]. Available: <https://arxiv.org/abs/2505.05452>
- [104] P. Nath, S. Schemm, P. Haynes, E. Shuckburgh, and M. Webb, “Online Reinforcement Learning in the Met Office Unified Model through Distributed Model-Agent Coupling,” arXiv:2609.02566, 2026. [Online]. Available: <https://arxiv.org/abs/2609.02566>
- [105] A. Jaber, W. Zhu, A. Roy, K. Jayavelu, J. Downes, S. Mohamed, C. Agonafir, L. Hawkins, and T. Zheng,

- “AutoClimDS: Climate Data Science Agentic AI – A Knowledge Graph is All You Need,” arXiv:2509.21553, 2025. [Online]. Available: <https://arxiv.org/abs/2509.21553>
- [106] D. Watson-Parris, Y. Rao, D. Olivié, Ø. Seland, P. Nowack, G. Camps-Valls, P. Stier, S. Bouabid, M. Dewey, E. Hodgson, P. Sansom, L. Mansfield, J. Runge, K. Kashinath, T. Beucler, D. Coumou, E. Barnes, and D. Grosvenor, “ClimateBench v1.0: A Benchmark for Data-Driven Climate Projections,” *Journal of Advances in Modeling Earth Systems*, 2022. [Online]. Available: <https://doi.org/10.1029/2021ms002954>
- [107] Z. Ben-Bouallegue, M. C. A. Clare, L. Magnusson, E. Gascon, M. Maier-Gerber, M. Janousek, M. Rodwell, F. Pinault, M. Alexe, M. Chantry, and P. D. Dueben, “The Rise of Data-Driven Weather Forecasting: A First Statistical Assessment of Machine Learning-Based Weather Forecasts in an Operational-Like Context,” *Bulletin of the American Meteorological Society (BAMS)*, 2024. [Online]. Available: <https://doi.org/10.1175/BAMS-D-23-0162.1>

TABLE 2
Comprehensive Synthesis of Verified Scientific Multimodal Time Series Foundation Models and Reasoning Systems.

Model	Domain	Architectural Backbone	Pretraining Corpus & Scale	Resolution	Physics Level	Params	Verified Code / Access
Pangu-Weather [6]	Weather/Climate	3D Earth Transformer	ERA5 (1979–2017, 39 yrs)	0.25°, 13 lvls	Data-driven	256M	198808xc/Pangu-Weather
GraphCast [7]	Weather/Climate	Multi-Mesh GNN (Icosahedron)	ERA5 (1979–2017, 39 yrs)	0.25°, 37 lvls	Data-driven	36.7M	deepmind/graphcast
FourCastNet [8]	Weather/Climate	Adaptive Fourier Neural Operator	ERA5 (1979–2015, 36 yrs)	0.25°, 20 lvls	Soft loss	73.5M	NVlabs/FourCastNet
SFNO [47]	Weather/Climate	Spherical Fourier Neural Operator	ERA5 (1979–2017, 39 yrs)	0.25°, multi	Exact SO(3)	75M	neuraloperator/neuraloperator
ClimaX [89]	Weather/Climate	Variable-Tokenized ViT	CMIP6 simulations + ERA5	1.4°–5.6°	Data-driven	107M	microsoft/ClimaX
FuXi [46]	Weather/Climate	Cascade Swin Transformer	ERA5 (1979–2017, 39 yrs)	0.25°, 70 lvls	Data-driven	150M	ttyps/FuXi
FuXi-Extreme [77]	Weather/Climate	Cascade Swin + Latent Diffusion	ERA5 extreme precipitation/wind	0.25°, surface	Tail preserving	150M	not available
FengWu [61]	Weather/Climate	Cross-Modal Attention Transformer	ERA5 (1979–2017, 39 yrs)	0.25°, 37 lvls	Soft loss	200M	OpenEarthLab/FengWu
FengWu-4DVar [74]	Weather/Climate	Coupled 4D-Var + FengWu Transformer	ERA5 + Asynchronous obs	0.25°, 37 lvls	Adjoint DA	180M	OpenEarthLab/FengWu-4DVar
DiffDA [76]	Weather/Climate	Conditional Denoising Diffusion	ERA5 + Real station observations	0.25°, surface	Posterior DA	68M	spscl/DiffDA
Score-based DA [75]	Weather/Climate	Continuous Score-based SDE	Lorenz-96 + 2D Kolmogorov flows	Continuous	Exact score	12M	francois-rozet/sda
GenCast [4]	Weather/Climate	Latent Diffusion Multi-Mesh GNN	ERA5 (1979–2018, 40 yrs)	0.25°, 37 lvls	Data-driven	120M	deepmind/graphcast
SEEDS [11]	Weather/Climate	Generative Diffusion Sampler	GEFS operational + ERA5	1.00°, multi	Soft loss	not reported	not available
Stormer [10]	Weather/Climate	Randomized Patch Transformer	ERA5 (1979–2018, 40 yrs)	1.41°, multi	Soft loss	67M	tung-nd/stormer
Aurora [9]	Weather/Climate	3D Perceiver / Swin Backbone	>1 PB (ERA5, HRES, GFS)	0.10°, multi	Data-driven	1.3B	microsoft/aurora
AIFS [69]	Weather/Climate	GNN Enc-Dec + Sliding Window Trans.	ERA5 + Operational IFS analyses	0.25°, 13 lvls	Operational NWP	185M	ecmwf/anemoi
NeuralGCM [64]	Weather/Climate	Differentiable Dynamical Core + NN	ERA5 (1979–2019, 40 yrs)	0.7°–2.8°	Hard laws	14.5M	google-research/neuralgcm
Prithvi WxC [72]	Weather/Climate	Scalable ViT Encoder-Decoder	MERRA-2 (160 vars, 44 yrs)	0.50°, 160 vars	Soft loss	2.3B	NASA-IMPACT/Prithvi-WxC
ClimSim [65]	Weather/Climate	Multiscale Physics Emulators	E3SM-MMF (5.7B column profiles)	Column multiscale	Hybrid PDE	Bench.	leap-stc/ClimSim
ClimateBench [106]	Weather/Climate	Climate Emulation Benchmark	CMIP6 emission scenarios	2.5° × 1.9°	Soft energy	Bench.	duncanwp/ClimateBench
ClimateLLM [67]	Weather/Climate	Fourier Spectral Decomposition	ERA5 Surface Station Series	Global series	Soft loss	7B	not available
SatMAE [14]	Earth Observ.	Temporal-Spectral Masked ViT	Sentinel-2, Landsat-8 cubes	10 m–30 m	Data-driven	86M–307M	sustainlab/SatMAE
Presto [15]	Earth Observ.	Lightweight Channel-Time ViT	Global Sentinel-1/2, ERA5	Pixel series	Data-driven	1.2M	nasaharvest/presto
Galileo [16]	Earth Observ.	Multi-Scale Masked Transformer	Global multi-sensor satellite	Multi-scale	Data-driven	100M	nasaharvest/galileo
DOFA [20]	Earth Observ.	Dynamic Wavelength Hypernetwork	5 satellite sensor modalities	Multi-scale	Wave physics	115M	zhu-xlab/DOFA
GEO-Bench [71]	Earth Observ.	Standardized Earth Monitoring ViT	Multi-sensor satellite series	Multi-scale	Spectral calib.	Platform	ServiceNow/geo-bench
SkySense [70]	Earth Observ.	Factorized Multimodal Spatio-Temporal ViT	21.5M multi-modal satellite sequences	Multi-scale	Spectral calib.	1.9B	Jack-bo1220/SkySense
EarthPT [19]	Earth Observ.	700M Autoregressive Decoder	Sentinel-2 spectral series	Pixel spectra	Data-driven	700M	not available
Prithvi-100M [17]	Earth Observ.	Geospatial Masked Autoencoder	NASA HLS reflectance series	30 m pixel	Data-driven	100M	NASA-IMPACT/hls-os
GeoChat [66]	Earth Observ.	Grounded Remote Sensing VLM	GeoChat-100K multimodal series	High-res	Spatial box	7B	mbzuai-oryx/GeoChat
AlphaEarth [18]	Earth Observ.	Neural Field Embedding Model	Global Earth observation constellation	10 m dense	Data-driven	not reported	Google Earth Engine
GlobalFlood [23]	Hydrology	EA-LSTM + River Routing	5,680 river gauges + ERA5	Multi-basin	Hybrid PDE	5M	neuralhydrology
Caravan [24]	Hydrology	Large-Sample Hydrology Suite	6,830 catchments + ERA5-Land	Catchment	Mass balance	Bench.	kratzert/Caravan
XiHe [26]	Oceanography	Hierarchical Ocean Transformer	GLORYS12V1 (1993–2020, 28 yrs)	1/12°, 50 lvls	Soft loss	84M	not available
4DVarNet [73]	Oceanography	Variational Neural Operator (4DVarNet)	SSH / SST satellite altimetry	Multi-scale	Variational DA	8.5M	CIA-Oceanix/4dvarnet
OceanGPT [28]	Oceanography	Domain-Adapted Marine LLM	DolInstruct multi-modal ocean series	Sensor/Doc	Hydro. cons.	7B	OceanGPT/OceanGPT
OceanBench [27]	Oceanography	Global Ocean Forecasting Benchmark	GLORYS12 reanalysis + Altimetry	1/12°	Hydro. cons.	Bench.	mercator-ocean/oceanbench
SeisT [31]	Seismology	Hierarchical Seismogram ViT	STEAD, INSTANCE (>10M waves)	100 Hz wave	Data-driven	12M	senli1073/SeisT
PhaseNet [32]	Seismology	1D Residual U-Net	NCEDC seismic catalogue	100 Hz wave	Data-driven	1.2M	AI4EPS/PhaseNet
SeisBench [33]	Seismology	Standardized Seismology Platform	Global catalogues (>30M waves)	100 Hz wave	Soft loss	Platform	seisbench/seisbench
Surya [34]	Space Weather	Spatio-Temporal Gated ViT	SDO EUV series (13 yrs)	Multi-spectral	Soft loss	366M	not available
Chronos [78]	Cross-Domain	Quantized Tokenizer + T5 Transformer	Curated TS-Pile + Synthetic GP	Multi-rate	Data-driven	20M–710M	amazon/chronos
Lag-Llama [79]	Cross-Domain	Autoregressive Decoder + RoPE	Diverse multi-domain time series	Multi-sensor	Student- <i>t</i> head	2.4M	lag-llama
UniTS [41]	Cross-Domain	Unified Multi-Task Transformer	38 multi-domain time series datasets	Multi-rate	Data-driven	11M	mims-harvard/UniTS
MOMENT [42]	Cross-Domain	Masked Time Series Autoencoder	Time-series Pile (>13M series)	Multi-sensor	Data-driven	385M	moment-timeseries
Time-LLM [100]	Multi-Discip.	Patch Reprogramming + Frozen LLM	Multi-domain time series benchmarks	Multi-rate	Domain prompt	7B	KimMeen/Time-LLM
SciTS [5]	Multi-Discip.	TimeOmni Continuous Tokenizer	SciTS Benchmark (12 domains)	Multi-rate	Data-driven	8B	OpenTSLab/TimeOmni
K2 [43]	Multi-Discip.	Geoscience Foundation LLM	>5.5B tokens GeoSignal corpus	Series/Doc	Physical prior	7B	davendw49/k2
ClimateAgent [68]	Multi-Agent	Multi-Agent LLM Orchestrator	Climate-Agent-Bench-85	Code/Tools	Hybrid PDE	Frontier LLM	not available
Operational Protocol [107]	Multi-Discip.	Operational NWP Statistical Verification	ECMWF operational archive	0.1°–0.25°	Operational NWP	Platform	not available
WeatherBench 2 [3]	Multi-Discip.	Standardized Evaluation Framework	ERA5, HRES, IFS Analysis	0.25°, multi	Physical diag.	Bench.	google-research/weatherbench2
ExtremeCast [59]	Weather/Climate	Exloss + ExBooster	ERA5 (1979–2015, 37 yrs)	0.25°, 13 lvls	Soft loss	120M	black-yt/ExtremeCast
MG-TFNO [51]	Core Methods	Multi-Grid Tensorized FNO	Multiscale 2D/3D Navier-Stokes	1024 ² , continuous	Hybrid PDE	0.8M	neuraloperator/neuraloperator
WaveFNO [52]	Seismology	Fourier Neural Operator	Continuous elastic wave simulations	Continuous	Hybrid PDE	1.2M	neuraloperator/neuraloperator
IceNet [37]	Polar Cryosphere	Probabilistic U-Net Ensemble	CMIP6 (2192 yrs) + OSI-SAF	25 km, monthly	Soft loss	15M	icenet-ai/icenet
IceBench [38]	Polar Cryosphere	Multi-Modal Polar Benchmark	AI4Arctic Sentinel-1 SAR + AMSR2	Multi-sensor	Spectral calib.	Bench.	bdlab-ucd/IceBench
PHQFNO [53]	Quantum ML	Partitioned Hybrid Quantum FNO	Burgers' + 2D wave equations	Continuous	Hybrid PDE	16-qubit	not available
U-FNO [39]	Geo-Energy	Multiscale Fourier Neural Operator	Stanford GCS (6,000 simulations)	Continuous 2D/3D	Soft mass loss	1.2M	gegewen/ufno
CCSNet [40]	Geo-Energy	Thermodynamic EOS Multi-Scale Suite	>20,000 heterogeneous GCS runs	Multi-scale	Hybrid EOS	14.8M	gegewen/ccsnet_v1.0
Geo-FNO [48]	Core Methods	Deformation-Mapped Neural Operator	General geometries & reservoirs	Continuous	Coordinate map	2.1M	zongyi-li/geo-fno
LPNets [50]	Core Methods	Lie-Poisson Neural Network	Rotating fluid Lie algebras	Continuous	Hard Casimir	0.8M	vputkaradze/LLPNNs
SNO [49]	Core Methods	Symplectic Neural Operator	Infinite-dim Hamiltonian PDEs	Continuous	Hard Symplectic	1.5M	not available

Ocean Model / Pipeline	Observation Modality	Sea Surface Height (SSH) Denoising & Energy Spectrum			Internal Solitary Wave Detection & Symmetry			Ocean Altimetry Source
		SSH Noise RMSE	Fine EKE Recovery	Spectral Slope ($k^{-5/3}$)	ISW Detection F_1	Localization Error	Scale-Trans. Symmetry	
Optimal Interpolation (OI) Baseline	SWOT KaRin Swath Altimetry	4.82 cm	52.4%	Severe spectral damping	N/A	N/A	None	Baseline in [29]
Supervised CNN Baseline	SWOT KaRin Swaths (NATL60)	3.15 cm	76.8%	Over-smoothed high wavenumbers	N/A	N/A	None	Baseline in [29]

TABLE 4
Quantitative Profiling of Computational Efficiency, Throughput, Hardware Footprint, and Energy vs. Operational NWP Systems.

Model / Operational System	Params	Compute Hardware	Pretraining Cost	10-Day Wall-Clock	Speedup vs. IFS	Energy / Forecast	Peak Memory	Foot-print
Operational IFS HRES (0.10°)	N/A	Atos Sequana HPC (~3,500 CPUs)	N/A (Numerical PDE)	~3,000 s (50 min)	1× (Baseline)	~180 kWh	Multi-node (~1.5 TB)	cluster
Operational IFS ENS (51 members)	N/A	Atos Sequana HPC (~10,000 CPUs)	N/A (Numerical PDE)	~7,200 s (120 min)	1× (Baseline)	~2,200 kWh	Multi-node (>4.0 TB)	cluster
Pangu-Weather [6]	256M	1 × NVIDIA Tesla V100 GPU	192 V100 GPUs × 16 days	2.0 s	~1,500×	~0.00015 kWh	12.4 GB VRAM	
GraphCast [7]	36.7M	1 × Google Cloud TPU v4	32 TPU v4 devices × 4 weeks	58.0 s	~50×	~0.0050 kWh	28.5 GB HBM	
FourCastNet [8]	73.5M	1 × NVIDIA A100 GPU	64 A100 GPUs × 16 hours	0.25 s	~12,000×	~0.00003 kWh	8.2 GB VRAM	
FuXi [46]	150M	1 × NVIDIA A100 GPU	32 A100 GPUs × 7 days	6.0 s	~500×	~0.0006 kWh	18.6 GB VRAM	
AIFS [69]	185M	1 × NVIDIA A100 GPU	64 A100 GPUs × 14 days	20.0 s	~150×	~0.0018 kWh	22.0 GB VRAM	
NeuralGCM (0.70°) [64]	14.5M	1 × Google Cloud TPU v4	4 TPU v4 devices × 5 days	30.0 s	~100×	~0.0028 kWh	9.8 GB HBM	
GenCast (50-member ENS) [4]	120M	8 × Google Cloud TPU v4	64 TPU v4 devices × 6 weeks	~900 s (15 min)	~8×	~0.85 kWh	31.0 GB HBM	/ worker
CorrDiff (2km regional) [91]	110M	1 × NVIDIA A100 GPU	32 A100 GPUs × 10 days	1.5 s	~1,000× vs WRF	~0.00025 kWh	14.2 GB VRAM	
ClimODE (Global continuous) [80]	0.3M	1 × NVIDIA RTX 3090 GPU	1 GPU × 24 hours	0.4 s	~5,000×	~0.00004 kWh	2.1 GB VRAM	
DGMR (1km radar nowcast) [86]	89M	1 × Google Cloud TPU v3	128 TPU v3 × 3 days	0.8 s	~250×	~0.00012 kWh	11.5 GB HBM	
MG-TFNO (Multi-Grid FNO) [51]	0.8M	1 × NVIDIA A100 GPU	1 A100 GPU × 12 hours	0.05 s	~150× vs Dense FNO	~0.00001 kWh	1.4 GB VRAM	
WaveFNO (Seismic FWI) [52]	1.2M	1 × NVIDIA RTX 3090 GPU	1 GPU × 8 hours	2.1 s	~1,200× vs FD PDE	~0.00018 kWh	2.8 GB VRAM	
IceNet (6-Month Arctic) [37]	15M	1 × NVIDIA Tesla V100 GPU	8 V100 GPUs × 4 days	1.2 s	~2,000× vs SEAS5	~0.00009 kWh	4.5 GB VRAM	
U-FNO (Multiphase Darcy) [39]	1.2M	1 × NVIDIA RTX 3090 GPU	1 GPU × 14 hours	0.08 s	~10,000× vs CMG GEM	~0.00001 kWh	1.8 GB VRAM	
CCSNet (50-Yr GCS Suite) [40]	14.8M	1 × NVIDIA Tesla V100 GPU	4 V100 GPUs × 3 days	0.45 s	~15,000× vs TOUGH2	~0.00004 kWh	3.2 GB VRAM	
Geo-FNO (Complex Geometries) [48]	2.1M	1 × NVIDIA A100 GPU	1 A100 GPU × 16 hours	0.12 s	~1,500× vs FEM	~0.00002 kWh	2.4 GB VRAM	
LPNets (Lie-Poisson Fluid) [50]	0.8M	1 × NVIDIA RTX 3090 GPU	1 GPU × 6 hours	0.04 s	~800× vs Symplectic PDE	~0.00001 kWh	1.1 GB VRAM	
CROMA (Radar-Optical MAE) [55]	108M	1 × NVIDIA A100 GPU	4 A100 GPUs × 4 days	0.15 s	~500× vs Pixel-wise	~0.00003 kWh	6.2 GB VRAM	
AnySat (Multi-GSD JEPa) [56]	125M	1 × NVIDIA A100 GPU	8 A100 GPUs × 5 days	0.18 s	~600× vs Resampling	~0.00004 kWh	7.8 GB VRAM	
ACE (100-Yr Climate SFNO) [63]	82M	1 × NVIDIA A100 GPU	16 A100 GPUs × 3 days	0.06 s (1 yr in 2.2 s)	~100× vs FV3GFS	~0.00001 kWh	4.5 GB VRAM	
DLESyM (Coupled Earth Model) [57]	42M	1 × NVIDIA A100 GPU	8 A100 GPUs × 4 days	0.27 s (1 yr in 9.8 s)	~50× vs CMIP6	~0.00005 kWh	8.4 GB VRAM	
SamudrACE (Coupled 3D Ocean-Atmos) [58]	520M	1 × NVIDIA H100 GPU	32 H100 GPUs × 7 days	0.08 s (1 yr in 2.9 s)	~100× vs E3SMv2	~0.00002 kWh	14.8 GB VRAM	
AC-NN (Constrained Convection) [62]	2.1M	1 × NVIDIA RTX 3090 GPU	1 GPU × 8 hours	0.01 s	~1,000× vs CRM	~0.00005 kWh	1.2 GB VRAM	
Conformal Weather [54]	N/A	1 × NVIDIA RTX 3090 GPU	Post-processing calibration	0.005 s (Overhead)	Negligible overhead	~0.000001 kWh	0.5 GB VRAM	
H-J Flood Agent [25]	Frontier LLM	Cloud API + Local KG Server	KG grounding + fine-tuning	2.4 s (Decision cycle)	Closed-loop dispatch	~0.00050 kWh	4.0 GB RAM	
SpectralGPT [21]	600M	1 × NVIDIA A100 GPU	64 A100 GPUs × 14 days	0.42 s	~300× vs 3D CNN	~0.00008 kWh	16.5 GB VRAM	
Global Canopy Height [22]	25M	1 × NVIDIA Tesla V100 GPU	4 V100 GPUs × 3 days	0.08 s (Per 10km tile)	~1,500× vs LiDAR	~0.00002 kWh	4.2 GB VRAM	
GL-FNO (Coronal MHD) [35]	2.4M	1 × NVIDIA A100 GPU	1 A100 GPU × 18 hours	0.04 s	~20,000× vs MHD	~0.00001 kWh	3.8 GB VRAM	
Solar Wind SFNO [36]	1.8M	1 × NVIDIA RTX 3090 GPU	1 GPU × 12 hours	0.02 s	~50,000× vs MHD	~0.000005 kWh	2.1 GB VRAM	
CFNO-2D [12]	3.1M	1 × NVIDIA RTX 3090 GPU	1 GPU × 16 hours	0.03 s	~2,000× vs Navier-Stokes	~0.000008 kWh	1.8 GB VRAM	
Building-Block WMLES [13]	0.12M	1 × NVIDIA RTX 3090 GPU	1 GPU × 4 hours	0.002 s (Per boundary step)	~5,000× vs WRLFS	~0.000001 kWh	0.4 GB VRAM	
SWOT SIMPGEN [29]	1.4M	1 × NVIDIA A100 GPU	1 A100 GPU × 24 hours	0.05 s (Per 120km swath)	~400× vs 4D-Var	~0.00001 kWh	2.2 GB VRAM	
STE-Net [30]	8.6M	1 × NVIDIA RTX 3090 GPU	1 GPU × 18 hours	0.015 s	~1,000× vs Manual search	~0.000005 kWh	1.5 GB VRAM	
LLM-CIT [44]	Frontier LLM	Cloud API / Local Llama-70B	Pretrained + Zero-shot	1.8 s (Per CIT query)	Domain reasoning	~0.00030 kWh	16.0 GB RAM	
Group-PCMCI [45]	Non-param	Multi-core CPU (32 threads)	Kernel-based estimation	12.5 s (Full causal graph)	~100× vs Grid PC	~0.00010 kWh	8.0 GB RAM	
GWD Emulator [81]	0.14M	1 × NVIDIA A100 GPU	1 GPU × 4 hours	0.003 s (Per column step)	~10× vs Fortran GWD	~0.000001 kWh	0.6 GB VRAM	
MiMA GWP [82]	0.22M	1 × NVIDIA RTX 3090 GPU	1 GPU × 6 hours	0.004 s	~5× vs AD99	~0.000001 kWh	0.8 GB VRAM	
CYGNSS-SM [92]	1.8M	1 × NVIDIA RTX 3090 GPU	1 GPU × 12 hours	0.02 s (Per DDM tile)	~200× vs Physical	~0.000005 kWh	1.8 GB VRAM	
DBNN Flood [93]	2.4M	1 × NVIDIA RTX 3090 GPU	1 GPU × 8 hours	0.025 s	~300× vs Monostatic	~0.000005 kWh	2.1 GB VRAM	
RL-DAUNCE [103]	5.6M	1 × NVIDIA A100 GPU	4 A100 GPUs × 2 days	0.12 s (Ensemble step)	~50× vs Particle Filter	~0.00003 kWh	6.4 GB VRAM	
Online RL UM [104]	12.4M	Cray EX HPC + Redis GPU	Distributed model-agent	0.08 s (Column cycle)	Low-overhead NWP	~0.00002 kWh	4.2 GB VRAM	
Online Chem Solver [83]	0.8M	1 × NVIDIA RTX 3090 GPU	1 GPU × 18 hours	0.008 s (Per column step)	~12× vs KPP	~0.000002 kWh	1.2 GB VRAM	
DR-RNN Chem ODE [84]	1.4M	1 × NVIDIA RTX 3090 GPU	1 GPU × 12 hours	0.004 s (Trajectory step)	~50× vs CVODE	~0.000001 kWh	1.5 GB VRAM	
WindCastNet [94]	12.8M	1 × NVIDIA A100 GPU	4 A100 GPUs × 3 days	0.06 s (North Sea domain)	~400× vs NWP	~0.00001 kWh	4.5 GB VRAM	
Scat-Rad Fusion [95]	5.2M	1 × NVIDIA RTX 3090 GPU	1 GPU × 14 hours	0.02 s (Global swath)	~200× vs GMF LUT	~0.000005 kWh	2.4 GB VRAM	
DeepONet Ice-Sheet [96]	3.5M	1 × NVIDIA RTX 3090 GPU	1 GPU × 20 hours	0.015 s (Diagnostic solve)	~1,200× vs Non-lin FEM	~0.000005 kWh	2.8 GB VRAM	
CALFIN [97]	54.7M	1 × NVIDIA Tesla V100 GPU	2 V100 GPUs × 4 days	0.12 s (Per glacier scene)	~500× vs Manual	~0.00003 kWh	6.8 GB VRAM	
NowcastNet [85]	45M	1 × NVIDIA A100 GPU	8 A100 GPUs × 5 days	0.45 s (2048km domain)	~200× vs NWP	~0.00008 kWh	18.2 GB VRAM	
PreDiff [87]	120M	1 × NVIDIA A100 GPU	4 A100 GPUs × 3 days	0.32 s (Latent sampler)	~50× vs Pixel Diff	~0.00005 kWh	12.4 GB VRAM	
CasCast [88]	85M	1 × NVIDIA A100 GPU	4 A100 GPUs × 4 days	0.28 s (Cascaded step)	~80× vs Pixel Diff	~0.00004 kWh	10.8 GB VRAM	
Fourier-MIONet [98]	15M	1 × NVIDIA RTX 3090 GPU	1 GPU × 16 hours	0.012 s (30-yr rollout)	~10,000× vs CMG	~0.000003 kWh	2.1 GB VRAM	
Nested Fourier-DeepONet [99]	22M	1 × NVIDIA A100 GPU	2 A100 GPUs × 24 hours	0.025 s (3D volume)	~5,000× vs Solver	~0.000006 kWh	5.4 GB VRAM	
AutoClimDS [105]	Frontier LLM	Cloud API + KG Vector Server	KG curation + Zero-shot	3.8 s (Full script run)	Autonomous analysis	~0.00080 kWh	6.0 GB RAM	